1. **1.** Să se determine punctele de extrem local pentru fiecare dintre următoarele funcţii:

a) $f(x,y) = y\left(x^2 + (\ln y)^2\right)$;

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 2yx = 0 \\ x^2 + (\ln y)^2 + y\left(2\ln y \cdot \frac{1}{y}\right) = 0 \end{cases} \Leftrightarrow \begin{cases} xy = 0 \\ x^2 + (\ln y)^2 + 2\ln y = 0 \end{cases}$$

Cazul I. $x = 0 \Rightarrow (\ln y)^2 + 2\ln y = 0 \Leftrightarrow \ln y(\ln y + 2) = 0$

$\Rightarrow \ln y \in \{-2, 0\} \Rightarrow y \in \{e^{-2}, 1\}$

Cazul II. $y = 0 \Rightarrow$ Nu dau bine logaritmii

$\Rightarrow$ fie $A(0, e^{-2})$, $B(0,1)$

$$H_f(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

$$\begin{cases} \frac{\partial^2 f}{\partial x^2} = 2y \\ \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 2x \\ \frac{\partial^2 f}{\partial y^2} = 2\ln y \cdot \frac{1}{y} + 2\frac{1}{y} = \frac{2}{y}(\ln y + 1) \end{cases}$$

$$\Rightarrow H_f(x,y) = \begin{pmatrix} 2y & 2x \\ 2x & \frac{2}{y}(\ln y + 1) \end{pmatrix}$$

$$H_f(0, e^{-2}) = \begin{pmatrix} 2e^{-2} & 0 \\ 0 & \frac{2}{e^{-2}}(\ln e^{-2} + 1) \end{pmatrix} = \begin{pmatrix} \underbrace{\frac{2}{e^2}}_{\Delta_1} & 0 \\ 0 & \underbrace{2e^2 \cdot (-1)}_{\Delta_2} \end{pmatrix}$$

$\Delta_1 > 0$

$\Delta_2 = \frac{2}{e^2} \cdot 2e^2(-1) = -4 < 0 \Rightarrow$ NU e pct de extrem

$$Hf(0,1) = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

$\Delta_1 > 0$

$\Delta_2 > 0$

$\Rightarrow B(0,1)$ pct de minime

b) $f(x,y) = (x+y)^3 - (x-y)(x-5y)$;

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 3(x+y)^2 - (x-5y) - (x-y) = 0 \\ 3(x+y)^2 + (x-5y) + 5(x-y) = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} 3(x+y)^2 - x + 5y - x + y = 0 \\ 3(x+y)^2 + x - 5y + 5x - 5y = 0 \end{cases} \Leftrightarrow \begin{cases} 3(x+y)^2 - 2x + 6y = 0 \\ 3(x+y)^2 + 6x - 10y = 0 \end{cases}$$

$$R_2 - R_1 \Rightarrow \begin{cases} 3(x+y)^2 - 2x + 6y = 0 \\ 8x - 16y = 0 \Leftrightarrow x = 2y \end{cases} \Rightarrow \begin{cases} 3(3y)^2 - 2(2y) + 6y = 0 \\ x = 2y \end{cases}$$

$$\Leftrightarrow \begin{cases} 3 \cdot 9y^2 - 4y + 6y = 0 \\ x = 2y \end{cases} \Leftrightarrow \begin{cases} 27y^2 + 2y = 0 \\ x = 2y \end{cases} \Leftrightarrow \begin{cases} y(27y+2) = 0 \\ x = 2y \end{cases}$$

I. $y = 0 \Rightarrow x = 0 \Rightarrow A(0,0)$

II. $y = -\frac{2}{27} \Rightarrow x = -\frac{4}{27} \Rightarrow B\left(-\frac{4}{27}, -\frac{2}{27}\right)$

$$Hf(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

$$\begin{cases} \frac{\partial^2 f}{\partial x^2} = 6(x+y) - 2 \\ \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 6(x+y) + 6 \\ \frac{\partial^2 f}{\partial y^2} = 6(x+y) - 10 \end{cases}$$

$Hf(0,0) = \begin{pmatrix} -2 & 6 \\ 6 & -10 \end{pmatrix} \rightarrow \Delta_1 < 0$

$\Delta_2 = 20 - 36 < 0 \Rightarrow$ NU e pct extrem

$Hf\left(-\frac{4}{27}, \frac{-2}{27}\right) = \begin{pmatrix} 6 \cdot \frac{-6}{27} - 2 & 6 \cdot \frac{-6}{27} + 6 \\ 6 \cdot \frac{-6}{27} + 6 & 6 \cdot \frac{-6}{27} - 10 \end{pmatrix} = \begin{pmatrix} \frac{-36-54}{27} & \frac{-36+27 \cdot 6}{27} \\ \frac{-36+27 \cdot 6}{27} & \frac{-36-270}{27} \end{pmatrix}$

$\Delta_1 < 0$

$\Delta_2 = \frac{-10}{3} \cdot \frac{-34}{3} - \frac{14}{3} \cdot \frac{14}{3} = \frac{340}{3} - \frac{196}{3} > 0$

$\Rightarrow B\left(-\frac{4}{27}, -\frac{2}{27}\right)$ pct maxim local

c) $f(x,y,z) = 6x - 4y - 2z - x^2 - y^2 - z^2;$

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \\ \frac{\partial f}{\partial z} = 0 \end{cases} \Leftrightarrow \begin{cases} 6 - 2x = 0 \\ -4 - 2y = 0 \\ -2 - 2z = 0 \end{cases} \Leftrightarrow \begin{cases} x = 3 \\ y = -2 \\ z = -1 \end{cases} \Rightarrow A(3, -2, -1)$$

$$Hf(x,y,z) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial z \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial z \partial y} \\ \frac{\partial^2 f}{\partial x \partial z} & \frac{\partial^2 f}{\partial y \partial z} & \frac{\partial^2 f}{\partial z^2} \end{pmatrix} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix} = Hf(3,-2,-1)$$

(Δ_1: -2; Δ_2: $\begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$; Δ_3: the whole matrix)

$\Delta_1 < 0$

$\Delta_2 > 0 \Rightarrow A(3,-2,-1)$ pct maxim

$\Delta_3 < 0$

d) $f(x,y,z) = -x^2 - y^2 - z^2 - x + xy + 2z;$

$$\Rightarrow \begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \\ \frac{\partial f}{\partial z} = 0 \end{cases} \Leftrightarrow \begin{cases} -2x - 1 + y = 0 \\ -2y + x = 0 \\ -2z + 2 = 0 \end{cases} \Leftrightarrow \begin{cases} y = 2x + 1 \\ x = 2y \\ z = 1 \end{cases} \Leftrightarrow \begin{cases} y = 4y + 1 \\ x = 2y \\ z = 1 \end{cases}$$

$\Leftrightarrow \begin{cases} y = -\frac{1}{3} \\ x = -\frac{2}{3} \\ z = 1 \end{cases} \Rightarrow A\left(-\frac{1}{3}, -\frac{2}{3}, 1\right)$

$$Hf(x,y,z) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial z \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial z \partial y} \\ \frac{\partial^2 f}{\partial x \partial z} & \frac{\partial^2 f}{\partial y \partial z} & \frac{\partial^2 f}{\partial z^2} \end{pmatrix} = \begin{pmatrix} -2 & 1 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix} = Hf\left(-\frac{1}{3}, -\frac{2}{3}, 1\right)$$

(Δ_1, Δ_2, Δ_3 mark the leading principal minors)

$\Delta_1 < 0$
$\Delta_2 > 0$ $\Rightarrow A\left(-\frac{1}{3}, -\frac{2}{3}, 1\right)$ pct max
$\Delta_3 < 0$

e) $f(x,y,z) = x^3 + y^2 + z^2 + 12xy + 2z;$

$$\Rightarrow \begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \\ \frac{\partial f}{\partial z} = 0 \end{cases} \Leftrightarrow \begin{cases} 3x^2 + 12y = 0 \\ 2y + 12x = 0 \\ 2z + 2 = 0 \end{cases} \Leftrightarrow \begin{cases} 3x^2 + 12y = 0 \\ 2y = -12x \\ z = -1 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} x^2 + 4y = 0 \\ y = -6x \\ z = -1 \end{cases} \Leftrightarrow \begin{cases} x^2 - 24x = 0 \\ y = -6x \\ z = -1 \end{cases} \Leftrightarrow \begin{cases} x(x-24) = 0 \\ y = -6x \\ z = -1 \end{cases}$$

$\Rightarrow A(0, 0, -1)$, $B(24, -144, -1)$

$$Hf(x,y,z) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial z \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial z \partial y} \\ \frac{\partial^2 f}{\partial x \partial z} & \frac{\partial^2 f}{\partial y \partial z} & \frac{\partial^2 f}{\partial z^2} \end{pmatrix} = \begin{pmatrix} 6x & 12 & 0 \\ 12 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\Rightarrow Hf(0,0,-1) = \begin{pmatrix} 0 & 12 & 0 \\ 12 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Δ_1
$\Delta_2 < 0$ NU e pct de minim

$$Hf(24,-144,-1) = \begin{pmatrix} 144 & 12 & 0 \\ 12 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$\Delta_1 > 0$

$\Delta_2 > 0 \Rightarrow A(24,-144,-1)$ pct minim

$\Delta_3 > 0$

f) $f(x,y,z) = x^3 + 2y^2 + z^2 + yz - 3x + 6y + 2.$

$$\Rightarrow \begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \\ \frac{\partial f}{\partial z} = 0 \end{cases} \Leftrightarrow \begin{cases} 3x^2 - 3 = 0 \\ 4y + z + 6 = 0 \\ 2z + y = 0 \end{cases} \Leftrightarrow \begin{cases} x^2 = 1 \\ 4y + z + 6 = 0 \\ y = -2z \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} x = \pm 1 \\ -8z + z + 6 = 0 \\ y = -2z \end{cases} \Leftrightarrow \begin{cases} x = \pm 1 \\ z = \frac{6}{7} \\ y = -\frac{12}{7} \end{cases} \Rightarrow A\left(-1, -\frac{12}{7}, \frac{6}{7}\right),$$

$$B\left(1, -\frac{12}{7}, \frac{6}{7}\right)$$

$$Hf(x,y,z) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial z \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial z \partial y} \\ \frac{\partial^2 f}{\partial x \partial z} & \frac{\partial^2 f}{\partial y \partial z} & \frac{\partial^2 f}{\partial z^2} \end{pmatrix} = \begin{pmatrix} 6x & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$Hf\left(-1, \frac{-12}{7}, \frac{6}{7}\right) = \begin{pmatrix} -6 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 1 & 0 \end{pmatrix} \Rightarrow \begin{matrix} \Delta_1 < 0 \\ \Delta_2 < 0 \\ \Delta_3 > 0 \end{matrix} \Rightarrow \text{nu e pct de extrem}$$

$$Hf\left(1, \frac{-12}{7}, \frac{6}{7}\right) = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 1 & 0 \end{pmatrix} \Rightarrow \begin{matrix} \Delta_1 > 0 \\ \Delta_2 > 0 \\ \Delta_3 < 0 \end{matrix} \Rightarrow B \text{ pct min}$$

2. a) Să se determine extremele funcţiei $f(x,y) = 4x + 3y$ cu legătura $x^2 + y^2 = 1$.

$f(x,y) = 4x+3y$

$g(x,y) = x^2+y^2-1 = 0$

$\begin{cases} \frac{\partial f}{\partial x} = 4 \\ \frac{\partial f}{\partial y} = 3 \end{cases}$ $\quad \begin{cases} \frac{\partial g}{\partial x} = 2x \\ \frac{\partial g}{\partial y} = 2y \end{cases}$

$\nabla f = \lambda \nabla g \iff (4,3) = \lambda(2x, 2y)$

$\iff \begin{cases} 4 = 2\lambda x \\ 3 = 2\lambda y \\ x^2+y^2 = 1 \end{cases} \iff \begin{cases} x = \frac{4}{2\lambda} \\ y = \frac{3}{2\lambda} \\ x^2+y^2 = 1 \end{cases} \iff \left(\frac{4}{2\lambda}\right)^2 + \left(\frac{3}{2\lambda}\right)^2 = 1$

$\iff \frac{16}{4\lambda^2} + \frac{9}{4\lambda^2} = 1 \iff \frac{25}{4\lambda^2} = 1 \iff 4\lambda^2 = 25 \iff \lambda^2 = \frac{25}{4} \iff \lambda = \pm\frac{5}{2}$

$x = \frac{4}{\pm 2 \cdot \frac{5}{2}} = \pm\frac{4}{5}$

$y = \frac{3}{\pm 2 \cdot \frac{5}{2}} = \pm\frac{3}{5}$

$\Rightarrow A\left(\frac{4}{5}, \frac{3}{5}\right), B\left(-\frac{4}{5}, -\frac{3}{5}\right)$

$f\left(\frac{4}{5}, \frac{3}{5}\right) = 4 \cdot \frac{4}{5} + 3 \cdot \frac{3}{5} = \frac{25}{5} = 5$

$f\left(-\frac{4}{5}, -\frac{3}{5}\right) = 4 \cdot \left(-\frac{4}{5}\right) + 3 \cdot \left(-\frac{3}{5}\right) = -\frac{25}{5} = -5$

$\Rightarrow A\left(\frac{4}{5}, \frac{3}{5}\right)$ pct de maxim conditionat

$B\left(-\frac{4}{5}, -\frac{3}{5}\right)$ pct de min conditionat

b) Să se determine extremele funcţiei $f(x,y) = 4x^3 + 3x^2y - 4y^3 - 48y$ cu legătura $x^2 + xy + y^2 = 3$.

$f(x,y) = 4x^3 + 3x^2y - 4y^3 - 48y$

$g(x,y) = x^2 + xy + y^2 - 3$

$\begin{cases} \frac{\partial f}{\partial x} = 12x^2 + 6yx \\ \frac{\partial f}{\partial y} = 3x^2 - 12y^2 - 48 \end{cases}$ $\quad \begin{cases} \frac{\partial g}{\partial x} = 2x + y \\ \frac{\partial g}{\partial y} = x + 2y \end{cases}$

$$\nabla f = \lambda \nabla g \iff (12x^2+6yx,\ 3x^2-12y^2-48) = \lambda(2x+y,\ x+2y)$$

$$\iff \begin{cases} 12x^2+6yx = \lambda(2x+y) \\ 3x^2-12y^2-48 = \lambda(x+2y) \\ x^2+xy+y^2-3=0 \end{cases} \iff \begin{cases} 12x^2+6yx = 2x\lambda+y\lambda \\ 3x^2-12y^2-48 = \lambda x+2y\lambda \\ x^2+xy+y^2-3=0 \end{cases}$$

$12x^2+6yx = 2x\lambda+y\lambda$

$3x^2-12y^2-48 = \lambda x+2y\lambda \quad |\cdot 4$

$12x^2+6yx = 2x\lambda+y\lambda$

$12x^2-48y^2-192 = 4\lambda x+8\lambda y \quad -$

$6xy+48y^2+192 = -2\lambda x-7\lambda y$

$\iff 6(3-x^2-y^2)+48y^2+192 = -2\lambda x-7\lambda y$

$\iff 18-6x^2-6y^2+48y^2+192 = -2\lambda x-7\lambda y$

$\iff 18-6x^2+42y^2+192 = -2\lambda x-7\lambda y$

$\iff 18-(6x^2-24y^2-96)+18y^2+96 = -2\lambda x-7\lambda y$

$\iff 18-2(\lambda x+2y\lambda)+18y^2+96 = -2\lambda x-7\lambda y$

$\iff 18-\cancel{2\lambda x}-4\lambda y+18y^2+96 = -\cancel{2\lambda x}-7\lambda y$

$\iff 18y^2+3\lambda y+114=0$

$$\iff \begin{cases} 12x^2+6yx = \lambda(2x+y) \\ 3x^2-12y^2-48 = \lambda(x+2y) \\ x^2+xy+y^2-3=0 \end{cases} \iff \begin{cases} 6x(2x+y)-\lambda(2x+y)=0 \\ \dots \\ \dots \end{cases}$$

$$\Rightarrow \lambda = \frac{12x^2+6xy}{2x+y} = \frac{3x^2-12y^2-48}{x+2y}$$

$\iff (x+2y)\,6x(2x+y) = (2x+y)(3x^2-12y^2-48)$

$\iff (2x+y)(6x^2+12xy-3x^2+12y^2+48)=0$

$\iff (2x+y)(3x^2+12xy+12y^2+48)=0$

$\Leftrightarrow (2x+y)(-9x^2+12(x^2+xy+y^2)+48)=0$

$\Leftrightarrow (2x+y)(-9x^2+12\cdot 3+48)=0$

$\Rightarrow$ Caz I. $2x+y=0 \Rightarrow y=-2x$

Caz II. $9x^2=84$ ✗

$\Rightarrow x^2+xy+y^2-3=0 \Leftrightarrow x^2+x(-2x)+(-2x)^2-3=0$

$\Leftrightarrow x^2-2x^2+4x^2=3 \Leftrightarrow 3x^2=3 \Leftrightarrow x=\pm 1$

$\Rightarrow y=\mp 2$

$f(1,-2)=4+3(-2)-4(-2)^3-48(-2)=4-6+32+96=126$

$f(-1,2)=-4+3\cdot 2-4\cdot 2^3-48\cdot 2=-126$

$\Rightarrow A(1,-2)$ pct maxim condițional, $B(-1,2)$ pct minim condițional

c) Să se determine extremele absolute ale funcţiei $f(x,y)=x^2+3y^2$ pe mulţimea $M=\{(x,y)\in\mathbb{R}^2 \mid x^2-2x+y^2-3\leq 0\}$.

$f(x,y)=x^2+3y^2$

$D: x^2-2x+y^2-3\leq 0 \Leftrightarrow x^2-2x+1+y^2-4\leq 0 \Leftrightarrow (x-1)^2+y^2\leq 4$

I. $\text{Int } D: (x-1)^2+y^2<4$

$\begin{cases} \frac{\partial f}{\partial x}: 2x=0 \\ \frac{\partial f}{\partial y}: 6y=0 \end{cases} \Rightarrow A(0,0)\in \text{Int } D \Rightarrow$ pct critic

$f(0,0)=0$

II. $\text{Fr } D: (x-1)^2+y^2=4$

fie $g(x,y)=(x-1)^2+y^2-4=0$

$\nabla f=\lambda \nabla g \Leftrightarrow (2x,6y)=\lambda(2(x-1),2y)$

$\Leftrightarrow \begin{cases} 2x=\lambda(2x-2) \\ 6y=\lambda 2y \\ (x-1)^2+y^2-4=0 \end{cases} \Leftrightarrow \begin{cases} x=\lambda(x-1) \\ 3y=\lambda y \\ (x-1)^2+y^2-4=0 \end{cases} \Leftrightarrow \begin{cases} x-\lambda x+\lambda=0 \\ 3y-\lambda y=0 \\ (x-1)^2+y^2-4=0 \end{cases}$

$$\Leftrightarrow \begin{cases} x(1-\lambda)+\lambda=0 \\ y(3-\lambda)=0 \\ (x-1)^2+y^2-4=0 \end{cases} \Leftrightarrow \begin{cases} x=\frac{\lambda}{\lambda-1} \\ 3-\lambda=0 \Rightarrow \lambda=3 \Rightarrow x=\frac{3}{2} \\ (x-1)^2+y^2-4=0 \end{cases}$$

$$\Rightarrow \left(\frac{1}{2}\right)^2+y^2=4 \Leftrightarrow y^2=\frac{15}{4} \Rightarrow y=\pm\frac{\sqrt{15}}{2}$$

$$f\left(\frac{3}{2},\pm\frac{\sqrt{15}}{2}\right)=\left(\frac{3}{2}\right)^2+\left(\pm\frac{\sqrt{15}}{2}\right)^2\cdot 3=\frac{9}{4}+\frac{15}{4}\cdot 3=\frac{54}{4}=\frac{27}{2}$$

$\Rightarrow A(0,0)$ pct minim

$B\left(\frac{3}{2},\frac{\sqrt{15}}{2}\right), C\left(\frac{3}{2},-\frac{\sqrt{15}}{2}\right)$ pct max

d) Să se determine extremele absolute ale funcţiei $f(x,y,z)=xyz$ pe mulţimea $M=\{(x,y,z)\in\mathbb{R}^3 \mid x^2+4y^2+9z^2\leq 3\}$.

$f(x,y,z)=xyz$

$D: x^2+4y^2+9z^2\leq 3$

I. $\mathrm{Int}\, D: x^2+4y^2+9z^2<3$

$$\begin{cases} \frac{\partial f}{\partial x}: yz=0 \\ \frac{\partial f}{\partial y}: xz=0 \\ \frac{\partial f}{\partial z}: xy=0 \end{cases} \Rightarrow A(0,0,0)$$

II. $\mathrm{Fr}\, D: x^2+4y^2+9z^2=3$

fie $g(x,y)=x^2+4y^2+9z^2-3=0$

$\nabla f=\lambda\nabla g \Leftrightarrow (yz,xz,xy)=\lambda(2x,8y,18z)$

$$\Leftrightarrow \begin{cases} yz=\lambda 2x \\ xz=\lambda 8y \\ xy=\lambda 18z \\ x^2+4y^2+9z^2=3 \end{cases}$$

$\Rightarrow (xyz)^2 = \lambda^3 288\, xyz \quad | : xyz \neq 0$ (dacă e una 0 sunt toate 0 și nu ne pe Fr. D)

$\Leftrightarrow xyz = 288\lambda^3$

$\underbrace{xyz}_{\lambda zx}$

$18 \cdot 16 = 108 \cdot 18 = 288$

$\Leftrightarrow \begin{cases} 2\lambda x^2 = 288\lambda^3 \mid :2\lambda \neq 0 \\ y \cdot \lambda 8y = 288\lambda^3 \mid :8\lambda \neq 0 \\ \lambda 18 z z = 288\lambda^3 \mid :18\lambda \neq 0 \end{cases} \Leftrightarrow \begin{cases} x^2 = 144\lambda^2 \\ y^2 = 36\lambda^2 \\ z^2 = 16\lambda^2 \end{cases}$

$\Rightarrow 144\lambda^2 + 4 \cdot 36\lambda^2 + 9 \cdot 16\lambda^2 = 3 \Leftrightarrow (144+144+144)\lambda^2 = 3 \quad | : (3 \cdot 144)$

$\Leftrightarrow \lambda^2 = \frac{1}{144}$

$\Rightarrow \begin{cases} x^2 = 1 \\ y^2 = \frac{1}{4} \\ z^2 = \frac{1}{9} \end{cases} \Rightarrow \begin{cases} x = \pm 1 \\ y = \pm \frac{1}{2} \\ z = \pm \frac{1}{3} \end{cases}$

$f(x,y,z) = xyz \Rightarrow f\left(1, \frac{1}{2}, \frac{1}{3}\right) = f\left(-1, -\frac{1}{2}, \frac{1}{3}\right) = f\left(-1, \frac{1}{2}, -\frac{1}{3}\right) =$

$= f\left(1, -\frac{1}{2}, -\frac{1}{3}\right) = \frac{1}{6} \Rightarrow$ val max

$f\left(-1, \frac{1}{2}, \frac{1}{3}\right) = f\left(1, -\frac{1}{2}, \frac{1}{3}\right) = f\left(1, \frac{1}{2}, -\frac{1}{3}\right) =$

$= f\left(-1, -\frac{1}{2}, -\frac{1}{3}\right) = -\frac{1}{6} \Rightarrow$ val min

3. Să se calculeze următoarele integrale improprii:

a) $\int_0^\infty \frac{dx}{1+x^4}$; $= \frac{1}{4} \int_0^\infty \frac{y^{-\frac{3}{4}}}{y+1} dy = I$

Not $x^4 = y \Leftrightarrow x = y^{\frac{1}{4}} \Leftrightarrow dx = \frac{1}{4} y^{-\frac{3}{4}} dy$

$\Rightarrow \begin{cases} p-1 = -\frac{3}{4} \\ p+q = 1 \end{cases} \Leftrightarrow \begin{cases} p = \frac{1}{4} \\ q = \frac{3}{4} \end{cases} \Rightarrow I = \frac{1}{4} B\left(\frac{1}{4}, \frac{3}{4}\right) = \frac{1}{4} \cdot \frac{\Gamma\left(\frac{1}{4}\right) \cdot \Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{1}{4}+\frac{3}{4}\right)} =$

$= \frac{1}{4} \cdot \frac{\frac{\pi}{\sin \frac{\pi}{4}}}{\Gamma(1)} = \frac{1}{2^4} \cdot \frac{\pi}{\frac{\sqrt{2}}{2}} = \frac{\pi}{2\sqrt{2}}$

Recap Integrale Eulериene

$$\Gamma(\alpha)=\int_0^{\infty} x^{\alpha-1}e^{-x}\,dx \quad,\ \alpha>0$$

$$B(p,q)=\int_0^1 x^{p-1}(1-x)^{q-1}\,dx=\int_0^{\infty}\frac{y^{p-1}}{(y+1)^{p+q}}\,dy \qquad p,q>0$$

Prop: $\Gamma(\alpha+1)=\alpha\Gamma(\alpha) \Rightarrow \Gamma(n+1)=n! \quad \forall \alpha, n>0$

$\Gamma(\alpha)\cdot\Gamma(1-\alpha)=\frac{\pi}{\sin\alpha\pi} \quad \forall \alpha\in(0,1) \Rightarrow \Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$

$$B(p,q)=\frac{\Gamma(p)\cdot\Gamma(q)}{\Gamma(p+q)}$$

b) $\int_0^{\infty}\frac{dx}{\sqrt[3]{x}(1+x^2)}$; $=\frac{1}{2}\int_0^{\infty}\frac{y^{-\frac{1}{2}}}{y^{\frac{1}{6}}(1+y)}\,dy=\frac{1}{2}\int_0^{\infty}\frac{y^{-\frac{2}{3}}}{y+1}\,dy=I$

Not $x^2=y \Leftrightarrow x=\sqrt{y} \Leftrightarrow dx=\frac{1}{2\sqrt{y}}\,dy$

$$\Rightarrow\begin{cases}p-1=-\frac{2}{3}\\ p+q=1\end{cases}\Leftrightarrow\begin{cases}p=\frac{1}{3}\\ q=\frac{2}{3}\end{cases}\Rightarrow I=\frac{1}{2}B\left(\frac{1}{3},\frac{2}{3}\right)=$$

$$=\frac{1}{2}\cdot\frac{\Gamma\left(\frac{1}{3}\right)\cdot\Gamma\left(\frac{2}{3}\right)}{\Gamma(1)}$$

$$=\frac{1}{2}\cdot\frac{\pi}{\sin\frac{\pi}{3}}=\frac{\pi}{\sqrt{3}}$$

c) $\int_0^{\infty}\frac{dx}{\sqrt[3]{x^2}(1+x^3)}$; $=\frac{1}{3}\int_0^{\infty}\frac{y^{-\frac{2}{3}}}{y^{\frac{2}{3}}(1+y)}\,dy=\frac{1}{3}\int_0^{\infty}\frac{y^{-\frac{8}{9}}}{1+y}\,dy=I$

Not $x^3=y \Leftrightarrow x=y^{\frac{1}{3}} \Leftrightarrow dx=\frac{1}{3}y^{-\frac{2}{3}}\,dy$

$$\begin{cases}p-1=-\frac{8}{9}\\ p+q=1\end{cases}\Leftrightarrow\begin{cases}p=\frac{1}{9}\\ q=\frac{8}{9}\end{cases}\Rightarrow I=\frac{1}{3}B\left(\frac{1}{9},\frac{8}{9}\right)=\frac{1}{3}\cdot\frac{\Gamma\left(\frac{1}{9}\right)\Gamma\left(\frac{8}{9}\right)}{\Gamma(1)}$$

$$=\frac{1}{3}\cdot\frac{\pi}{\sin\frac{\pi}{9}}$$

d) $\int_0^\infty \frac{\sqrt[4]{x}}{(1+x)^2}\,dx;$ $= \int_0^\infty \frac{x^{\frac{1}{4}}}{(1+x)^2}dx = I$

$\Rightarrow \begin{cases} p-1=\frac{1}{4} \\ p+q=2 \end{cases} \Leftrightarrow \begin{cases} p=\frac{5}{4} \\ q=\frac{3}{4} \end{cases} \Rightarrow I = B\left(\frac{5}{4},\frac{3}{4}\right) = \frac{\Gamma\left(\frac{5}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right)}{\Gamma(2)} =$

$= \frac{1}{4}\cdot\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right) =$

$= \frac{1}{4}\cdot\frac{\pi}{\sin\frac{\pi}{4}} = \frac{\pi}{2\sqrt{2}}$

e) $\int_0^\infty x^p e^{-x^q}\,dx,$

Not $x^2 = y \Leftrightarrow x = y^{\frac{1}{2}} \Leftrightarrow dx = \frac{1}{2}y^{\frac{1-2}{2}}dy$

$= \int_0^\infty y^{\frac{p}{2}}\cdot e^{-y}\cdot\frac{1}{2}\cdot y^{\frac{1-2}{2}}dy = \frac{1}{2}\int_0^\infty y^{\frac{p+1-2}{2}}\cdot e^{-y}dy$

$\Rightarrow \alpha - 1 = \frac{p+1-2}{2} \Leftrightarrow \alpha = \frac{p+1}{2} \Rightarrow = \frac{1}{2}\Gamma\left(\frac{p+1}{2}\right)$

f) $\int_0^\infty e^{-x^2}\,dx;$

Not $x^2 = y \Rightarrow x = y^{\frac{1}{2}} \Leftrightarrow dx = \frac{1}{2}y^{-\frac{1}{2}}dy$

$= \frac{1}{2}\int_0^\infty y^{-\frac{1}{2}}\cdot e^{-y}dy = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi}$

g) $\int_0^{\frac{\pi}{2}} \sin^6 x\cos^4 x\,dx;$

Not $\sin^2 x = y \Leftrightarrow dx = 2\sin x\cos x\,dx$

x	0	$\frac{\pi}{2}$
y	0	1

$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin^5 x\cdot\cos^3 x\cdot 2\sin x\cos x\,dx = \frac{1}{2}\int_0^1 y^{\frac{5}{2}}\cdot(1-y)^{\frac{3}{2}}dy$

$\Rightarrow \begin{cases} p-1=\frac{5}{2} \\ q-1=\frac{3}{2} \end{cases} \Rightarrow \begin{cases} p=\frac{7}{2} \\ q=\frac{5}{2} \end{cases}$

$= \frac{1}{2}B\left(\frac{7}{2},\frac{5}{2}\right) = \frac{1}{2}\frac{\Gamma\left(\frac{7}{2}\right)\cdot\Gamma\left(\frac{5}{2}\right)}{\Gamma(6)}$

$= \frac{1}{2\cdot 5!}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\cdot\sqrt{\pi}\cdot\frac{3}{2}\cdot\frac{1}{2}\cdot\sqrt{\pi}$

$= \frac{\pi}{2^6}\cdot\frac{3}{2\cdot 4} = \frac{3\pi}{2^9}$

h) $\int_0^1 \left(\ln \frac{1}{x}\right)^p \, dx$, $p > -1$.

Not $\ln \frac{1}{x} = y \Leftrightarrow \frac{1}{x} = e^y \Leftrightarrow x = e^{-y} \Leftrightarrow dx = -e^{-y} dy$

x	0	1
y	∞	0

$$= \int_\infty^0 y^p \cdot (-e^{-y}) dy = \int_0^\infty y^p \cdot e^{-y} dy = \Gamma(p+1)$$

5. Să se calculeze lungimile următoarelor curbe:

a) $C : \begin{cases} x = a(t - \sin t), \\ y = a(1 - \cos t), \end{cases} t \in [0, 2\pi]$, cu $a > 0$;

$$\mathcal{L}_C = \int_C 1 \, ds = \int_a^b \sqrt{x'(t)^2 + y'(t)^2} \, dt$$

$$\Rightarrow \mathcal{L}_C = \int_0^{2\pi} \sqrt{(a(1-\cos t))^2 + (a \sin t)^2} \, dt = \int_0^{2\pi} \sqrt{a^2(1 - 2\cos t + \underbrace{\cos^2 t + \sin^2 t}_{=1})} \, dt$$

$$= \int_0^{2\pi} |a| \sqrt{2 - 2\cos t} \, dt = a \int_0^{2\pi} 2 \cdot \sqrt{\frac{1-\cos t}{2}} \, dt =$$

$$= 2a \int_0^{2\pi} \left|\sin \frac{t}{2}\right| dt$$

pt $t \in [0, 2\pi] \Rightarrow \frac{t}{2} \in [0, \pi] \Rightarrow \sin \frac{t}{2} \geq 0$

$$= 2a \int_0^{2\pi} \sin \frac{t}{2} \, dt = -2a \cdot \cos \frac{t}{2} \cdot 2 \Big|_0^{2\pi} = -4a \cdot \cos \frac{t}{2} \Big|_0^{2\pi} =$$

$$= -4a \cdot \cos \pi + 4a \cos 0 = 4a + 4a = 8a$$

b) $C : \begin{cases} x = a(2\cos t - \cos 2t), \\ y = a(2\sin t - \sin 2t), \end{cases} t \in [0, 2\pi]$, cu $a > 0$;

$$\mathcal{L}_C = \int_C 1 \, ds = \int_0^{2\pi} \sqrt{a^2(-2\sin t + 2\sin 2t)^2 + a^2(2\cos t - 2\cos 2t)^2} \, dt =$$

$$= \int_0^{2\pi} |a| \sqrt{4\sin^2 t - 8 \sin t \sin 2t + 4\sin^2 2t + 4\cos^2 t - 8\cos t \cos 2t + 4\cos^2 2t} \, dt$$

$$= a \int_0^{2\pi} \sqrt{8 - 8\sin t \sin 2t - 8\cos t \cos 2t} \, dt$$

$$= 2\sqrt{2}\,a \int_0^{2\pi} \sqrt{1-\sin t\cdot 2\sin t\cos t - \cos t(1-2\sin^2 t)}\,dt$$

$$= 2\sqrt{2}\,a \int_0^{2\pi} \sqrt{1-2\sin^2 t\cos t - \cos t + 2\sin^2 t\cos t}\,dt$$

$$= 2\sqrt{2}\,a \int_0^{2\pi} \sqrt{1-\cos t}\,dt = 2\sqrt{2}\,a \int_0^{2\pi} \sqrt{2}\left|\sin\frac{t}{2}\right| dt$$

$$= 4a\int_0^{2\pi} \sin\frac{t}{2}\,dt = -4a\cos\frac{t}{2}\cdot 2\Big|_0^{2\pi} = 16a$$

c) $C:\begin{cases} x = ae^{mt}\cos t, \\ y = ae^{mt}\sin t, \end{cases}$ $t\in[0,2\pi]$, cu $a, m > 0$.

$$\mathcal{L}_C = \int_C 1\,ds = \int_0^{2\pi} \sqrt{a^2(me^{mt}\cos t - e^{mt}\sin t)^2 + a^2(me^{mt}\sin t + e^{mt}\cos t)^2}\,dt$$

$$= \int_0^{2\pi} \sqrt{a^2e^{2mt}\left((m\cos t - \sin t)^2 + (m\sin t + \cos t)^2\right)}\,dt$$

$$= |a|\int_0^{2\pi} e^{mt}\sqrt{m^2\cos^2 t - 2m\sin t\cos t + \sin^2 t + m^2\sin^2 t + 2m\sin t\cos t + \cos^2 t}\,dt$$

$$= |a|\int_0^{2\pi} e^{mt}\sqrt{m^2+1}\,dt = a\frac{\sqrt{m^2+1}}{m}\int_0^{2\pi} me^{mt}\,dt =$$

Not $mt = x \Leftrightarrow m\,dt = dx$

t	0	2π
x	0	$2\pi m$

$$= \frac{a\sqrt{m^2+1}}{m}\int_0^{2\pi m} e^x\,dx = \frac{a\sqrt{m^2+1}}{m}\left(e^{2\pi m}-1\right)$$

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6. Să se calculeze următoarele integrale curbilinii de speţa I:

a) $\int_C ye^{-x}\,ds$, unde $C:\begin{cases} x=\ln\left(1+t^2\right), \\ y=2\operatorname{arctg} t-t, \end{cases}$ $t\in[0,1]$;

$$\int_C ye^{-x}\,ds = \int_0^1 (2\operatorname{arctg} t - t)\, e^{-\ln(1+t^2)} \sqrt{\left(\frac{2}{1+t^2}-1\right)^2+\left(\frac{2t}{1+t^2}\right)^2}\,dt$$

$$= \int_0^1 (2\operatorname{arctg} t - t) e^{-\ln(1+t^2)} \sqrt{\frac{4}{(1+t^2)^2} - \frac{4}{1+t^2} + 1 + \frac{4t^2}{(1+t^2)^2}}\,dt$$

$$= \int_0^1 (2\operatorname{arctg} t - t) e^{-\ln(1+t^2)} \sqrt{\frac{-4t^2}{(1+t^2)^2} + 1 + \frac{4t^2}{(1+t^2)^2}}\,dt$$

$$= \int_0^1 (2\operatorname{arctg} t - t) e^{-\ln(1+t^2)}\,dt = \int_0^1 \frac{2\operatorname{arctg} t - t}{\ln(1+t^2)}\,dt =$$

$$= \int_0^1 \frac{2\operatorname{arctg} t - t}{1+t^2}\,dt = 2\int_0^1 \frac{\operatorname{arctg} t}{1+t^2}\,dt - \frac{1}{2}\int_0^1 \frac{2t}{1+t^2}\,dt$$

Not $\operatorname{arctg} t = x$
$\Leftrightarrow \frac{1}{1+t^2}\,dt = dx$

t	0	1
x	0	$\frac{\pi}{4}$

Not $1+t^2 = x$
$\Leftrightarrow 2t\,dt = dt$

t	0	1
x	1	2

$$= 2\int_0^{\frac{\pi}{4}} x\,dx - \frac{1}{2}\int_1^2 \frac{1}{x}\,dx$$

$$= 2\,\frac{x^2}{2}\Big|_0^{\frac{\pi}{4}} - \frac{1}{2}\ln 2 = \frac{\pi^2}{16} - \ln\sqrt{2}$$

b) $\int_C \left(x^2+y^2\right)ds$, unde C este segmentul $[AB]$, cu $A(a,a)$ şi $B(b,b)$, pentru $b>a$;

$$C:\begin{cases} x=t \\ y=t \end{cases},\ t\in[a,b]$$

$$\Rightarrow \int_C (x^2+y^2)\,ds = \int_a^b (t^2+t^2)\cdot\sqrt{1^2+1^2}\,dt = \int_a^b 2t^2\cdot\sqrt{2}\,dt$$

$= 2\sqrt{2} \cdot \frac{t^3}{3}\Big|_a^b$

$= \frac{2\sqrt{2}}{3}(b^3 - a^3)$

c) $\int_C xy\,ds$, unde $C: \begin{cases} x = a\cos^2 t, \\ y = a\cos t, \end{cases}$ $t \in [0, \pi]$, cu $a > 0$;

$$\int_C xy\,ds = \int_0^\pi a\cos^2 t \cdot a\cos t \cdot \sqrt{(2a\cos t \cdot (-\sin t))^2 + (-a\sin t)^2}\,dt$$

$$= \int_0^\pi a^2\cos^3 t \cdot \sqrt{4a^2\cos^2 t \sin^2 t + a^2\sin^2 t}\,dt$$

$$= \int_0^\pi a^2\cos^3 t \cdot a\sin t\sqrt{4\cos^2 t + 1}\,dt$$

$$= -\int_0^\pi -a^3 \cdot \sin t \cdot \cos^3 t\sqrt{4\cos^2 t + 1}\,dt$$

Not $\cos t = x \Leftrightarrow -\sin t\,dt = dx$

t	0	π
x	1	-1

$$= a^3\int_{-1}^{1} x^3\sqrt{4x^2+1}\,dx \quad \Big| \Rightarrow I = 0$$

(funcție impară)

d) $\int_C x\sqrt{x^2 - y^2}\,ds$, unde C este definită de $(x^2 + y^2)^2 = a^2(x^2 - y^2)$ şi $x \geq 0$, cu $a > 0$;

$$C: (x^2+y^2)^2 = a^2(x^2-y^2) \Leftrightarrow x^2 - y^2 = \frac{(x^2+y^2)^2}{a^2}$$

$$\int_C x\sqrt{x^2-y^2}\,ds = \int_C x \cdot \frac{x^2+y^2}{a^2}$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases}$$

$$(x^2+y^2)^2 = a^2(x^2-y^2) \Leftrightarrow r^4 = a^2r^2(\cos^2 t - \sin^2 t) \;\Big|: r^2 \neq 0$$

$$\Leftrightarrow r^2 = a^2(\cos^2 t - \sin^2 t) = \cos 2t$$

$\Rightarrow t = a\sqrt{\cos^2 t - \sin^2 t} \Rightarrow \begin{cases} x = a\sqrt{\cos^2 t - \sin^2 t} \cdot \cos t \\ y = a\sqrt{\cos^2 t - \sin^2 t} \cdot \sin t \end{cases}$ ..... calculo crazy

e) $\int_C \left(2z - \sqrt{x^2+y^2}\right) ds$, unde $C : \begin{cases} x = t\cos t, \\ y = t\sin t, \quad t \in [0, 2\pi]; \\ z = t, \end{cases}$

$$\int_C \left(2z - \sqrt{x^2+y^2}\right) ds = \int_0^{2\pi} (2t - t)\sqrt{(\cos t - t\sin t)^2 + (\sin t + t\cos t)^2 + 1}\, dt$$

$$= \int_0^{2\pi} t\sqrt{\cos^2 t - 2t\cos t\sin t + t^2\sin^2 t + \sin^2 t + 2t\sin t\cos t + t^2\cos^2 t + 1}\, dt$$

$$= \frac{1}{2}\int_0^{2\pi} 2t\sqrt{t^2+2}\, dt$$

Not $t^2 + 2 = x \Rightarrow 2t\,dt = dx$

t	0	2π
x	2	$4\pi^2+2$

$$= \frac{1}{2}\int_2^{4\pi^2+2} x^{\frac{1}{2}}\, dx = \frac{1}{2} \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\Bigg|_2^{4\pi^2+2} = \frac{1}{3}\left((4\pi^2+2)^{\frac{3}{2}} - 2^{\frac{3}{2}}\right) =$$

$$= \frac{2\sqrt{2}}{3}\left((2\pi^2+1)^{\frac{3}{2}} - 1\right)$$

f) $\int_C \frac{ds}{(x^2+y^2+z^2)^3}$, unde C este definită de $x^2+y^2+z^2 = 2a^2$ şi $z = a > 0$;

$$\begin{cases} x^2+y^2+z^2 = 2a^2 \\ z = a \end{cases} \Rightarrow x^2+y^2 = a^2$$

$$\Rightarrow C : \begin{cases} x = a\cos t \\ y = a\sin t \\ z = a \end{cases}, t \in [0, 2\pi]$$

$$\Rightarrow \int_C \frac{ds}{(x^2+y^2+z^2)^3} = \int_0^{2\pi} \frac{1}{8a^6} \sqrt{a^2\sin^2 t + a^2\cos^2 t}\, dt$$

$$= \frac{1}{8a^6}\int_0^{2\pi} a\, dt = \frac{1}{8a^5} \cdot t\Big|_0^{2\pi} = \frac{\pi}{4a^5}$$

g) $\int_C ze^{\sqrt{x^2+y^2}}\, ds$, unde C este definită de $x^2+y^2+z^2=a^2$ şi $x+y=0$, cu $a>0$.

$x^2+y^2+z^2=a^2$

$x+y=0 \Leftrightarrow x=-y$

$\Rightarrow 2y^2+z^2=a^2 \Leftrightarrow z^2=a^2-2y^2 \Rightarrow z=\sqrt{a^2-2y^2}$

$\Rightarrow C: \begin{cases} x=t \\ y=-t \\ z=\sqrt{a^2-2t^2} \end{cases}, t\in\left[-\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right]$

$$\int_C ze^{\sqrt{x^2+y^2}}\, ds = \int_{-\frac{a}{\sqrt{2}}}^{\frac{a}{\sqrt{2}}} \sqrt{a^2-2t^2}\cdot e^{\sqrt{2t^2}} \sqrt{1^2+(-1)^2+\left(\frac{1}{2\sqrt{a^2-2t^2}}\cdot(-4t)\right)^2}\, dt$$

$$= \int_{-\frac{a}{\sqrt{2}}}^{\frac{a}{\sqrt{2}}} \sqrt{a^2-2t^2}\cdot e^{\sqrt{2t^2}} \sqrt{2+\frac{4t^2}{a^2-2t^2}}\, dt$$

$$= \int_{-\frac{a}{\sqrt{2}}}^{\frac{a}{\sqrt{2}}} \sqrt{a^2-2t^2}\cdot e^{\sqrt{2t^2}} \sqrt{\frac{2a^2}{a^2-2t^2}}\, dt$$

$$= \int_{-\frac{a}{\sqrt{2}}}^{\frac{a}{\sqrt{2}}} \sqrt{a^2-2t^2}\cdot e^{\sqrt{2t^2}} \cdot \frac{\sqrt{2}a}{\sqrt{a^2-2t^2}}\, dt$$

$$= \sqrt{2}a \int_{-\frac{a}{\sqrt{2}}}^{\frac{a}{\sqrt{2}}} e^{\sqrt{2}|t|}\, dt = 2a \int_0^{\frac{a}{\sqrt{2}}} \sqrt{2}e^{\sqrt{2}t}\, dt = 2a\int_0^a e^x\, dx$$

↳ f para

$\sqrt{2}t = x \Leftrightarrow \sqrt{2}\,dt = dx$

t	0	$\frac{a}{\sqrt{2}}$
x	0	a

$$= 2a\, e^x\Big|_0^a = 2a(e^a-1)$$

Definiţia 2.19

Integrala curbilinie de speţa a II-a a 1-formei diferenţiale $\omega = P(x,y)\,dx + Q(x,y)\,dy$ de-a lungul lui C este

$$\int_C \omega = \int_a^b \left[P(x(t),y(t))\,x'(t) + Q(x(t),y(t))\,y'(t)\right] dt.$$

7. Să se calculeze următoarele integrale curbilinii de speţa a II–a:

a) $\int_C (2a-y)\,dx + x\,dy$, unde $C: \begin{cases} x = a(t-\sin t), \\ y = a(1-\cos t), \end{cases} t \in [0, 2\pi]$;

$$\int_C (2a-y)dx + x\,dy = \int_0^{2\pi} \Big((2a - a(1-\cos t)) \cdot a(1-\cos t) + a(t-\sin t)\cdot a\sin t \Big)\, dt$$

$$= \int_0^{2\pi} \Big((2a - a + a\cos t)(a - a\cos t) + (at - a\sin t)\cdot a\sin t \Big)\, dt$$

$$(a + a\cos t)(a - a\cos t)$$

$$= \int_0^{2\pi} \left(a^2 - a^2\cos^2 t + a^2 t\sin t - a^2\sin^2 t \right) dt$$

$$= \int_0^{2\pi} a^2 - a^2 + a^2 t\sin t\, dt = a^2 \int_0^{2\pi} t\sin t\, dt$$

$f' = \sin t \Rightarrow f = -\cos t$

$g = t \Rightarrow g' = 1$

$$= a^2 \left(-t\cos t \Big|_0^{2\pi} + \int_0^{2\pi} \cos t\, dt \right)$$

$$= a^2 \left(-2\pi + \sin t \Big|_0^{2\pi} \right)$$

$$= -2\pi a^2$$

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b) $\int_C \frac{x^3\,dy - y^3\,dx}{x^{\frac{5}{3}} + y^{\frac{5}{3}}}$, unde C este arcul $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$, cuprins în primul cadran;

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} \Leftrightarrow x^{\frac{2}{3}} = a^{\frac{2}{3}} - y^{\frac{2}{3}} \Leftrightarrow x = \left(a^{\frac{2}{3}} - y^{\frac{2}{3}}\right)^{\frac{3}{2}}$$

$$\Rightarrow \begin{cases} x = a\cos^3 t \\ y = a\sin^3 t \end{cases}, \quad t \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow \int_C \frac{x^3\,dy - y^3\,dx}{x^{\frac{5}{3}} + y^{\frac{5}{3}}} = \int_C -\frac{y^3}{x^{\frac{5}{3}} + y^{\frac{5}{3}}}\,dx + \frac{x^3}{x^{\frac{5}{3}} + y^{\frac{5}{3}}}\,dy$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{-a^3 \sin^9 t}{a^{\frac{5}{3}}(\cos^5 t + \sin^5 t)} \cdot a \cdot 3\cos^2 t (-\sin t) + \frac{a^3 \cos^9 t}{a^{\frac{5}{3}}(\cos^5 t + \sin^5 t)} \cdot a \cdot 3\sin^2 t \cos t \right) dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{3a^4 \sin^{10} t \cdot \cos^2 t + 3a^4 \cos^{10} t \sin^2 t}{a^{\frac{5}{3}}(\cos^5 t + \sin^5 t)} dt$$

$$= \frac{3a^4}{a^{\frac{5}{3}}} \int_0^{\frac{\pi}{2}}$$

c) $\int_C (y^2+z^2)\,dx + (z^2+x^2)\,dy + (x^2+y^2)\,dz$, unde C este curba de intersecţie a suprafeţelor $x^2+y^2+z^2 = 4ax$ şi $x^2+y^2 = ax$, cu $z \geq 0$;

$$\begin{cases} x^2+y^2+z^2 = 4ax \\ x^2+y^2 = ax \end{cases}$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r^2 = ar\cos t \;|: r \neq 0 \Rightarrow r = a\cos t$$

$$\Rightarrow \begin{cases} x = a\cos^2 t \\ y = a\sin t\cos t \\ z^2 = 3ax = 3a \cdot a\cos^2 t \\ z \geq 0 \end{cases} \Rightarrow \begin{cases} x = a\cos^2 t \\ y = a\sin t\cos t \\ z = \sqrt{3}a\cos t \end{cases} \quad t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$(\sin t \cos t)' = \cos^2 t - \sin^2 t$$

$$\int_C (y^2+z^2)\,dx + (z^2+x^2)\,dy + (x^2+y^2)\,dz =$$

$$= \int_0^{\pi} \Big((a^2\sin^2 t\cos^2 t + 3a^2\cos^2 t) \cdot 2a\cos t(-\sin t) + (3a^2\cos^2 t + a^2\cos^4 t)(\cos^2 t - \sin^2 t) + (a^2\cos^4 t + a^2\sin^2 t\cos^2 t)\sqrt{3}a(-\sin t) \Big) dt$$

$$I = \int_C (y^2+z^2)\,dx + (z^2+x^2)\,dy + (x^2+y^2)\,dz =$$

$$= \int_C (x^2+y^2+z^2)(dx+dy+dz) - \int_C x^2\,dx + y^2\,dy + z^2\,dz$$

C inclusă (derivatele sunt toate 0) $\Rightarrow \int = 0$

$$= \int_C 4ax\,(dx+dy+dz) = 4a \int_C x\,dx + x\,dy + x\,dz =$$

$$= 4a \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} a\cos^2 t \left(2a\cos t(-\sin t) + a(\cos^2 t - \sin^2 t) + \sqrt{3}a(-\sin t)\right) dt$$

$$= 4a^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \Big(\underbrace{-2\cos^3 t \sin t}_{\text{impară}} + \cos^4 t - \cos^2 t \sin^2 t - \sqrt{3}a \underbrace{\cos^2 t \sin t}_{\text{impară}}\Big) dt$$

$$= 4a^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 t(\cos^2 t - \sin^2 t) = 4a^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 t(1-2\sin^2 t)\,dt$$

$$= 4a^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos^2 t - 2\cos^2 t \sin^2 t)\,dt = 4a^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1+\cos 2t}{2} - \frac{\sin^2 2t}{2}\right) dt$$

$$= 2a^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 + \cos 2t - \frac{1-\cos 4t}{2}\,dt = 2a^3 \left(t + \frac{\sin 2t}{2} - \frac{t}{2} + \frac{\sin 4t}{4}\right)\Bigg|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= 2a^3\left(\pi + \underbrace{\sin\pi}_{0} - \frac{\pi}{2}\right) = a^3\pi$$

d) $\int_C y^2\,dx + z^2\,dy + x^2\,dz$, unde C este curba de intersecţie a suprafeţelor $x^2+y^2+z^2=a^2$ şi $x^2+y^2=ax$, cu $z \geq 0$;

$x = r\cos t$
$y = r\sin t$ $\Rightarrow r^2 = ar\cos t \mid :r \Rightarrow r = a\cos t \Rightarrow x = a\cos^2 t$
$\Rightarrow y = a\sin t\cos t$

$\Rightarrow z^2 = a^2 - a^2\cos^2 t = a^2(1-\cos^2 t) = a^2\sin^2 t \Rightarrow z = a|\sin t|$

$\Rightarrow C_1: \begin{cases} x = a\cos^2 t \\ y = a\sin t\cos t \\ z = -a\sin t \end{cases}, t \in [-\frac{\pi}{2}, 0]$ şi $C_2: \begin{cases} x = a\cos^2 t \\ y = a\sin t\cos t \\ z = a\sin t \end{cases}$ $t \in [0, \frac{\pi}{2}]$

$\Rightarrow \int_C = \int_{C_1} + \int_{C_2}$

$$= \int_{-\frac{\pi}{2}}^{0} \Big(a^2 \sin^2 t \cos^2 t \cdot 2a\cos t\,(-\sin t) + a^2 \sin^2 t \cdot a(\cos^2 t - \sin^2 t) + a\cos^4 t \cdot (-a\cos t) \Big) dt$$

$$+ \int_{0}^{\frac{\pi}{2}} \Big(a^2 \sin^2 t \cos^2 t \cdot 2a\cos t\,(-\sin t) + a^2 \sin^2 t \cdot a(\cos^2 t - \sin^2 t) + a\cos^4 t \cdot a\cos t \Big) dt$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \Big(a^2 \sin^2 t \cos^2 t \cdot 2a\cos t\,(-\sin t) + a^2 \sin^2 t \cdot a(\underbrace{\cos^2 t - \sin^2 t}_{\cos 2t}) \Big) dt$$

$$- \int_{-\frac{\pi}{2}}^{0} a^3 \cos^5 t + \int_{0}^{\frac{\pi}{2}} a^3 \cos^5 t\, dt \quad \Big\{ \text{sch v} \quad \begin{array}{l} t = -x \\ dt = -dx \end{array} \quad \begin{array}{c|ccc} t & -\frac{\pi}{2} & & 0 \\ \hline x & \frac{\pi}{2} & & 0 \end{array} \Rightarrow -\int_{0}^{\frac{\pi}{2}} a^3 \cos^5 x\, dx + \int_{0}^{\frac{\pi}{2}} a^3 \cos^5 t\, dt$$

$$= a^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \Big(-2\sin^3 t \cos^3 t + \sin^2 t(\cos^2 t - \sin^2 t) \Big) dt$$

$$= a^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \Big(-\frac{1}{4} 8\sin^3 t \cos^3 t + \frac{1-\cos 2t}{2} \cos 2t \Big) dt$$

$$= \frac{a^3}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \Big(\sin^3 2t + 2(1-\cos 2t)\cos 2t \Big) dt = \frac{a^3}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin^3 2t + 2\cos 2t - 2\cos^2 2t)\, dt$$

$$= \frac{a^3}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \Big(\sin 2t(1-\cos^2 2t) + 2\cos 2t - \frac{1-\cos 4t}{} \Big) dt$$

$$= \frac{a^3}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \Big(\sin 2t - \underbrace{\cos^2 2t \sin 2t}_{\text{sch v}} + 2\cos 2t - 1 - \cos 4t \Big) dt$$

$$= \frac{a^3}{4} \Big(\frac{\cos 2t}{2} - \frac{\cos^3 2t}{6} + \sin 2t - t - \frac{\sin 4t}{4} \Big) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= \frac{a^3}{4} \Big(0 \quad -\pi - 0 \Big) = -\frac{\pi a^3}{4}$$

.... restul subpunctelor (dacă mai am timp și/sau chef)

8. a) Să se arate că $\int_C \left(e^{-x^2+y^2}\cos 2xy\right)\mathrm{d}x + \left(e^{-x^2+y^2}\sin 2xy\right)\mathrm{d}y = 0$, pentru orice curbă netedă, simplă şi închisă, C.

$\frac{\partial P}{\partial y} = e^{-x^2+y^2}\cdot 2y\cdot\cos 2xy + e^{-x^2+y^2}\cdot 2x(-\sin 2xy)$

$\frac{\partial Q}{\partial x} = e^{-x^2+y^2}\cdot(-2x)\cdot\sin 2xy + e^{-x^2+y^2}\cdot 2y\cdot\cos 2xy$

$\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ $\Rightarrow \omega$ f. închisă

$\mathbb{R}^2$ m. stelată $\Rightarrow \omega$ f. exactă

ω f. exactă
C curbă netedă, închisă $\Rightarrow \int_C \omega = 0$

b) Să se calculeze $\int_C \underbrace{y}_{P}\,\mathrm{d}x + \underbrace{x}_{Q}\,\mathrm{d}y$, unde C este o curbă netedă, simplă, parametrizată, arbitrară, ce unește punctele $A(2,1)$ şi $B(1,3)$, în această ordine.

$\frac{\partial P}{\partial y} = 1$, $\frac{\partial Q}{\partial x} = 1$ $\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ $\Rightarrow \omega$ f. închisă

$\mathbb{R}^2$ mulțime stelată $\Rightarrow \omega$ exactă

ω exactă $\Rightarrow \exists f \in \mathcal{C}^1$ a.î. $\omega = df \Leftrightarrow \begin{cases} P = \frac{\partial f}{\partial x} = y \\ Q = \frac{\partial f}{\partial y} = x \end{cases} \overset{\int dy}{\Rightarrow}$

$\Rightarrow f(x,y) = xy + c(x) \Rightarrow \frac{\partial f}{\partial x} = y + c'(x)$

dar $\frac{\partial f}{\partial x} = y$ $\Rightarrow c'(x) = 0$ $\Rightarrow c(x) = c_1$

$\Rightarrow f(x,y) = xy + c_1$

ω exactă, C netedă, parametrizată $\overset{\text{TLD}}{\Rightarrow} \int_C \omega = f(x_B,y_B) - f(x_A,y_A) = f(1,3) - f(2,1) = 1\cdot 3 + c_1 - 2\cdot 1 - c_1 = 1$

c) Să se determine valoarea integralei

$$\int_C 2x\left(\frac{y}{x^2+1}-1\right)dx+\left[\ln\left(x^2+1\right)-2\right]dy,$$

unde C este o curbă netedă, simplă, parametrizată, arbitrară, ce unește punctele $A(-1,1)$ și $B(1,2)$, în această ordine.

$P = 2x\left(\frac{y}{x^2+1}-1\right)$, $Q = \ln(x^2+1)-2$

$\frac{\partial P}{\partial y} = 2x\cdot\frac{1}{x^2+1}$

$\frac{\partial Q}{\partial x} = \frac{1}{x^2+1}\cdot 2x$

$\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ $\Rightarrow$ ω închisă

$\mathbb{R}^2$ deschisă $\overset{T.P.}{\Rightarrow}$ ω exactă

$\Rightarrow \exists f \in \mathcal{C}^1$ a.î. $\omega = df \Rightarrow \begin{cases} P = \frac{\partial f}{\partial x} = 2x\left(\frac{y}{x^2+1}-1\right) \\ Q = \frac{\partial f}{\partial y} = \ln(x^2+1)-2 \end{cases} \overset{\int dy}{\Rightarrow}$

$\Rightarrow f(x,y) = \int(\ln(x^2+1)-2)\,dy = (\ln(x^2+1)-2)y + c(x)$

$\Rightarrow \frac{\partial f}{\partial x} = y\frac{1}{x^2+1}\cdot 2x + c'(x) = P$

$\Rightarrow c'(x) = 2x \Leftrightarrow c(x) = x^2 + c_1 \Rightarrow f(x,y) = (\ln(x^2+1)-2)y + x^2 + c_1$

ω exactă

C netedă, parametrizată $\overset{T.N.L.}{\Rightarrow} \int_C \omega = f(x_B', y_B^2) - f(\bar{x}_A, \hat{y}_A) =$

$= (\ln(2)-2)2 + 1 + c_1 - (\ln(2)-2)\cdot 1 - 1 - c_1$

$= \ln 2 - 2$

2.18. Să se calculeze valoarea integralei $I = \int_C (x^2+3y)\,ds$, unde C este segmentul care unește punctele $A(1,-1)$ și $B(2,1)$, în această ordine.

→ din culegere /$\frac{7\sqrt{5}}{3}$

$\frac{x-x_A}{x_B-x_A} = \frac{y-y_A}{y_B-y_A} \Leftrightarrow x-1 = \frac{y+1}{2} \Leftrightarrow y = 2x-3 \Rightarrow \begin{cases} x = x \\ y = 2x-3 \end{cases}$

$\Rightarrow \int_C (x^2+3y)\,ds = \int_1^2 (x^2+3(2x-3))\cdot\sqrt{1^2+2^2}\,dx =$

$= \int_1^2 (x^2+6x-9)\cdot\sqrt{5}\,dx = \sqrt{5}\left(\frac{x^3}{3}+\frac{6x^2}{2}-9x\right)\Big|_1^2$

$= \sqrt{5}\left(\frac{8}{3}-\frac{1}{3}+12-3-18+9\right) = \frac{7\sqrt{5}}{3}$

9. Să se calculeze $\int_C \frac{y\,dx+x\,dy}{1+xy}$ pe un drum arbitrar cu capetele $A(-1,-1)$ şi $B(3,3)$, ce nu intersectează hiperbola de ecuaţie $xy=-1$.

$$\frac{x-x_A}{x_B-x_A}=\frac{y-y_A}{y_B-y_A} \iff \frac{x+1}{3+1}=\frac{y+1}{3+1} \iff 4(x+1)=4(y+1) \mid :4 \quad -1$$

$$\iff x=y$$

$$\Rightarrow \begin{cases} x=x \\ y=x \end{cases}$$

$$\Rightarrow \int_C \frac{y}{1+xy}\,dx+\frac{x}{1+xy}\,dy=\int_{-1}^{3}\frac{x}{1+x^2}+\frac{x}{1+x^2}\,dx=$$

$$=\int_{-1}^{3}\frac{2x}{1+x^2}\,dx=\int_2^{10}\frac{1}{t}\,dt=\ln|t|\Big|_2^{10}=\ln 10-\ln 2=\ln 5$$

Not $1+x^2=t \iff 2x\,dx=dt$

x	-1	3
t	2	10

10. Să se calculeze circulaţia câmpului $\mathbf{V}(x,y)=\frac{3x^3+xy^2}{\sqrt{x^2+y^2}}\,\mathbf{i}-\frac{3y^3+x^2y}{\sqrt{x^2+y^2}}\,\mathbf{j}$ de-a lungul curbei $r=a(1+\cos t)$, cu $t\in\left[0,\frac{\pi}{2}\right]$.

$$\begin{cases} x=a(1+\cos t)\cos t \\ y=a(1+\cos t)\sin t \end{cases} \qquad t\in\left[0,\frac{\pi}{2}\right]$$

Teorema 2.16

Lucrul mecanic efectuat de forţa $\mathbf{F}$ în deplasarea unei particule de-a lungul curbei C este

$$W=\int_C \mathbf{F}\cdot d\mathbf{r}=\int_a^b \left[P\left(x(t),y(t)\right)x'(t)+Q\left(x(t),y(t)\right)y'(t)\right]dt.$$

Mai general, dacă $\mathbf{F}(x,y)=P(x,y)\,\mathbf{i}+Q(x,y)\,\mathbf{j}$ este un câmp vectorial ce acţionează asupra fiecărui punct din plan, atunci integrala $\int_C \mathbf{F}\cdot d\mathbf{r}$ reprezintă *circulaţia* câmpului vectorial $\mathbf{F}$ de-a lungul curbei C, pe care o vom nota cu $\mathcal{C}_C(\mathbf{F})$.

$$\mathcal{C}_C(\vec{F})=\int_C \vec{F}\,d\vec{r}=\int_0^{\frac{\pi}{2}}\left(\frac{3\left(a(1+\cos t)\cos t\right)^3+a^3(1+\cos t)^3\cos t\sin^2 t}{\sqrt{a^2(1+\cos t)^2\cos^2 t+a^2(1+\cos t)^2\sin^2 t}}\cdot\left(-a\sin t\cos t\right.\right.$$

$$\left.\left.-a(1+\cos t)\sin t\right)\right.$$

$$+\left.\frac{3\left(a(1+\cos t)\sin t\right)^3+a^3(1+\cos t)^3\cos^2 t\sin t}{\sqrt{a^2(1+\cos t)^2\cos^2 t+a^2(1+\cos t)^2\sin^2 t}}\cdot\left(-a\sin^2 t+a(1+\cos t)\cos t\right)\right)dt$$

$$\sqrt{a^2(1+\cos t)^2}=a(1+\cos t)$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{3a^3(1+\cos t)^3\cos^3 t + a^3(1+\cos t)^3\cos t\sin^2 t}{a(1+\cos t)} \cdot (-2a\sin t\cos t - a\sin t) \right.$$

$$\left. + \frac{3a^3(1+\cos t)^3\sin^3 t + a^3(1+\cos t)^3\cos^2 t\sin t\,(a\cos t + a(\cos^2 t - \sin^2 t))}{a(1+\cos t)} \right) dt$$

$$= \int_0^{\frac{\pi}{2}} \Big(\big(3a^3(1+\cos t)^2\cos^3 t + a^3(1+\cos t)^2\cos t\sin^2 t\big)(-\sin 2t - \sin t) +$$

$$+ \big(3a^3(1+\cos t)^2\sin^3 t + a^3(1+\cos t)^2\cos^2 t\sin t\big)(\cos t + \cos 2t) \Big)\, dt$$

$$= a^3 \int_0^{\frac{\pi}{2}} \text{calcule făcute -----}$$

11. Să se calculeze următoarele integrale duble:

a) $\iint_D e^{x^2+y^2}\,dx\,dy$, unde $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 1\}$;

↳ disc cu centrul $O(0,0)$ $R=1$

$\Rightarrow x \in [-1,1]$, $\quad y^2 \le 1-x^2 \Rightarrow |y| \le \sqrt{1-x^2} \Rightarrow -\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ t \in [0, 2\pi] \end{cases} \Big| \Rightarrow (r\cos t)^2 + (r\sin t)^2 \le 1 \Leftrightarrow r^2 \le 1 \Rightarrow r \in [0,1]$$

$$\Rightarrow I = \iint_D e^{x^2+y^2}\,dx\,dy = \int_0^{2\pi}\int_0^1 e^{r^2} \cdot \underbrace{r}_{\text{jacobianul}}\,dr\,dt = \frac{1}{2}\int_0^{2\pi}\int_0^1 2r \cdot e^{r^2}\,dr\,dt$$

Not $r^2 = x \Rightarrow 2r\,dr = dx$ $\quad \begin{array}{c|cc} r & 0 & 1 \\ \hline x & 0 & 1 \end{array}$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^1 e^x\,dx\,dt = \frac{1}{2}\int_0^{2\pi}(e-1)\,dt = \frac{1}{2}(e-1)\cdot t\Big|_0^{2\pi} = \frac{(e-1)\cdot 2\pi}{2} = \pi(e-1)$$

b) $\iint_D \sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}}\,dx\,dy$, unde $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 1, x, y \ge 0\}$;

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Big| \Rightarrow r^2 \le 1 \Leftrightarrow r \in [0,1]$$

$x, y \geq 0 \Rightarrow \cos t, \sin t \geq 0 \Rightarrow t \in [0, \frac{\pi}{2}]$

$$\Rightarrow \iint_D \sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}}\,dx\,dy = \int_0^{\frac{\pi}{2}} \int_0^1 \sqrt{\frac{1-r^2}{1+r^2}} \cdot r\,dr\,dt =$$

$$= \int_0^{\frac{\pi}{2}} \int_0^1 \frac{1}{\sqrt{r^2+1}} \cdot \sqrt{r^2-1} \cdot r\,dr\,dt$$

Not $r^2 = x \Rightarrow 2r\,dr = dx \Rightarrow$

r	0	1
x	0	1

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_0^1 \sqrt{\frac{1-x}{1+x}}\,dx\,dt$$

$x = \cos 2y$

$dx = -2\sin 2y\,dy$

x	0	1
y	$\frac{\pi}{4}$	0

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_{\pi/4}^0 \sqrt{\frac{1-\cos 2y}{1+\cos 2y}} \cdot (-2\sin 2y)\,dy\,dt = \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_{\pi/4}^0 \sqrt{\frac{1-\cos 2y}{2} \cdot \frac{2}{1+\cos 2y}} \cdot (-2\sin 2y)\,dy\,dt$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_{\pi/4}^0 \sqrt{\frac{\sin^2 y}{\cos^2 y}} (-2\sin 2y)\,dy\,dt = \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_{\pi/4}^0 \left|\frac{\sin y}{\cos y}\right| (-2\sin 2y)\,dy\,dt$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_{\pi/4}^0 \frac{\sin y}{\cos y} \cdot -2\sin y \cos y\,dy\,dt$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{\pi/4} \sin^2 y\,dy\,dt = \int_0^{\frac{\pi}{2}} \int_0^{\pi/4} \frac{1-\cos 2y}{2}\,dy\,dt$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_0^{\pi/4} (1-\cos 2y)\,dy\,dt = \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(y\Big|_0^{\pi/4} - \frac{\sin 2y}{2}\Big|_0^{\pi/4}\right) dt$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{4} - \frac{1}{2}\right) dt = \frac{1}{4}\left(\frac{\pi}{2} - 1\right) \cdot \frac{\pi}{2} = \frac{\pi}{8}\left(\frac{\pi}{2} - 1\right)$$

mai de unde e $\cdot \frac{1}{2}$ ☹

c) $\iint_D \frac{xy\,dx\,dy}{\sqrt{x^2+y^2}}$, unde $D = \left\{(x,y) \in \mathbb{R}^2 \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1, x, y \ge 0\right\}$;

$$\Rightarrow \begin{cases} x = ar\cos t \\ y = br\sin t \end{cases} \qquad r \in [0,1]$$

$$\Rightarrow J = abr$$

$$x, y \ge 0 \Rightarrow t \in \left[0, \frac{\pi}{2}\right]$$

$$\iint_D \frac{xy}{\sqrt{x^2+y^2}}\,dx\,dy = \int_0^{\frac{\pi}{2}} \int_0^1 \frac{ar\cos t \cdot br\sin t}{\sqrt{a^2r^2\cos^2 t + b^2r^2\sin^2 t}} abr\,dr\,dt$$

$$= \int_0^{\frac{\pi}{2}} \int_0^1 \frac{a^2b^2r^2\cos t\sin t}{\sqrt{a^2\cos^2 t + b^2\sin^2 t}}\,dr\,dt = a^2b^2 \int_0^{\frac{\pi}{2}} \int_0^1 \frac{r^2\cos t\sin t}{\sqrt{a^2\cos^2 t + b^2\sin^2 t}}\,dr\,dt$$

$$= a^2b^2 \int_0^{\frac{\pi}{2}} \frac{\cos t\sin t}{\sqrt{a^2\cos^2 t + b^2\sin^2 t}} \cdot \frac{r^3}{3}\Big|_0^1 dt = \frac{a^2b^2}{3} \int_0^{\frac{\pi}{2}} \frac{\cos t\sin t}{\sqrt{a^2\cos^2 t + b^2\sin^2 t}}\,dt$$

$$= \frac{a^2b^2}{3} \int_0^{\frac{\pi}{2}} \frac{\cos t\sin t}{\sqrt{a^2(1-\sin^2 t) + b^2\sin^2 t}}\,dt = \frac{a^2b^2}{3} \int_0^{\frac{\pi}{2}} \frac{\cos t\sin t}{\sqrt{a^2 + \sin^2 t(b^2-a^2)}}\,dt$$

t	0	$\frac{\pi}{2}$
x	0	1

Not $\sin t = x \Leftrightarrow \cos t\,dt = dx$

$$= \frac{a^2b^2}{3} \int_0^1 \frac{x}{\sqrt{a^2 + x^2(b^2-a^2)}}\,dx = \frac{a^2b^2}{3} \int_0^1 \frac{x}{\sqrt{b^2-a^2} \cdot \sqrt{x^2 + \frac{a^2}{b^2-a^2}}}\,dx$$

$$= \frac{a^2b^2}{3} \cdot \frac{1}{\sqrt{b^2-a^2}} \sqrt{x^2 + \frac{a^2}{b^2-a^2}}\Bigg|_0^1$$

$$= \frac{a^2b^2}{3} \cdot \frac{1}{\sqrt{b^2-a^2}} \left(\sqrt{\frac{b^2}{b^2-a^2}} - \sqrt{\frac{a^2}{b^2-a^2}}\right)$$

$$= \frac{a^2b^2}{3} \cdot \underbrace{\frac{1}{\sqrt{b^2-a^2}} \cdot \frac{b-a}{\sqrt{b^2-a^2}}}_{a+b} = \frac{a^2b^2}{3(a+b)}$$

d) $\iint_D e^{-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)}\,dx\,dy$, unde D este exteriorul elipsei $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$; $\Rightarrow \frac{x^2}{a^2}+\frac{y^2}{b^2}>1$

$$\begin{cases} x = a r\cos t \\ y = b r \sin t \\ t\in[0,2\pi] \end{cases} \Rightarrow r^2>1,\ r\geq 0 \Rightarrow r\in(1,\infty) \qquad J = abr$$

$$\int_0^{2\pi}\int_1^{\infty} e^{-r^2}\cdot abr\,dr\,dt = -\frac{ab}{2}\int_0^{2\pi}\int_1^{\infty} -2r\cdot e^{-r^2}\,dr\,dt$$

Not $-r^2=x \Leftrightarrow -2r\,dr=dx \Leftrightarrow$

r	1	∞
x	-1	$-\infty$

$$= \frac{ab}{2}\int_0^{2\pi}\int_{-\infty}^{-1} e^x\,dx\,dt = \frac{ab}{2}\int_0^{2\pi} e^x\Big|_{-\infty}^{-1}\,dt = \frac{ab}{2}\int_0^{2\pi} e^{-1}\,dt$$

$$= \frac{ab}{2}\,e^{-1}t\Big|_0^{2\pi} = \frac{\pi ab}{e}$$

e) $\iint_D \frac{dx\,dy}{\sqrt{x^2+y^2}(1+x^2+y^2)^2}$, unde D este interiorul cercului $x^2+y^2=a^2$;

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ t\in[0,2\pi] \end{cases} \Rightarrow r^2\leq a^2 \Rightarrow r\in[0,a] \qquad \Rightarrow J = r$$

$$\iint_D \frac{dx\,dy}{\sqrt{x^2+y^2}(1+x^2+y^2)^2} = \int_0^{2\pi}\int_0^{a} \frac{1}{\sqrt{r^2}(1+r^2)^2}\cdot r\,dr\,dt$$

$$= \int_0^{2\pi}\int_0^{a} \frac{1}{(1+r^2)^2}\,dr\,dt$$

Not $r=\operatorname{tg} x \Leftrightarrow dr=(1+\operatorname{tg}^2 x)\,dx$

r	0	a
x	0	$\operatorname{arctg} a$

$$= \int_0^{2\pi}\int_0^{\operatorname{arctg} a} \frac{1}{1+\operatorname{tg}^2 x}\,dx\,dt = \frac{1}{a}\int_0^{2\pi}\int_0^{\operatorname{arctg} a} \cos^2 x\,dx\,dt = \int_0^{2\pi}\int_0^{\operatorname{arctg} a} \frac{1+\cos 2x}{2}\,dx\,dt$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\operatorname{arctg} a}(1+\cos 2x)\,dx\,dt = \frac{1}{2}\int_0^{2\pi}\left(x+\frac{\sin 2x}{2}\right)\Big|_0^{\operatorname{arctg} a}\,dt$$

$$= \frac{1}{2}\int_0^{2\pi} \operatorname{arctg} a + \frac{\sin(2\operatorname{arctg} a)}{2}\, dt =$$

$$= \frac{1}{2}\left(\operatorname{arctg} a + \frac{\sin(2\operatorname{arctg} a)}{2}\right)\cdot t \Big|_0^{2\pi} =$$

$$= \pi\left(\operatorname{arctg} a + \frac{\sin(2\operatorname{arctg} a)}{2}\right) = \pi\left(\operatorname{arctg} a + \frac{a}{a^2+1}\right)$$

$$\frac{\sin(2\operatorname{arctg} a)}{2} = \frac{2\sin(\operatorname{arctg} a)\cdot\cos(\operatorname{arctg} a)}{2} =$$

$$= \frac{a}{\sqrt{a^2+1}}\cdot\frac{1}{\sqrt{a^2+1}}$$

f) $\iint_D xy\sqrt{x^2+4y^2}\,\mathrm{d}x\,\mathrm{d}y$, unde D este sfertul din primul cadran al discului unitar;

se refera la primu cadran gen...

$$x^2+y^2 \le 1$$

$$\Rightarrow \begin{cases} x = r\cos t \\ y = r\sin t \end{cases}, \ r\in[0,1], \ t\in\left[0,\frac{\pi}{2}\right], \ j = r$$

$$\iint xy\sqrt{x^2+4y^2}\,dx\,dy = \int_0^{\pi/2}\int_0^1 r^2\sin t\cos t\cdot\sqrt{r^2\cos^2 t + 4r^2\sin^2 t}\cdot r\,dr\,dt$$

$$= \int_0^{\pi/2}\sin t\cos t\sqrt{\cos^2 t+4\sin^2 t}\int_0^1 r^4\,dr\,dt = \frac{1}{5}\int_0^{\frac{\pi}{2}}\sin t\cos t\sqrt{\underbrace{\cos^2 t}_{1-\sin^2 t}+4\sin^2 t}\,dt$$

$$= \frac{1}{5}\int_0^{\frac{\pi}{2}}\sin t\sqrt{1+3\sin^2 t}\cos t\,dt$$

Not $\sin t = x \Leftrightarrow \cos t\,dt = dx$

t	0	$\frac{\pi}{2}$
x	0	1

$$= \frac{1}{30}\int_0^1 6x\sqrt{1+3x^2}\,dx = \frac{1}{30}\int_1^4\sqrt{t}\,dt = \frac{1}{30}\cdot\frac{t^{\frac{3}{2}}}{\frac{3}{2}}\Big|_1^4$$

Not $1+3x^2 = t \Leftrightarrow 6x\,dx = dt$

x	0	1
t	1	4

$$= \frac{1}{45}(8-1) = \frac{7}{45}$$

g) $\iint_D \left(1+\sqrt{x^2+y^2}\right) dx\,dy$, unde $D=\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\le y, x\ge 0\}$;

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r^2 \le r\sin t \;\Big|:r>0 \iff r\le \sin t$$

$$\Rightarrow r\in[0,\sin t] \overset{CE}{\Rightarrow} \sin t\ge 0$$

$$x\ge 0 \Rightarrow \cos t\ge 0$$

$$\Downarrow \quad t\in\left[0,\frac{\pi}{2}\right]$$

$$j = r$$

$$\Rightarrow \iint_D \left(1+\sqrt{x^2+y^2}\right)dx\,dy = \int_0^{\frac{\pi}{2}}\int_0^{\sin t}(1+r)\,r\,dr\,dt = \int_0^{\frac{\pi}{2}}\int_0^{\sin t}(r+r^2)\,dr\,dt$$

$$= \int_0^{\frac{\pi}{2}}\left(\frac{r^2}{2}+\frac{r^3}{3}\;\Big|_0^{\sin t}\right)dt = \int_0^{\frac{\pi}{2}}\left(\frac{\sin^2 t}{2}+\frac{\sin^3 t}{3}\right)dt$$

$$= \int_0^{\frac{\pi}{2}}\left(\frac{1-\cos 2t}{4}+\sin t\cdot\frac{(1-\cos^2 t)}{3}\right)dt = \int_0^{\frac{\pi}{2}}\frac{1}{4}\,dt+\int_0^{\frac{\pi}{2}}\frac{\cos 2t}{4}+\sin t\cdot\frac{1-\cos^2 t}{3}\,dt$$

$$= \frac{1}{4}t\Big|_0^{\frac{\pi}{2}} + \frac{1}{2}\left(\int_0^{\frac{\pi}{2}}\frac{\cos 2t}{2}\,dt + \frac{1}{3}\int_0^{\frac{\pi}{2}}\sin t\cdot(1-\cos^2 t)\,dt\right)$$

$$= \frac{\pi}{8} + \frac{1}{2}\cdot\underbrace{\frac{\sin 2t}{4}\Big|_0^{\frac{\pi}{2}}}_{=0} + \frac{1}{3}\int_0^{\frac{\pi}{2}}\sin t(1-\cos^2 t)\,dt$$

Not $\cos t = x \iff -\sin t\,dt = dx \iff$

t	0	$\frac{\pi}{2}$
x	1	0

$$= \frac{\pi}{8} + \frac{1}{3}\int_0^1 (1-x^2)\,dt = \frac{\pi}{8}+\frac{1}{3}\cdot\left(x-\frac{x^3}{3}\right)\Big|_0^1 = \frac{\pi}{8}+\frac{1}{3}\left(1-\frac{1}{3}\right) = \frac{\pi}{8}+\frac{2}{9}$$

h) $\iint_D \frac{x^2+y^2}{x+\sqrt{x^2+y^2}}\,dx\,dy$, unde D este definit de $y\ge 0$ şi $0\le r\le a(1+\cos t)$, cu $a>0$;

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \qquad r\in[0,\, a(1+\cos t)]$$

$$\Rightarrow a(1+\cos t)\ge 0 \iff \cos t\ge -1 \text{ adev } \forall t$$

$$y\ge 0 \Rightarrow \sin t\ge 0 \iff t\in[0,\pi]$$

$$j = r$$

$$\iint_D \frac{x^2+y^2}{x+\sqrt{x^2+y^2}}\,dx\,dy = \int_0^{\pi}\int_0^{a(1+\cos t)} \frac{r^2}{r\cos t + r}\cdot r\,dr\,dt =$$

$$= \int_0^{\pi}\int_0^{a(1+\cos t)} \frac{r^2}{\cos t+1}\,dr\,dt = \int_0^{\pi} \frac{1}{\cos t+1}\cdot \frac{r^3}{3}\Big|_0^{a(1+\cos t)} = \frac{1}{3}\int_0^{\pi} \frac{a^3(1+\cos t)^3}{1+\cos t}\,dt$$

$$= \frac{a^3}{3}\int_0^{\pi}(1+\cos t)^2\,dt = a^3\int_0^{\pi} 1+2\cos t+\cos^2 t\,dt =$$

$$= \frac{a^3}{3}\left(t\Big|_0^{\pi} + 2\sin t\Big|_0^{\pi} + \int_0^{\pi}\cos^2 t\,dt\right)$$

$$= \frac{a^3}{3}\cdot\pi + \frac{a^3}{3}\int_0^{\pi}\frac{1+\cos 2t}{2}\,dt = \frac{a^3}{3}\pi + \frac{a^3}{6}\left(t\Big|_0^{\pi} + \frac{\sin 2t}{2}\Big|_0^{\pi}\right)$$

$$= \frac{a^3}{3}\left(\pi+\frac{\pi}{2}\right) = \frac{a^3\cdot 3\pi}{3\cdot 2} = \frac{\pi a^3}{2}$$

i) $\iint_D (4x-3-x^2-y^2)\,dx\,dy$, unde D este definit de $x^2+y^2-4x-3\le 0$;

$$x^2+y^2-4x-3\le 0 \iff x^2-4x+4+y^2-7\le 0$$

$$\iff (x-2)^2+y^2\le 7$$

$$\begin{cases} x = 2+r\cos t \\ y = r\sin t \end{cases} \quad t\in[0,2\pi] \quad \Big| \Rightarrow r^2\le 7 \iff r\in[0,\sqrt{7}]$$

$$\Rightarrow \iint_D (4x-3-x^2-y^2)\,dx\,dy = \int_0^{2\pi}\int_0^{\sqrt{7}} \left(4(2+r\cos t)-3-(2+r\cos t)^2-(r\sin t)^2\right) r\,dr\,dt$$

$$= \int_0^{2\pi}\int_0^{\sqrt{7}} \left(8+4r\cos t-3-4-4r\cos t-r^2\cos^2 t-r^2\sin^2 t\right) r\,dr\,dt$$

$$= \int_0^{2\pi}\int_0^{\sqrt{7}} (r-r^3)\,dr\,dt = \int_0^{2\pi} \frac{r^2}{2}-\frac{r^4}{4}\Big|_0^{\sqrt{7}}\,dt = \int_0^{2\pi} 4\frac{1}{2}-\frac{49}{4}\,dt =$$

$$= -\frac{35\cdot 2\pi}{4_2} = \frac{-35\pi}{2}$$

dacă mă gândesc că ar trebui să fie $D: x^2+y^2-4x+3 \le 0$

$\Leftrightarrow x^2+y^2-4x+3 \le 0 \Leftrightarrow x^2-4x+4+y^2-1 \le 0 \Leftrightarrow (x-2)^2+y^2 \le 1$

$$\begin{cases} x = 2+r\cos t \\ y = r\sin t \end{cases} \Rightarrow r^2 \in [0,1], \quad t \in [0, 2\pi], \quad J = r$$

$$\Rightarrow \iint_D (4x-3-x^2-y^2)\,dx\,dy = \int_0^{2\pi}\int_0^1 \left(4(2+r\cos t)-3-(2+r\cos t)^2-(r\sin t)^2\right) r\,dr\,dt$$

$$= \int_0^{2\pi}\int_0^1 \left(8+4r\cos t-3-4-4r\cos t-r^2\cos^2 t-r^2\sin^2 t\right) r\,dr\,dt$$

$$= \int_0^{2\pi}\int_0^1 (r-r^3)\,dr\,dt = \int_0^{2\pi} \frac{r^2}{2}-\frac{r^4}{4}\Big|_0^1 dt = \int_0^{2\pi} \frac{2/1}{2}-\frac{1}{4}\,dt = \frac{1}{4}\cdot 2\pi = \frac{\pi}{2}$$

j) $\iint_D \sqrt{a^2-x^2-y^2}\,dx\,dy$, unde D este definit de $x^2+y^2 \le ax$, cu $a>0$;

$x^2+y^2 \le ax$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ J = r \end{cases} \Rightarrow r^2 \le a r\cos t \;|:r\neq 0 \Leftrightarrow r \le a\cos t$$

$$\Rightarrow r \in [0, a\cos t]$$

$$\overset{CE}{\Rightarrow} a\cos t \ge 0 \;\Big|_{a>0} \Rightarrow \cos t \ge 0 \Rightarrow t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\iint_D \sqrt{a^2-x^2-y^2}\,dx\,dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^{a\cos t} \sqrt{a^2-r^2}\cdot r\,dr\,dt$$

Not $a^2-r^2 = x \Leftrightarrow -2r\,dr = dx \Leftrightarrow$

r	0	$a\cos t$
t	a^2	$a^2-a^2\cos^2 t = a^2\sin^2 t$

$$= -\frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_{a^2}^{a^2\sin^2 t} \sqrt{x}\,dx = -\frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\Big|_{a^2}^{a^2\sin^2 t} = -\frac{1}{3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(a^3|\sin t|^3-a^3\right) dt$$

$$= \frac{a^3}{3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(1-|\sin t|^3\right) dt = \frac{a^3\cdot\pi}{3} - \frac{a^3}{3}\int_{-\frac{\pi}{2}}^{0} \underbrace{-\sin^3 t}_{(1-\cos^2 t)\sin t}\,dt + \frac{a^3}{3}\int_0^{\frac{\pi}{2}} -\sin^3 t\,dt$$

Not $\cos t = x \Leftrightarrow -\sin t\,dt = dx$

① $\begin{array}{c|c|c} t & -\frac{\pi}{2} & 0 \\ \hline x & 0 & 1 \end{array}$ ② $\begin{array}{c|c|c} t & 0 & \frac{\pi}{2} \\ \hline x & 1 & 0 \end{array}$

$$= \frac{a^3\pi}{3} - \frac{2a^3}{3}\int_0^1 (1-x^2)\,dx$$

$$= \frac{a^3\pi}{3} - \frac{2a^3}{3}\cdot\left(x - \frac{x^3}{3}\right)\Big|_0^1 = \frac{a^3\pi}{3} - \frac{2a^3}{3}\cdot\frac{2}{3} = \frac{a^3}{3}\left(\pi - \frac{4}{3}\right)$$

k) $\iint_D \left(x^2+y^2-\frac{x+y}{2}\right)\mathrm{d}x\,\mathrm{d}y$, unde D este definit de $x^2+y^2\le 1$;

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r^2 \le 1 \Leftrightarrow r\in[0,1]$$

$t \in [0,2\pi]$

$j = r$

$$\Rightarrow \iint_D \left(x^2+y^2-\frac{x+y}{2}\right)dx\,dy = \int_0^{2\pi}\int_0^1 \left(r^2 - \frac{r\cos t + r\sin t}{2}\right) r\,dr\,dt$$

$$\int_0^{2\pi}\left(\int_0^1 r^3\,dr - \frac{\cos t+\sin t}{2}\int_0^1 r^2\,dr\right)dt$$

$$\int_0^{2\pi}\left(\frac{r^4}{4}\Big|_0^1 - \frac{\cos t+\sin t}{2}\cdot\frac{r^3}{3}\Big|_0^1\right)dt = \int_0^{2\pi}\left(\frac{1}{4} - \frac{1}{6}(\cos t+\sin t)\right)dt$$

$$= \frac{2\pi}{4} - \frac{1}{6}\left(\int_0^{2\pi}\cos t\,dt + \int_0^{2\pi}\sin t\,dt\right) = \frac{\pi}{2} - \frac{1}{6}\left(\underbrace{\sin t\Big|_0^{2\pi}}_{=0} - \underbrace{\cos t\Big|_0^{2\pi}}_{=0}\right) = \frac{\pi}{2}$$

l) $\iint_D y\,\mathrm{d}x\,\mathrm{d}y$, unde $D = \{(x,y)\in\mathbb{R}^2 \mid x^2+y^2 \le 2x+2y-1\}$.

$x^2+y^2 \le 2x+2y-1 \Leftrightarrow x^2-2x+1+y^2-2y+1 \le 1 \Leftrightarrow (x-1)^2+(y-1)^2 \le 1$

$$\Rightarrow \begin{cases} x = 1+r\cos t \\ y = 1+r\sin t \end{cases} \Rightarrow r^2 \le 1 \Rightarrow r\in[0,1] \quad , j = r$$

$t\in[0,2\pi]$

$$\Rightarrow \iint_D y\,dx\,dy = \int_0^{2\pi}\int_0^1 (1+r\sin t)\cdot r\,dr\,dt = \int_0^{2\pi}\int_0^1 (r + r^2\sin t)\,dr\,dt =$$

$$= \int_0^{2\pi}\left(\frac{r^2}{2}\Big|_0^1 + \sin t\cdot\frac{r^3}{3}\Big|_0^1\right)dt = \int_0^{2\pi}\left(\frac{1}{2} + \frac{\sin t}{3}\right)dt = \frac{1}{2}t\Big|_0^{2\pi} - \underbrace{\frac{\cos t}{3}\Big|_0^{2\pi}}_{=0}$$

$$= \pi$$

12. Să se calculeze ariile domeniilor limitate de curbele: vezi pag 232

a) $(x^2+y^2)^2 = 2a^2xy$, cu $a>0$;

Se deduce că $(x^2+y^2)^2 \le 2a^2xy$

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r^4 \le 2a^2 r^2 \sin t\cos t \mid :r^2$$

$$\Rightarrow r^2 \le 2a^2 \sin t\cos t \Leftrightarrow r^2 \le a^2 \sin 2t$$

$$\Leftrightarrow r \in [0, a\sqrt{\sin 2t}]$$

$J = r$

CE $\sin 2t \ge 0 \Rightarrow t \in [0, \frac{\pi}{4}]$

$t \in [0, \frac{\pi}{2}]$

$$\Rightarrow A(D) = \iint_D 1\,dxdy = \int_0^{\frac{\pi}{4}}\int_0^{a\sqrt{\sin 2t}} r\,dr\,dt = \int_0^{\frac{\pi}{4}} \frac{r^2}{2}\Big|_0^{a\sqrt{\sin 2t}} dt =$$

$$= \int_0^{\frac{\pi}{4}} \frac{a^2\sin 2t}{2}\,dt = \frac{a^2}{2}\int_0^{\frac{\pi}{4}} \sin 2t\,dt = \frac{a^2}{2}\cdot\frac{-\cos 2t}{2}\Big|_0^{\frac{\pi}{4}}$$

$$= \frac{a^2}{2}\cdot\frac{1}{2} = \frac{a^2}{4}$$

$$\Rightarrow A_{tot} = 4A(D) = a^2$$

b) $(x^2+y^2)^2 = 2ax^3$, cu $a>0$;

Se deduce că $(x^2+y^2)^2 \le 2ax^3$

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r^4 \le 2a\, r^3\cos^3 t \mid :r^3 > 0 \Leftrightarrow r \le 2a\cos^3 t$$

$J = r$

$$\Rightarrow r \in [0, 2a\cos^3 t]$$

CE $\Rightarrow \cos^3 t \ge 0 \Rightarrow t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$$\Rightarrow A(D) = \iint_D 1\,dxdy \Leftrightarrow \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^{2a\cos^3 t} r\,dr\,dt = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{r^2}{2}\Big|_0^{2a\cos^3 t} dt =$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2a^2\cos^6 t\,dt = 2a^2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^6 t\,dt = 4a^2\int_0^{\frac{\pi}{2}} \left(\frac{1+\cos 2t}{2}\right)^3 dt$$

$$= \frac{a^2}{2}\int_0^{\frac{\pi}{2}} (1+\cos 2t)^3\,dt = \frac{a^2}{2}\int_0^{\frac{\pi}{2}} (1+3\cos 2t + 3\cos^2 2t + \cos^3 2t)\,dt$$

$$= \frac{a^2}{2}\int_0^{\frac{\pi}{2}} \left(1+3\cos 2t + 3\cdot\frac{1+\cos 4t}{2} + \frac{1+\cos 4t}{2}\cos 2t\right)dt$$

$$= \frac{a^2}{2}\left(\left(t+3\cdot\frac{\sin 2t}{2}+\frac{3}{2}\left(t+\frac{\sin 4t}{8}\right)\right)\Big|_0^{\frac{\pi}{2}} + \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos 2t + \cos 2t\cdot\cos 4t\right)dt$$

$$= \frac{a^2}{2}\left(\frac{\pi}{2} + \frac{3}{2}\cdot 0 + \frac{3}{2}\left(\frac{\pi}{2}+0\right) + \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos 2t + \frac{1}{2}(\cos 2t + \cos 6t)\,dt\right)$$

$$= \frac{a^2}{2}\left(\frac{5\pi}{4} + \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{3\cos 2t}{2} + \frac{\cos 6t}{2}\,dt\right)$$

$$= \frac{a^2}{2}\left(\frac{5\pi}{4} + \frac{3}{4}\cdot\underbrace{\frac{\sin 2t}{2}}_{=0} + 0\right)$$

$$\int_0^{\frac{\pi}{2}} \cos^6 t\,dt = \frac{1}{2}\int_0^1 \frac{2x\cdot x^5}{\sqrt{1-x^2}}\,dx$$

Not $\cos t = x \Leftrightarrow t = \arccos x \Leftrightarrow dt = -\frac{1}{\sqrt{1-x^2}}\,dx$

t	0	$\frac{\pi}{2}$
x	1	0

Not $x^2 = t \Leftrightarrow 2x\,dx = dt$

x	0	1
t	0	1

$$= \frac{1}{2}\int_0^1 t^{\frac{5}{2}}\cdot(1-t)^{-\frac{1}{2}}\,dt = \frac{1}{2}B\left(\frac{7}{2},\frac{1}{2}\right) = \frac{1}{2}\,\frac{\Gamma\left(\frac{7}{2}\right)\cdot\Gamma\left(\frac{1}{2}\right)}{\Gamma(4)}$$

$$= \frac{1}{2}\,\frac{\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\cdot\left(\Gamma\left(\frac{1}{2}\right)\right)^2}{2!} = \frac{5\pi}{2^5} = \frac{5\pi}{32}$$

$$\Rightarrow A(\Delta) = 4a^2\,\frac{5\pi}{32} = \frac{5\pi a^2}{8}$$

c) $(x^2+y^2)^2 = a^2\left(x^2-y^2\right)$, cu $a>0$.

Logic $(x^2+y^2)^2 \le a^2(x^2-y^2)$

$\Rightarrow \begin{cases} x = r\cos t \\ y = r\sin t \end{cases}$ $\Rightarrow r^4 \le a^2 r^2(\cos^2 t - \sin^2 t) \mid :r^2>0$

$\Leftrightarrow r^2 \le a^2\cos 2t \Rightarrow r\in[0, a\sqrt{\cos 2t}]$

$j = r$

Cf $a\sqrt{\cos 2t} \ge 0 \Rightarrow t\in[-\frac{\pi}{4}, \frac{\pi}{4}] \Big/ \Rightarrow t\in[0,\frac{\pi}{4}]$ deci $t\in C_1$

$$\Rightarrow \iint_D 1\,dxdy = \int_0^{\frac{\pi}{4}}\int_0^{a\sqrt{\cos 2t}} r\,dr\,dt = \int_0^{\frac{\pi}{4}} \frac{r^2}{2}\Big|_0^{a\sqrt{\cos 2t}}\,dt =$$

$$= \int_0^{\frac{\pi}{4}} \frac{a^2\cos 2t}{2}\,dt = \frac{a^2}{2}\int_0^{\frac{\pi}{4}} \cos 2t\,dt = \frac{a^2}{2}\cdot\frac{\sin 2t}{2}\Big|_0^{\frac{\pi}{4}} = \frac{a^2}{4}$$

$\Rightarrow A_{tot} = 4A(D) = a^2$

30.01.2025

13. Să se calculeze următoarele integrale triple:

a) $\iiint_D \frac{dx\,dy\,dz}{(1+x+y+z)^3}$, unde D este definit de $x+y+z\le 1$, $x,y,z\ge 0$;

$\Rightarrow z < 1-x-y$

$$\Rightarrow \iiint_D \frac{dxdydz}{(1+x+y+z)^3} = \iint_{D_1}\left(\int_0^{1-x-y}\frac{dz}{(1+x+y+z)^3}\right)dxdy =$$

Not $1+x+y+z = t \Leftrightarrow dz = dt$

z	0	$1-x-y$
t	$1+x+y$	2

$$= \iint_{D_1}\int_{1+x+y}^{2}\frac{dt}{t^3}\,dxdy = \iint_{D_1}\frac{t^{-2}}{-2}\Big|_{1+x+y}^{2}\,dxdy = -\frac{1}{2}\iint_{D_1}\left(\frac{1}{4}-(1+x+y)^{-2}\right)dxdy$$

$$= -\frac{1}{2}\int_0^1\int_0^1\left(\frac{1}{4}-(1+x+y)^{-2}\right)dxdy = -\frac{1}{8}\int_0^1\int_0^1 dxdy + \frac{1}{2}\int_{0,0}^{1,1}(1+x+y)^{-2}\,dxdy$$

$$= -\frac{1}{8}\int_0^1 x\Big|_0^1\,dy + \frac{1}{2}\int_0^1(-1)(1+x+y)^{-1}\Big|_0^1\,dy = -\frac{1}{8}\cdot y\Big|_0^1 - \frac{1}{2}\int_0^1 (2+y)^{-1}-(1+y)^{-1}\,dy$$

$$= -\frac{1}{8} - \frac{1}{2}\left(\int_0^1\frac{1}{2+y}\,dy - \int_0^1\frac{1}{1+y}\,dy\right) = -\frac{1}{8} - \frac{1}{2}\left(\ln|2+y|_0^1 - \ln(1+y)\Big|_0^1\right)$$

$= -\frac{1}{8} - \frac{1}{2}\ln\frac{3}{4}$

b) $\iiint_D xyz\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, unde D este definit de $x^2+y^2+z^2\le 1$, $x,y,z\ge 0$;

$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases}$ $\Big|\Rightarrow r^2\le 1 \Leftrightarrow r\in[0,1]$, $\theta\in[0,\frac{\pi}{2}]$, $\varphi\in[0,\frac{\pi}{2}]$

$J = r^2\sin\theta$

$$\Rightarrow \iiint_D xyz\,dx\,dy\,dz = \int_0^{\pi/2}\int_0^{\pi/2}\int_0^1 r^3\sin^2\theta\cos\theta\cos\varphi\sin\varphi\, r^2\sin\theta\,dr\,d\theta\,d\varphi$$

$$= \int_0^{\pi/2}\int_0^{\pi/2}\sin^3\theta\cos\theta\cos\varphi\sin\varphi\int_0^1 r^5\,dr\;d\theta\,d\varphi$$

$$= \int_0^{\pi/2}\cos\varphi\sin\varphi\int_0^{\pi/2}\sin^3\theta\cos\theta\cdot\frac{r^6}{6}\Big|_0^1\,d\theta\,d\varphi$$

$\Big\{$ not $\sin\theta = t \Leftrightarrow \cos\theta\,d\theta = dt$; $\begin{array}{c|c|c}\theta & 0 & \pi/2\\ \hline t & 0 & 1\end{array}$

$$= \frac{1}{6}\int_0^{\pi/2}\cos\varphi\sin\varphi\int_0^1 t^3\,dt\,d\varphi = \frac{1}{6}\int_0^{2\pi}\cos\varphi\cdot\sin\varphi\cdot\frac{t^4}{4}\Big|_0^1 =$$

$$= \frac{1}{24}\int_0^{\pi/2}\cos\varphi\sin\varphi\,d\varphi = \frac{1}{24}\cdot\int_0^1 t\,dt = \frac{1}{24}\cdot\frac{t^2}{2}\Big|_0^1 = \frac{1}{48}$$

c) $\iiint_D (x^2+y^2+z^2)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, unde D este definit de $x^2+y^2+z^2\le 1$, $x,y,z\ge 0$;

$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases}$ $\Big|\Rightarrow r^2\le 1 \Leftrightarrow r\in[0,1]$, $\theta\in[0,\frac{\pi}{2}]$, $\varphi\in[0,\frac{\pi}{2}]$

$J = r^2\sin\theta$

$$\Rightarrow \int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}}\int_0^1 r^2\cdot r^2\sin\theta\,dr\,d\theta\,d\varphi = \int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}}\sin\theta\cdot\frac{r^5}{5}\Big|_0^1\,d\theta\,d\varphi = \frac{1}{5}\int_0^{\frac{\pi}{2}}-\cos\theta\Big|_0^{\frac{\pi}{2}}d\varphi$$

$$= -\frac{1}{5}\int_0^{\frac{\pi}{2}}-1\,d\varphi = \frac{1}{5}\cdot\frac{\pi}{2} = \frac{\pi}{10}$$

d) $\iiint_D \sqrt{R^2 - x^2 - y^2 - z^2}\,dx\,dy\,dz$, unde D este definit de $x^2 + y^2 + z^2 \leq R^2$;

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases} \quad \Big| \quad \Rightarrow r^2 \leq R^2 \Rightarrow r \in [0, R]$$

$$\theta \in [0, \tfrac{\pi}{2}]$$

$$\varphi \in [0, \tfrac{\pi}{2}]$$

$$j = r^2\sin\theta$$

$$\Rightarrow \int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}}\int_0^{R} \sqrt{R^2 - r^2}\cdot r^2\sin\theta\,dr\,d\theta\,d\varphi$$

$$\int_0^R \sqrt{R^2 - r^2}\cdot r^2\,dr = \int_0^R (R^2 - r^2)^{\frac{1}{2}}\cdot r^2\,dr = R^3\int_0^R \left(1 - \left(\frac{r}{R}\right)^2\right)^{\frac{1}{2}}\cdot\left(\frac{r}{R}\right)^2 dr$$

Not $\left(\frac{r}{R}\right)^2 = t \Leftrightarrow \frac{2r}{R^2}\,dr = dt \Leftrightarrow \frac{2}{R}\left(\frac{r}{R}\right)dr = dt$ $\Rightarrow \frac{r}{R} = t^{\frac{1}{2}}$

r	0	R
t	0	1

$$= \frac{R^4}{2}\int_0^R \left(1 - \left(\frac{r}{R}\right)^2\right)^{\frac{1}{2}}\cdot\frac{r}{R}\cdot\frac{2}{R}\cdot\frac{r}{R}\,dr = \frac{R^4}{2}\int_0^1 (1-t)^{\frac{1}{2}}\cdot t^{\frac{1}{2}}\,dt$$

$$= \frac{R^4}{2}\,\beta\left(\frac{3}{2}, \frac{3}{2}\right) = \frac{R^4}{2}\cdot\frac{\Gamma(\frac{3}{2})\cdot\Gamma(\frac{3}{2})}{\Gamma(3)} = \frac{R^4}{2}\cdot\frac{\frac{1}{2}\cdot\frac{1}{2}\cdot\pi}{2!} = \frac{\pi R^4}{16}$$

$$\Rightarrow \frac{\pi R^4}{16}\int_0^{2\pi}\int_0^{\pi}\sin\theta\,d\theta\,d\varphi = \frac{\pi R^4}{16}\cdot\int_0^{2\pi} -\cos\theta\Big|_0^{\pi} d\varphi = \frac{\pi R^4}{16}\cdot 2\int_0^{2\pi} d\varphi = \frac{\pi R^4}{8}\cdot 2\pi = \frac{\pi^2 R^4}{4}$$

e) $\iiint_D (x^2 + y^2 - z^2)\,dx\,dy\,dz$, unde D este definit de $x^2 + y^2 + z^2 \leq R^2$;

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases} \quad \Big| \quad \Rightarrow r^2 \leq R^2 \Leftrightarrow r \in [0, R]$$

$$\theta \in [0, \pi]$$

$$\varphi \in [0, 2\pi]$$

$$j = r^2\sin\theta$$

$$\iiint (x^2 + y^2 - z^2)\,dx\,dy\,dz = \int_0^{2\pi}\int_0^{\pi}\int_0^{R} (r^2\sin^2\theta - r^2\cos^2\theta)\cdot r^2\sin\theta\,dr\,d\theta\,d\varphi$$

$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{R} r^4(\sin^2\theta - \cos^2\theta)\sin\theta\,dr\,d\theta\,d\varphi = \int_0^{2\pi}\int_0^{\pi}(1 - 2\cos^2\theta)\sin\theta\cdot\frac{r^5}{5}\Big|_0^R d\theta\,d\varphi$$

$= \frac{R^5}{5} \int_0^{2\pi} \int_0^{\pi} (2\cos^2\theta - 1)(-\sin\theta)\, d\theta\, d\varphi$

Not $\cos\theta = t \Leftrightarrow -\sin\theta\, d\theta = dt$ $\quad \begin{array}{c|cc} \theta & 0 & \pi \\ \hline t & 1 & -1 \end{array}$

$= \frac{R^5}{5} \int_0^{2\pi} \int_1^{-1} (2t^2 - 1)\, d\varphi = -\frac{R^5}{5} \int_0^{2\pi} \int_{-1}^{1} (2t^2 - 1)\, d\varphi = -\frac{R^5}{5} \int_0^{2\pi} \left(\frac{2t^3}{3} - t \right) \Big|_{-1}^{1} d\varphi =$

$= -\frac{R^5}{5} \int_0^{2\pi} \frac{2}{3} - 1 - \left(-\frac{2}{3} + 1 \right) d\varphi = -\frac{R^5}{5} \int_0^{2\pi} \left(\frac{2}{3} - 1 + \frac{2}{3} - 1 \right) d\varphi = -\frac{R^5}{5} \cdot \left(-\frac{2}{3}\varphi \right) \Big|_0^{2\pi}$

$= \frac{4\pi R^5}{15}$

f) $\iiint_D \sqrt{x^2+y^2+z^2}\, dx\, dy\, dz$, unde D este definit de $x^2+y^2+z^2 \le z$;

$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases} \Rightarrow r^2 \le r\cos\theta \mid :r>0 \Leftrightarrow r \le \cos\theta$

$\Rightarrow r \in [0, \cos\theta]$ $\quad \text{CE} \Rightarrow \cos\theta > 0 \mid \theta \in [0,\pi] \Rightarrow \theta \in [0, \frac{\pi}{2}]$

$\varphi \in [0, 2\pi]$

$J = r^2 \sin\theta$

$\Rightarrow \iiint_D \sqrt{x^2+y^2+z^2}\, dx\, dy\, dz = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^{\cos\theta} r \cdot r^2 \sin\theta\, dr\, d\theta\, d\varphi = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin\theta \cdot \frac{r^4}{4} \Big|_0^{\cos\theta} d\theta\, d\varphi$

$= \frac{1}{4} \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin\theta \cdot \cos^4\theta\, d\theta\, d\varphi$

Not $\cos\theta = t \Leftrightarrow -\sin\theta\, d\theta = dt$ $\quad \begin{array}{c|cc} \theta & 0 & \frac{\pi}{2} \\ \hline t & 1 & 0 \end{array}$

$= \frac{1}{4} \int_0^{2\pi} \int_0^{1} t^4\, dt\, d\varphi = \frac{1}{4} \int_0^{2\pi} \frac{t^5}{5} \Big|_0^1 d\varphi = \frac{1}{4} \cdot \int_0^{2\pi} \frac{1}{5}\, d\varphi = \frac{2\pi}{20} = \frac{\pi}{10}$

g) $\iiint_D z e^{-(x^2+y^2+z^2)}\, dx\, dy\, dz$, unde D este definit de $x^2+y^2+z^2 \le a^2$, $z \ge 0$, cu $a > 0$;

$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \ge 0 \Rightarrow \theta \in [0, \frac{\pi}{2}] \end{cases}$

$\mid \Rightarrow r^2 \le a^2 \Rightarrow r \in [0, a]$

$J = r^2\sin\theta$ $\qquad \varphi \in [0, 2\pi]$

$$\Rightarrow \iiint_D z\, e^{-(x^2+y^2+z^2)}\,dx\,dy\,dz = \int_0^{2\pi}\int_0^{\pi/2}\int_0^{a} r\cos\theta\, e^{-r^2}\cdot r^2\sin\theta\,dr\,d\theta\,d\varphi$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\pi/2}\sin\theta\cos\theta\int_0^{a} e^{-r^2}(-r^2)\cdot(-2r)\,dr\,d\theta\,d\varphi$$

Not $-r^2 = t \Leftrightarrow -2r\,dr = dt$ $\quad \begin{array}{c|c|c} r & 0 & a \\ \hline t & 0 & -a^2 \end{array}$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\cdot\int_0^{-a^2} e^t\cdot t\,dt\,d\theta\,d\varphi$$

$f' = e^t \Rightarrow f = e^t$, $g = t \Rightarrow g' = 1$

$$= \frac{\text{cerc}a}{2}\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\,d\theta\,d\varphi$$

Not $\sin\theta = t \Leftrightarrow \cos\theta\,d\theta = dt$ $\quad \begin{array}{c|c|c} \theta & 0 & \frac{\pi}{2} \\ \hline t & 0 & 1 \end{array}$

$$\int_0^{-a^2} e^t\cdot t\,dt = te^t\Big|_0^{-a^2} - \int_0^{-a^2} e^t\,dt$$

$$= -a^2e^{-a^2} - e^t\Big|_0^{-a^2} = -a^2e^{-a^2} - e^{-a^2} + 1$$

$$= 1 - e^{-a^2}(a^2+1) < \text{cerc}a$$

$$= \frac{\text{cerc}a}{2}\int_0^{2\pi}\int_0^{1} t\,dt\,d\varphi$$

$$= \frac{\text{cerc}a}{2}\int_0^{2\pi}\frac{t^2}{2}\Big|_0^1 d\varphi = \frac{\text{cerc}a}{4}\int_0^{2\pi} d\varphi = \frac{\text{cerc}a\cdot\pi}{2} = \frac{\pi}{2}\left(1 - e^{-a^2}(a^2+1)\right)$$

h) $\iiint_D \sqrt{x^2+y^2+z^2}\,dx\,dy\,dz$, unde D este definit de $x^2+y^2+z^2 \le 2ax$, cu $a > 0$.

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases} \Rightarrow j = r^2\sin\theta$$

$\Rightarrow r^2 \le 2ar\sin\theta\cos\varphi \;|:r\ge 0 \Leftrightarrow r \le 2a\sin\theta\cos\varphi$

$\Rightarrow r \in [0, 2a\sin\theta\cos\varphi]$

$C \in \sin\theta\cos\varphi > 0$

$\theta \in [0,\pi] \Rightarrow \sin\theta > 0$

$\varphi \in [0, 2\pi]$

$\Rightarrow \cos\varphi > 0 \Rightarrow \varphi = [-\frac{\pi}{2}, \frac{\pi}{2}]$

$$\Rightarrow \iiint_D \sqrt{x^2+y^2+z^2}\,dx\,dy\,dz = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^{\pi}\int_0^{2a\sin\theta\cos\varphi} r\cdot r^2\sin\theta\,dr\,d\theta\,d\varphi$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^{\pi}\sin\theta\cdot\frac{r^4}{4}\Big|_0^{2a\sin\theta\cos\varphi} d\theta\,d\varphi = 4a^4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^{\pi}\sin\theta\cdot(1-\cos^2\theta)^2\cos^4\varphi\,d\theta\,d\varphi$$

Not $\cos\theta = t \Leftrightarrow -\sin\theta = t \Leftrightarrow$ $\quad \begin{array}{c|c|c} \theta & 0 & \pi \\ \hline t & 1 & -1 \end{array}$

$$= 4a^4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4\varphi\int_{-1}^{1}(1-t^2)^2\,dt\,d\varphi = 4a^4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4\varphi\int_{-1}^{1}(1-2t^2+t^4)\,dt\,d\varphi$$

$$= 4a^4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4\varphi \left(t - \frac{t^3}{3} + \frac{t^5}{5}\right)\Big|_{-1}^{1} d\varphi = 4a^4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4\varphi \cdot \underbrace{\left(1 - \frac{1}{3} + \frac{1}{5} + 1 - \frac{1}{3} + \frac{1}{5}\right)}_{\frac{16}{15}} d\varphi$$

$$= \frac{64a^4}{15} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4\varphi \, d\varphi = \frac{64a^4}{15} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1+\cos 2\varphi}{2}\right)^2 = \frac{16a^4}{15} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1+\cos 2\varphi)^2 d\varphi =$$

$$= \frac{16a^4}{15} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + 2\cos 2\varphi + \cos^2 2\varphi)\, d\varphi = \frac{16a^4}{15} \left(\varphi \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + 2 \cdot \underbrace{\frac{\sin 2\varphi}{2}\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}}_{0} + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1+\cos 4\varphi}{2} d\varphi \right)$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1+\cos 4\varphi}{2} d\varphi = \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + \cos 4\varphi)\, d\varphi = \frac{1}{2} \left(\varphi + \frac{\sin 4\varphi}{4}\right)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\pi}{2}$$

$$= \frac{16a^4}{15}\left(\pi + \frac{\pi}{2}\right) = \frac{\overset{8}{\cancel{16}}a^4}{\underset{15}{\cancel{15}}} \cdot \frac{3\pi}{\cancel{2}} = \frac{8a^4\pi}{5}$$

14. Să se calculeze volumele domeniilor limitate de suprafețele:

a) $x^2 + y^2 + z^2 = 8$ și $x^2 + y^2 - 2z = 0$; $\quad z = \frac{x^2+y^2}{2} > 0$

S_1 $\quad$ S_2

S_1: $x^2+y^2+z^2=8$ e o sferă (obviously)

$S \cap Ox \Rightarrow y = z = 0 \Rightarrow x = \pm 2\sqrt{2}$

$S \cap Oy \Rightarrow x = z = 0 \Rightarrow y = \pm 2\sqrt{2}$

$S \cap Oz \Rightarrow x = y = 0 \Rightarrow z = \pm 2\sqrt{2}$

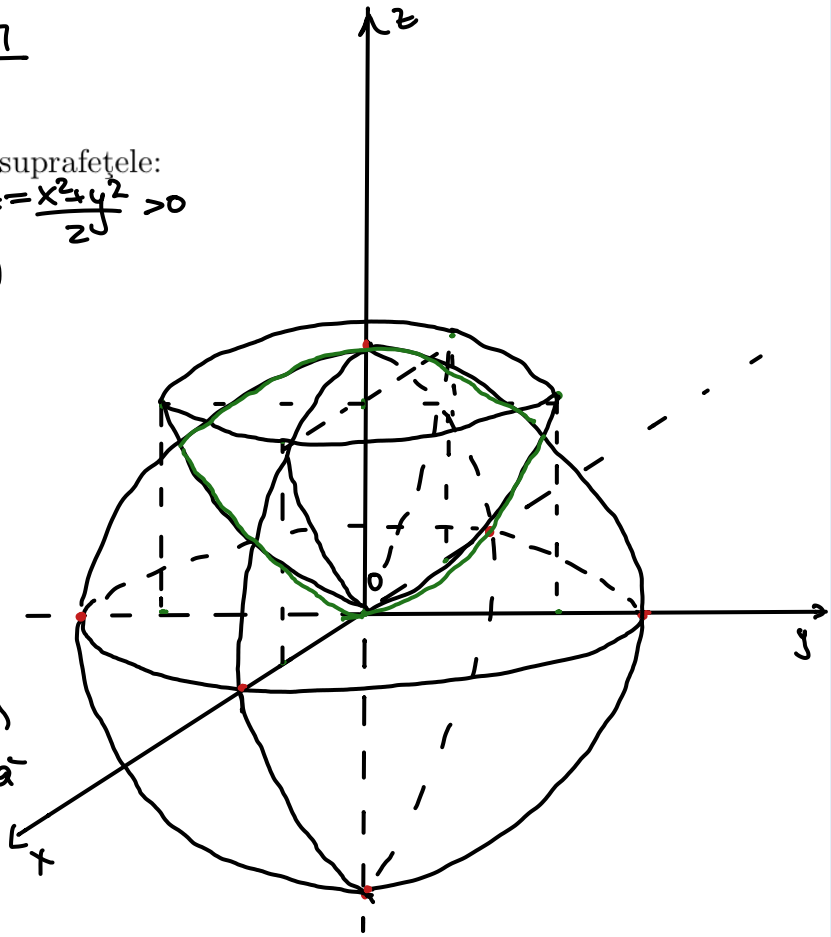
$S_2 \cap xOy \Rightarrow x^2 + y^2 = 0$ punct

$S_2 \cap xOz \Rightarrow x^2 - 2z = 0 \Rightarrow (-2,2)(2,2)$

$\Leftrightarrow z = \frac{x^2}{2}$ parabolă

$S_2 \cap yOz \Rightarrow y^2 = 2z \Rightarrow z = \frac{y^2}{2}$ parabolă $(-2,2)(2,2)$

$z = k \Rightarrow x^2 + y^2 = 2k$ cercuri



$$S_1 \cap S_2: \begin{cases} x^2+y^2+z^2=8 \\ x^2+y^2-2z=0 \end{cases} \Rightarrow 2z = 8 - z^2 \Leftrightarrow z^2 + 2z - 8 = 0 \Leftrightarrow z^2 + 4z - 2z - 8 = 0$$

$\Leftrightarrow (z+4)(z-2) = 0 \Leftrightarrow z \in \{-4, 2\}$ dar $z > 0$ $\Rightarrow z = 2$

$\Rightarrow x^2 + y^2 = 4$

$\Rightarrow D: x^2+y^2 \le 4$

$$\Rightarrow Vol(K) = \iiint_K 1\,dx\,dy\,dz = \iint_D \left(\int_{\frac{x^2+y^2}{2}}^{\sqrt{8-x^2-y^2}} dz \right) dx\,dy =$$

$$= \iint_D \left(\sqrt{8-x^2-y^2} - \frac{x^2+y^2}{2} \right) dx\,dy$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases} \Rightarrow r^2 \le 4 \Leftrightarrow r \in [0,2], \quad t \in [0, 2\pi]$$

Not $8-r^2 = x \Rightarrow -2r\,dr = dx$

x	0	4
r	$2\sqrt{2}$	0

$$= \int_0^{2\pi}\int_0^2 \left(\sqrt{8-r^2} - \frac{r^2}{2} \right) \cdot r\,dr\,dt = \int_0^{2\pi} \left(\int_0^2 \sqrt{8-r^2}\cdot r\,dr - \int_0^2 \frac{r^3}{2}\,dr \right) dt$$

$$= \int_0^{2\pi} \left(\frac{1}{2}\int_4^8 \sqrt{x}\,dx - \frac{r^4}{8}\Big|_0^2 \right) dt = \int_0^{2\pi} \left(\frac{1}{2}\cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\Big|_4^8 - 2 \right) dt =$$

$$= \left(\frac{\sqrt{8^3} - \sqrt{4^3}}{3} - 2 \right) 2\pi = \left(\frac{16\sqrt{2}-8}{3} - 2 \right) 2\pi = \frac{2\pi}{3}(16\sqrt{2}-8+12) = \frac{2\pi}{3}(16\sqrt{2}-14)$$

$$= \frac{4\pi}{3}(8\sqrt{2}-7)$$

b) $z = 4 - x^2 - y^2$ (S1) şi $2z = 5 + x^2 + y^2$ (S2); $\Rightarrow z > 0$

$S_1 \cap xOy$ ($z=0$) $\Rightarrow x^2+y^2=4$ cerc

$S_1 \cap xOz$ ($y=0$) $\Rightarrow z = 4-x^2$ parabola

$S_1 \cap yOz$ ($x=0$) $\Rightarrow z = 4-y^2$ parabola

paraboloid

$S_1 \cap Ox \Rightarrow y = z = 0 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$

$S_1 \cap Oy \Rightarrow x = z = 0 \Rightarrow y^2 = 4 \Rightarrow y = \pm 2$

$S_1 \cap Oz \Rightarrow x = y = 0 \Rightarrow z = 4 \Rightarrow (0,0,4)$ vf

$S_2 \cap xOy$ ($z=0$) $\Rightarrow x^2+y^2 = -5$ fals $\Rightarrow S_2 \cap xOy = \emptyset$

$S_2 \cap xOz$ ($y=0$) $\Rightarrow 2z = 5 + x^2$ parabola

$S_2 \cap yOz$ ($x=0$) $\Rightarrow 2z = 5 + y^2$ parabola

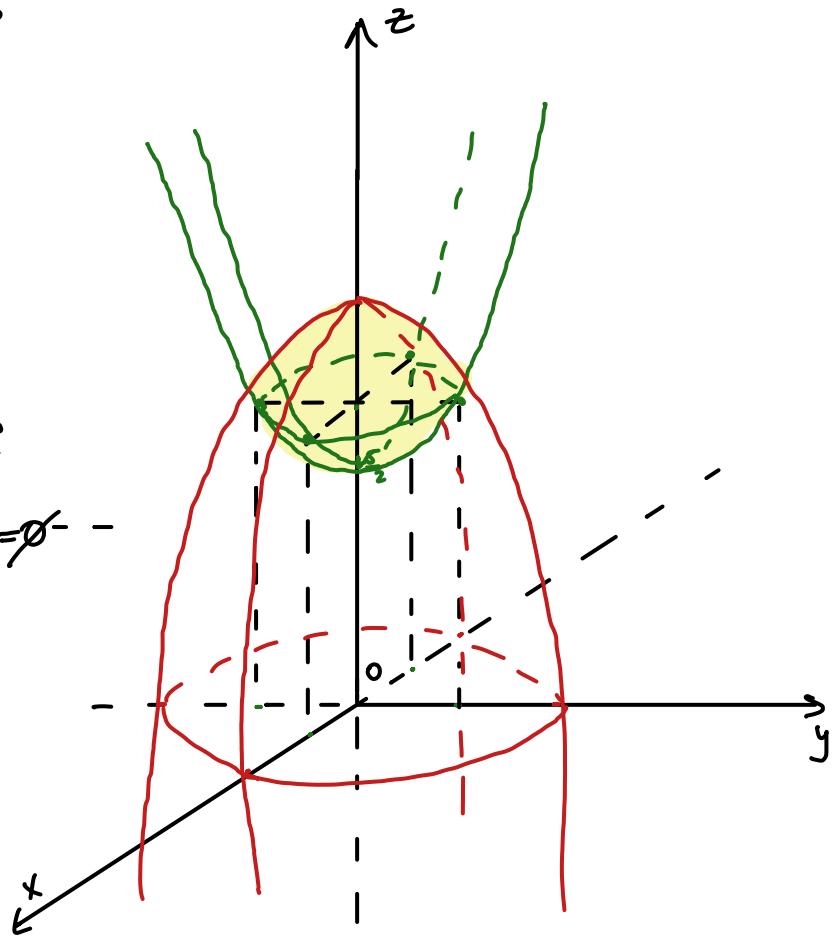
$(\pm 1, 0, 3)$

$(0, \pm 1, 3)$

$S_2 \cap Ox$ ($y=z=0$) $\Rightarrow x^2 = -5$ ok

$S_2 \cap Oy$ ($x=z=0$) $\Rightarrow y^2 = -5$ ok

$S_2 \cap Oz$ ($x=y=0$) $\Rightarrow z = \frac{5}{2} \Rightarrow (0,0,\frac{5}{2})$ vf



$$\begin{cases} z = 4-x^2-y^2 \\ 2z = 5+x^2+y^2 \end{cases} \Leftrightarrow \begin{cases} x^2+y^2 = 4-z \\ x^2+y^2 = 2z-5 \end{cases} \quad | \Rightarrow 4-z = 2z-5 \Rightarrow z = 3$$

$$\Rightarrow x^2+y^2 = 1 \Rightarrow D: x^2+y^2 \le 1$$

$$\Rightarrow Vol(K) = \iiint_K 1\,dxdydz = \iint_D \left(\int_{\frac{5+x^2+y^2}{2}}^{4-x^2-y^2} dz \right) dxdy = \iint_D \left(4-x^2-y^2-\frac{5+x^2+y^2}{2}\right) dxdy$$

$$= \iint_D \frac{8-2x^2-2y^2-5-x^2-y^2}{2}\,dxdy = \iint_D \frac{3-3(x^2+y^2)}{2}\,dxdy = \frac{3}{2}\iint_D (1-(x^2+y^2))\,dxdy$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ |J| = r \end{cases} \Rightarrow r^2 \le 1 \Rightarrow r \in [0,1], \quad t \in [0, 2\pi]$$

$$= \frac{3}{2}\int_0^{2\pi}\int_0^1 (1-r^2) r\,dr\,dt = \frac{3}{2}\int_0^{2\pi}\int_0^1 (r-r^3)\,dr\,dt = \frac{3}{2}\int_0^{2\pi} \left(\frac{r^2}{2}\Big|_0^1 - \frac{r^4}{4}\Big|_0^1 \right) dt$$

$$= \frac{3}{2}\cdot\left(\frac{2}{4}-\frac{1}{4}\right)\cdot 2\pi = \frac{3\pi}{4}$$

c) $2x^2 + y^2 + z^2 = 1$ şi $2x^2 + y^2 = z^2$, cu $z \ge 0$;

$S_1 \cap xOy \overset{z=0}{\Rightarrow} 2x^2+y^2=1$ elipsă

$S_1 \cap xOz \overset{y=0}{\Rightarrow} 2x^2+z^2=1$ tot elipsă

$S_1 \cap yOz \overset{x=0}{\Rightarrow} y^2+z^2=1$ cerc

$S_1 \cap Ox \overset{y=z=0}{\Rightarrow} 2x^2=1 \Leftrightarrow x = \pm\frac{1}{\sqrt{2}}$

$S_1 \cap Oy \overset{x=z=0}{\Rightarrow} y^2=1 \Leftrightarrow y = \pm 1$

$S_1 \cap Oz \overset{x=y=0}{\Rightarrow} z = \pm 1$

$z=k \Rightarrow 2x^2+y^2+k^2=1$ elipsă

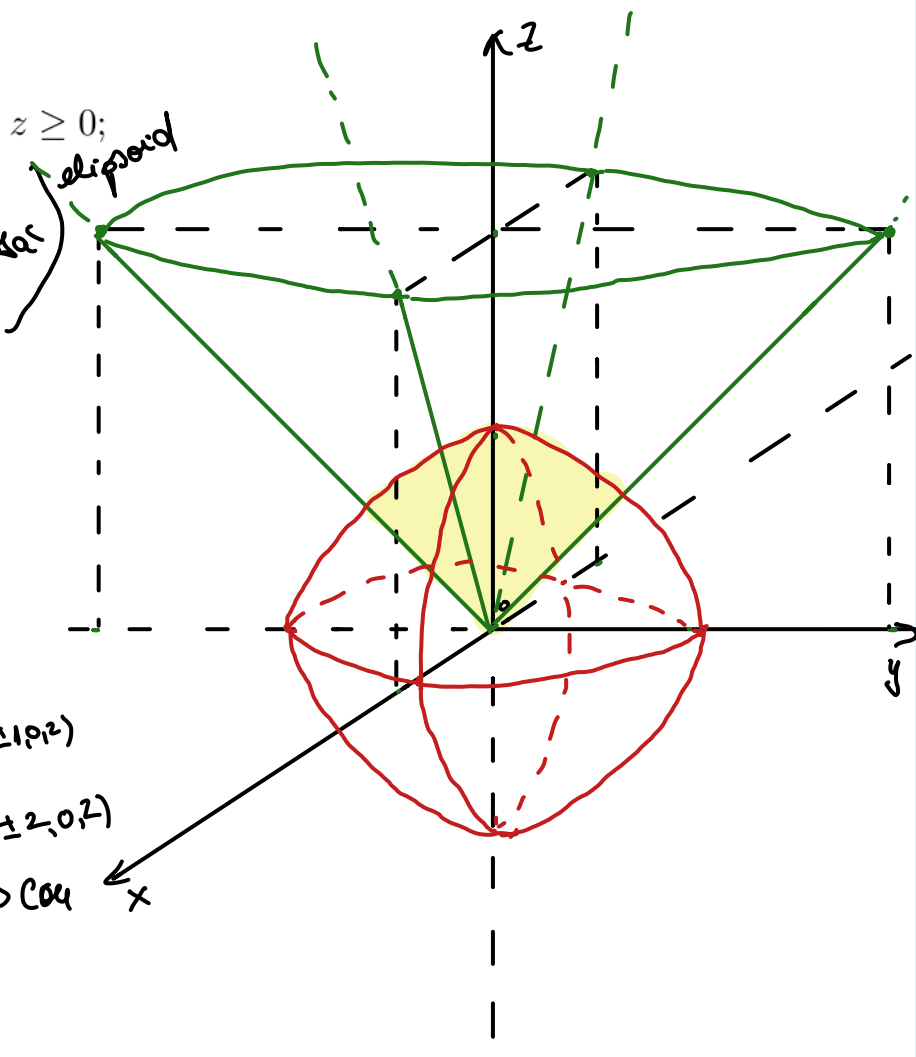
$S_2 \cap xOy \overset{z=0}{\Rightarrow} 2x^2+y^2=0$ punct (0,0,0) vârful

$S_2 \cap xOz \overset{y=0}{\Rightarrow} 2x^2=z^2 \Rightarrow z=\sqrt{2}|x| \Rightarrow$ 2 drepte $(\pm 1,0,\sqrt{2})$

$S_2 \cap yOz \overset{x=0}{\Rightarrow} y^2=z^2 \Rightarrow z=|y|$ 2 drepte $(\pm 2,0,2)$

$S_2 \cap Ox = S_2 \cap Oy = S_2 \cap Oz = \{O(0,0,0)\}$ $\Rightarrow$ con

$z=k \Rightarrow 2x^2+y^2=k^2$ elipsă



$$\begin{cases} 2x^2+y^2+z^2=1 \\ 2x^2+y^2=z^2 \end{cases} \Rightarrow 2x^2+y^2=1-z^2 \Rightarrow 1-z^2=z^2$$

$$\Rightarrow 2z^2=1 \Leftrightarrow z^2=\frac{1}{2} \Rightarrow z=\frac{1}{\sqrt{2}} \quad (z\geq 0)$$

$$\Rightarrow \left(\frac{x}{\frac{1}{\sqrt{2}}}\right)^2+y^2=\frac{1}{2} \rightarrow D:\left(\frac{x}{\frac{1}{\sqrt{2}}}\right)^2+y^2=\frac{1}{2} \quad (2x^2+y^2=\tfrac{1}{2})$$

$$\Rightarrow \mathrm{Vol}(K)=\iiint_K 1\,dxdydz=\iint_D\left(\int_{\sqrt{2x^2+y^2}}^{\sqrt{1-2x^2-y^2}} dz\right)dxdy=\iint_D \sqrt{1-2x^2-y^2}-\sqrt{2x^2+y^2}\,dxdy$$

$$\begin{cases} x=\frac{1}{\sqrt{2}} r\cos t \\ y=r\sin t \\ J=\frac{1}{\sqrt{2}} r \end{cases} \Rightarrow r^2\leq\frac{1}{2} \Leftrightarrow r\in\left[0,\frac{1}{\sqrt{2}}\right], \quad t\in[0,2\pi]$$

$$=\frac{1}{\sqrt{2}}\int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}}\left(\sqrt{1-r^2}-r\right)r\,drdt=\frac{1}{\sqrt{2}}\int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}} r\sqrt{1-r^2}-r^2\,dr\,dt$$

$$\begin{cases} \text{Not } 1-r^2=x \Leftrightarrow -2r\,dr=dx \\ \begin{array}{c|ccc} r & 0 & & \frac{1}{\sqrt{2}} \\ \hline x & 1 & & \frac{1}{2} \end{array} \end{cases}$$

$$=\frac{1}{\sqrt{2}}\int_0^{2\pi}\left(\frac{1}{2}\int_{\frac{1}{2}}^{1}\sqrt{x}\,dx-\frac{r^3}{3}\Big|_0^{\frac{1}{\sqrt{2}}}\right)dt$$

$$=\frac{1}{\sqrt{2}}\int_0^{2\pi}\left(\frac{1}{2}\cdot\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\Big|_{\frac{1}{2}}^{1}-\frac{1}{6\sqrt{2}}\right)dt$$

$$=\frac{1}{\sqrt{2}}\int_0^{2\pi}\left(\frac{1-\frac{1}{2\sqrt{2}}}{3}-\frac{1}{6\sqrt{2}}\right)dt=\frac{1}{\sqrt{2}}\cdot 2\pi\cdot\frac{2\sqrt{2}-2}{6\sqrt{2}}=\frac{(\sqrt{2}-1)\pi}{3}$$

d) $x^2+y^2+z^2=4a^2$ şi $x^2+y^2=2ax$, cu $a>0$;

S_1 e o sferă (obviously)

$S_1\cap Ox \overset{y=z=0}{\Rightarrow} x=\pm 2a$

$S_1\cap Oy \overset{x=z=0}{\Rightarrow} y=\pm 2a$

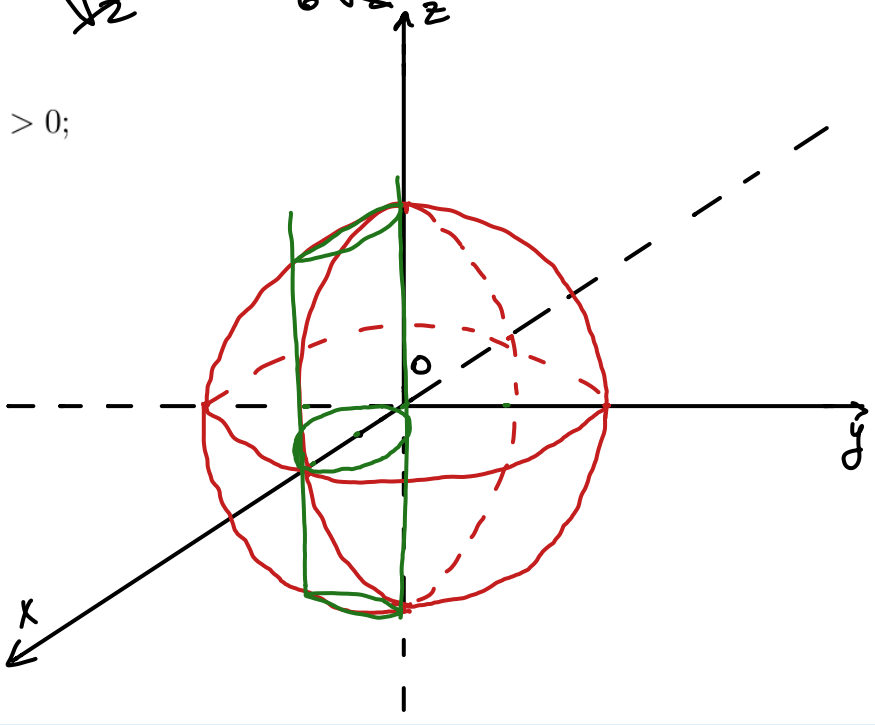
$S_1\cap Oz \overset{x=y=0}{\Rightarrow} z=\pm 2a$

$S_2: x^2+y^2=2ax \Leftrightarrow x^2-2ax+a^2+y^2=a^2$

$\Leftrightarrow (x-a)^2+y^2=a^2$ un cilindru cu cerc de centru $(a,0)$

$S_2\cap Ox \overset{y=0}{\Rightarrow} (x-a)^2=a^2 \Rightarrow x-a=\pm a \Rightarrow x\in\{0,2a\}$

$S_2\cap Oy \overset{x=0}{\Rightarrow} y^2=0 \rightarrow y=0$



$\Rightarrow D: (x-a)^2+y^2 \le a^2 \Rightarrow Vol(K) = \iint_D \left(\int_0^{\sqrt{4a^2-x^2-y^2}} dz \right) dx\,dy$

... se calc integrala

e) $\frac{x^2}{4}+\frac{y^2}{9}=2z$ și $\frac{x^2}{4}+\frac{y^2}{9}=z^2$;

S_1 S_2

$S_1 \cap xOy \overset{z=0}{\Rightarrow}$ pct $O(0,0,0)$

$S_1 \cap xOz \overset{y=0}{\Rightarrow} x^2=8z$ parabolă $\rightarrow$ paraboloid $(\pm 2, 0, \frac{1}{2})$

$S_1 \cap yOz \overset{x=0}{\Rightarrow} y^2=18z$ parabolă $\rightarrow (0, \pm 3, \frac{1}{2})$

$z=k \Rightarrow \frac{x^2}{4}+\frac{y^2}{9}=2k$ elipsă

$S_2 \cap xOy \overset{z=0}{\Rightarrow} O(0,0,0)$

$S_2 \cap xOz \overset{y=0}{\Rightarrow} x^2=4z^2 \Rightarrow x=\pm 2z$ dude

$S_2 \cap yOz \overset{x=0}{\Rightarrow} y^2=9z^2 \Rightarrow y=\pm 3z$

$z=k \Rightarrow \frac{x^2}{4}+\frac{y^2}{9}=k^2$ elipsă

con eliptic ... se face desenul

$z^2=2z \Rightarrow z(z-2)=0 \Rightarrow z=2$ (dacă zau fi oau fi efectiv pct $(0,0,0)$)

$\Rightarrow \frac{x^2}{4}+\frac{y^2}{9}=4 \Rightarrow D: \left(\frac{x}{2}\right)^2+\left(\frac{y}{3}\right)^2 \le 4$

$\begin{cases} x = 2r\cos t \\ y = 3r\sin t \\ j = 6r \end{cases} \Rightarrow r^2 \le 4 \Rightarrow r \in [0,2]$, $t \in [0, 2\pi]$

trec să fie invers capetele

$\Rightarrow Vol(K) = \iiint_K dx\,dy\,dz = -\iint_D \left(\int_{\sqrt{\frac{x^2}{4}+\frac{y^2}{9}}}^{\frac{x^2}{4}+\frac{y^2}{9}} dz \right) dx\,dy = -\iint_D \left(\frac{\frac{x^2}{4}+\frac{y^2}{9}}{2} - \sqrt{\frac{x^2}{4}+\frac{y^2}{9}} \right) dx\,dy$

$= -\int_0^{2\pi}\int_0^2 \left(\frac{r^2}{2}-r\right) \cdot 6r\,dr\,dt = 6\int_0^{2\pi}\int_0^2 \left(\frac{r^3}{2}-r^2\right) dr\,dt = -6\int_0^{2\pi} \left(\frac{r^4}{8}-\frac{r^3}{3}\right)\Big|_0^2 dt$

$= 6 \cdot 2\pi \cdot \left(\frac{8}{3}-\frac{3}{2}\right) = 8\pi$

f) $x^2 + y^2 = 1$ şi $z^2 = x^2 + y^2$, cu $z \geq 0$.

S_1 = cilindru S_2

$S_2 \cap xOy \Rightarrow$ pct $O(0,0,0)$

$S_2 \cap xOz \Rightarrow x^2 = z^2 \Rightarrow z = |x| = \pm x$ (2 drepte)

$S_2 \cap yOz \Rightarrow y^2 = z^2$ (2 drepte) $\Rightarrow$ con

$z = k \Rightarrow x^2 + y^2 = k^2$ cerc

$$\begin{cases} x^2+y^2=1 \\ x^2+y^2=z^2 \end{cases} \Rightarrow z^2 = 1, \; z \geq 0 \Rightarrow z = 1$$

$D: x^2 + y^2 \leq 1$

$$\Rightarrow Vol(K) = \iiint_K dx\,dy\,dz = \iint_D \left(\int_1^{\sqrt{x^2+y^2}} dz \right) dx\,dy = \iint_D \sqrt{x^2+y^2}\, dx\,dy =$$

$$= \int_0^{2\pi} \int_0^1 r^2 \, dr\,dt = 2\pi \cdot \frac{r^3}{3}\Big|_0^1 = \frac{2\pi}{3}$$

15. Să se calculeze volumul domeniului mărginit de suprafaţa $(x^2 + y^2 + z^2)^2 = a^2(x^2 + y^2)$, cu $a > 0$.

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \\ J = r^2\sin\theta \end{cases} \Rightarrow r^4 = a^2 r^2 \sin^2\theta \;|:r^2>0 \Leftrightarrow r^2 = a^2\sin^2\theta \Leftrightarrow r = a\sin\theta$$

$\varphi \in [0, 2\pi] \Rightarrow r \in [0, a\sin\theta]$ cu $\theta \in [0, \frac{\pi}{2}]$

$$\Rightarrow Vol(K) = \iiint_K dx\,dy\,dz = \int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^{a\sin\theta} r^2\sin\theta \, dr\,d\theta\,dt = \int_0^{2\pi}\int_0^{\frac{\pi}{2}} \sin\theta \cdot \frac{r^3}{3}\Big|_0^{a\sin\theta} d\theta\,d\varphi$$

$$= \frac{1}{3}\int_0^{2\pi}\int_0^{\frac{\pi}{2}} \sin\theta \cdot a^3 \cdot \sin^3\theta \, d\theta\,d\varphi = \frac{a^3}{3}\int_0^{2\pi}\int_0^{\frac{\pi}{2}} \sin^4\theta \, d\theta\,d\varphi = \frac{a^3}{3}\int_0^{2\pi}\int_0^{\frac{\pi}{2}} \left(\frac{1-\cos 2\theta}{2}\right)^2 d\theta\,d\varphi$$

$$= \frac{a^3}{12}\int_0^{2\pi}\int_0^{\frac{\pi}{2}} (1 - 2\cos 2\theta + \cos^2 2\theta)\, d\theta\,d\varphi = \frac{a^3}{12}\int_0^{2\pi}\left(\left(t - \underbrace{\sin 2t}_{0}\right)\Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \frac{1+\cos 4\theta}{2}\, d\theta \right) d\varphi$$

$$= \frac{a^3}{12}\int_0^{2\pi}\left(\frac{\pi}{2} + \frac{1}{2}\int_0^{\pi/2} (1+\cos 4\theta)\, d\theta \right) d\varphi = \frac{a^3}{12}\int_0^{2\pi}\left(\frac{\pi}{2} + \frac{1}{2}\left(t + \underbrace{\frac{\sin 4\theta}{4}}_{0}\right)\Big|_0^{\frac{\pi}{2}} \right) d\varphi$$

$$= \frac{a^3}{24} \cdot 2\pi \cdot \frac{2\pi}{4} = \frac{\pi^2 a^3}{8}$$

16. Să se calculeze aria paraboloidului $z = x^2 + y^2$, cu $z \in [0, 9]$.

$S: z = x^2 + y^2$

$\Rightarrow \begin{cases} x = x \\ y = y \\ z = x^2 + y^2 \end{cases}$

$A(S) = \iint_D \| \vec{T}_x \times \vec{T}_y \| \, dx \, dy$

$\vec{T}_x = 1\vec{i} + 0\vec{j} + 2x\vec{k}$

$\vec{T}_y = 0\vec{i} + 1\vec{j} + 2y\vec{k}$

$\Rightarrow \vec{T}_x \times \vec{T}_y = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2x \\ 0 & 1 & 2y \end{vmatrix} =$

$= \vec{k} - 2x\vec{i} - 2y\vec{j}$

$\Rightarrow \| \vec{T}_x \times \vec{T}_y \| = \sqrt{(-2x)^2 + (-2y)^2 + 1^2} = \sqrt{4x^2 + 4y^2 + 1}$

$\Rightarrow A = \iint_D \sqrt{4x^2 + 4y^2 + 1} \, dx \, dy$

$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases}$

$\Rightarrow r^2 \le 9 \Rightarrow r \in [0, 3]$

$t \in [0, 2\pi]$

$J = r$

$z \in [0, 9] \Rightarrow x^2 + y^2 \le 9$

$\Rightarrow A = \int_0^{2\pi} \int_0^3 \sqrt{4r^2 + 1} \cdot r \, dr \, dt$

Not $4r^2 + 1 = x \Leftrightarrow 8r \, dr = dx$

r	0	3
x	1	37

$= \frac{1}{8} \int_0^{2\pi} \int_1^{37} \sqrt{x} \, dx \, dt = \frac{\pi}{2\pi} \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \Big|_1^{37} = \frac{\pi}{6} \left(37^{\frac{3}{2}} - 1 \right)$

17. Să se calculeze aria suprafeţei $x^2 + y^2 + z^2 = a^2$, situată în interiorul cilindrului $x^2 + y^2 = ax$, cu $a > 0$.

S_1 obviously sferă, centru $O(0,0,0)$

$x = y = z = \pm a$

$x^2+y^2=ax \Leftrightarrow x^2-ax+y^2=0 \mid +\frac{1}{4}$

$\Leftrightarrow \left(x-\frac{1}{2}\right)^2+y^2=\left(\frac{1}{2}\right)^2$

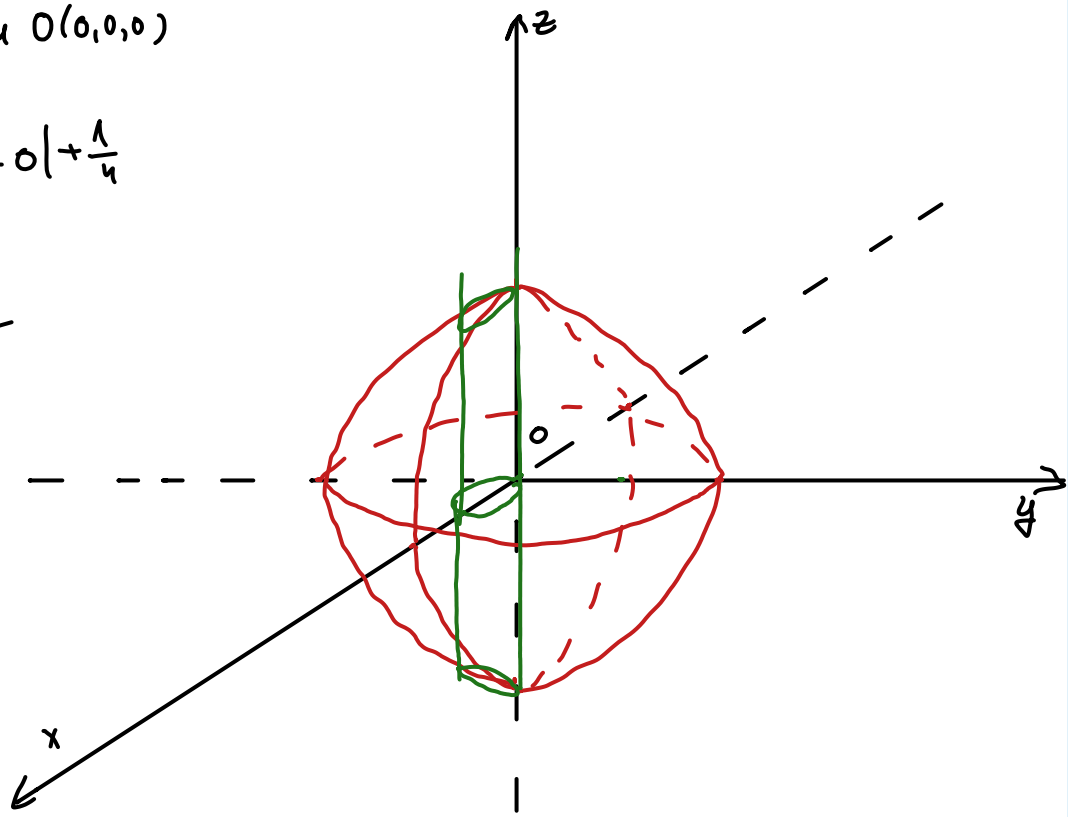
$S_1 \cap Ox \overset{y=z=0}{\Rightarrow} \left(x-\frac{1}{2}\right)^2=\left(\frac{1}{2}\right)^2$

$\Rightarrow x-\frac{1}{2}=\pm\frac{1}{2}$

$\Rightarrow x \in \{0,1\}$

$S_2 \cap Oy \Rightarrow y=0$

$S_2 \cap Oz$?



$$\begin{cases} x = x \\ y = y \\ z = \sqrt{a^2-x^2-y^2} \end{cases} \qquad D: x^2+y^2 \le ax$$

$$A = \iint_D \|\vec{T_u} \times \vec{T_v}\| \, du\,dv = \iint_D \|\vec{T_x} \times \vec{T_y}\| \, dx\,dy$$

$$\vec{T_x} = \vec{i} + \vec{k} \cdot \frac{-2x}{2\sqrt{a^2-x^2-y^2}}$$

$$\vec{T_y} = \vec{j} + \vec{k} \cdot \frac{-2y}{2\sqrt{a^2-x^2-y^2}}$$

$$\Rightarrow \vec{T_x} \times \vec{T_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{-x}{\sqrt{a^2-x^2-y^2}} \\ 0 & 1 & \frac{-y}{\sqrt{a^2-x^2-y^2}} \end{vmatrix}$$

$$= \vec{k} + \vec{i}\frac{x}{\sqrt{a^2-x^2-y^2}} + \vec{j}\frac{y}{\sqrt{a^2-x^2-y^2}}$$

$$\Rightarrow \|\vec{T_u} \times \vec{T_v}\| = \sqrt{1^2+\frac{x^2}{a^2-x^2-y^2}+\frac{y^2}{a^2-x^2-y^2}} = \sqrt{\frac{a^2-x^2-y^2+x^2+y^2}{a^2-x^2-y^2}} =$$

$$= \frac{a}{\sqrt{a^2-x^2-y^2}} \Rightarrow A = \iint_D \frac{a}{\sqrt{a^2-x^2-y^2}} \, dx\,dy$$

$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases}$ $\Big| \Rightarrow r^2 \le a r\cos t \;|:r>0 \Leftrightarrow r \le a\cos t$

$\overset{cs}{\Rightarrow} \cos t > 0 \Rightarrow t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$r \in [0, a\cos t]$

$$\Rightarrow A = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_{0}^{a\cos t} \frac{a}{\sqrt{a^2 - r^2}} \cdot r \, dr \, dt$$

Not $a^2 - r^2 = x \Leftrightarrow -2r\,dr = dx$ $\quad \begin{array}{c|ccc} r & 0 & & a\cos t \\ \hline x & a^2 & & a^2 - a^2\cos^2 t = a^2\sin^2 t \end{array}$

$$= \frac{a}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_{a^2\sin^2 t}^{a^2} \frac{1}{\sqrt{x}} \, dx \, dt = \frac{a}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^{\frac{1}{2}}}{\frac{1}{2}} \Big|_{a^2\sin^2 t}^{a^2} dt = a \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (a - a|\sin t|) \, dt$$

$$= a^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 - |\sin t|) \, dt = a^2 \left(t \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \int_{-\frac{\pi}{2}}^{0} -\sin t \, dt - \int_{0}^{\frac{\pi}{2}} \sin t \, dt \right)$$

$$= a^2 \left(\pi - \cos t \Big|_{-\frac{\pi}{2}}^{0} + \cos t \Big|_{0}^{\frac{\pi}{2}} \right) = a^2 \left(\pi - 1 - 0 + (0 - 1) \right) = a^2(\pi - 2)$$

$(\because t \in [-\frac{\pi}{2}, \frac{\pi}{2}] \Rightarrow \cos t > 0)$

$\Rightarrow \times 2$ (S_1)

18. Să se calculeze aria suprafeţei $z = \sqrt{x^2 + y^2}$, situată în interiorul cilindrului $x^2 + y^2 = 2x$. (S_2)

$z = \sqrt{x^2+y^2} \Rightarrow \begin{cases} z^2 = x^2 + y^2 \\ z \ge 0 \end{cases}$

$S_1 \cap xOy \Rightarrow O(0,0,0)$ pct

$S_1 \cap xOz \Rightarrow x^2 = z^2$

$\Rightarrow z = |x| = \pm x$

2 drepte

Analog $z = \pm y$

$z = k \Rightarrow x^2 + y^2 = k^2$ cercuri $\Big| \Rightarrow$ con

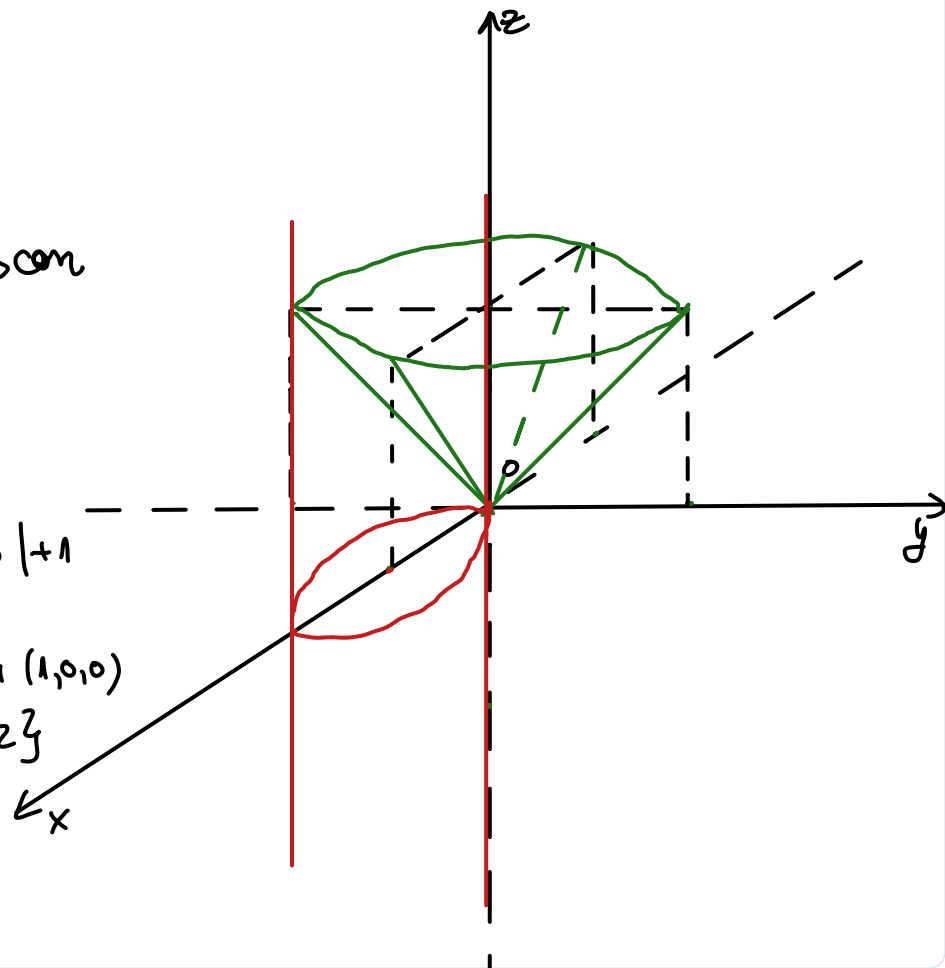
$S_2: x^2 + y^2 = 2x \Leftrightarrow x^2 - 2x + y^2 = 0 \;|+1$

$\Leftrightarrow (x-1)^2 + y^2 = 1$

$S_1 \cap xOy \Rightarrow$ cerc cu centrul în $(1,0,0)$

$S_1 \cap xOz \Rightarrow x - 1 = \pm 1 \Rightarrow x \in \{0, 2\}$

$S_1 \cap yOz \Rightarrow y = 0$



$S: \begin{cases} x = x \\ y = y \\ z = \sqrt{x^2+y^2} \end{cases}$ $\quad D: x^2+y^2 \le 2x$

$$A = \iint_D \|\overrightarrow{T_u} \times \overrightarrow{T_v}\| \, du\, dv$$

$\overrightarrow{T_x} = 1\cdot\vec{i} + 0\cdot\vec{j} + \frac{x}{\sqrt{x^2+y^2}}$; $\overrightarrow{T_y} = 0\vec{i} + 1\vec{j} + \frac{y}{\sqrt{x^2+y^2}}$

$$\overrightarrow{Tx} \times \overrightarrow{Ty} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{x}{\sqrt{x^2+y^2}} \\ 0 & 1 & \frac{y}{\sqrt{x^2+y^2}} \end{vmatrix} = \vec{k} - \frac{x}{\sqrt{x^2+y^2}}\vec{i} - \frac{y}{\sqrt{x^2+y^2}}\vec{j}$$

$$\Rightarrow \|\overrightarrow{Tx} \times \overrightarrow{Ty}\| = \sqrt{1^2 + \frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2}} = \sqrt{2}$$

$$\Rightarrow A = \iint_D \sqrt{2}\, dx\, dy$$

$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases}$ $\quad \Rightarrow r^2 \le 2r\cos t \;|: r>0 \Leftrightarrow r \le 2\cos t \Rightarrow r \in [0, 2\cos t]$

$\overset{\in}{\Rightarrow} t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$j = r$

$$\Rightarrow A = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\cos t} \sqrt{2}\, r\, dr\, dt = \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{r^2}{2}\Big|_0^{2\cos t} dt = \frac{\sqrt{2}}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4\cos^2 t\, dt$$

$$= 2\sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1+\cos 2t}{2} dt = 2\sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1+\cos 2t)\, dt = \sqrt{2}\left(t + \frac{\sin 2t}{2}\right)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \sqrt{2}\pi$$

($\sin 2t$ evaluat la 0)

19. Să se calculeze aria decupată de $(x^2+y^2)^2 = a^2(x^2-y^2)$ pe suprafaţa $x^2+y^2 = 2az$, cu $a > 0$.

S_2: paraboloid

$\begin{cases} x = x \\ y = y \\ z = \frac{x^2+y^2}{2a} \end{cases}$ $\quad \Rightarrow A = \iint_D \|\overrightarrow{Tx} \times \overrightarrow{Ty}\| \, dx\, dy$

$$\vec{T_x} = \vec{i} + \frac{2x}{2a}\vec{k} \qquad \vec{T_y} = \vec{j} + \frac{2y}{2a}\vec{k}$$

$$\Rightarrow \vec{T_x} \times \vec{T_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{x}{a} \\ 0 & 1 & \frac{y}{a} \end{vmatrix} = \vec{k} - \frac{x}{a}\vec{i} - \frac{y}{a}\vec{j}$$

$$\Rightarrow \|\vec{T_x} \times \vec{T_y}\| = \sqrt{1^2 + \frac{x^2}{a^2} + \frac{y^2}{a^2}} = \sqrt{\frac{a^2+x^2+y^2}{a^2}} = \frac{\sqrt{a^2+x^2+y^2}}{a}$$

$$\Rightarrow A = \iint_D \frac{\sqrt{a^2+x^2+y^2}}{a}\,dx\,dy \quad \text{dar} \begin{cases} (x^2+y^2)^2 \le a^2(x^2-y^2) \\ x, y \ge 0 \end{cases} \Rightarrow A(S) = 2A$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases} \Rightarrow r^4 \le a^2 r^2(\cos^2 t - \sin^2 t) \;|:r^2>0$$

$$\Leftrightarrow r^2 \le a^2\cos 2t \Leftrightarrow r \le a\sqrt{\cos 2t} \Rightarrow r \in [0, a\sqrt{\cos 2t}]$$

$$\overset{CE}{\Rightarrow} a\sqrt{\cos 2t} \ge 0 \Rightarrow \cos 2t \ge 0 \Rightarrow t \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$$

$$\Rightarrow A = 2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \int_0^{a\sqrt{\cos 2t}} \frac{\sqrt{a^2+r^2}}{a}\, r\, dr\, dt$$

Not $r^2 + a^2 = x \Leftrightarrow 2r\,dr = dx$

r	0	$a\sqrt{\cos 2t}$
x	a^2	$a^2 + a^2\cos 2t$

$$= \frac{1}{a}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \int_{a^2}^{a^2+a^2\cos 2t} \sqrt{x}\,dx\,dt = \frac{1}{a}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\Bigg|_{a^2}^{a^2+a^2\cos 2t} dt =$$

$$= \frac{2}{3a}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\left(a^2 + a^2\cos 2t\right)^{\frac{3}{2}} - a^3\right) dt = \frac{2}{3a}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\left(a^2(1+\cos 2t)\right)^{\frac{3}{2}} - a^3\right) dt$$

$$= \frac{2a^3}{3a}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\left(1+\cos 2t\right)^{\frac{3}{2}} - 1\right) dt = \frac{2a^2}{3}\left(2^{\frac{3}{2}}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\frac{1+\cos 2t}{2}\right)^{\frac{3}{2}} dt - \frac{\pi}{2}\right)$$

$$= \frac{2a^2}{3}\left(2^{\frac{3}{2}}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\cos^2 t\right)^{\frac{3}{2}} dt - \frac{\pi}{2}\right) = \frac{2a^2}{3}\left(2^{\frac{3}{2}}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} |\cos t|^3\, dt - \frac{\pi}{2}\right)$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} |\cos t|^3 dt = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\cos t)^3 dt = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos t \cdot (1-\sin^2 t)\, dt$$

Not $\sin t = x \Leftrightarrow \cos t\, dt = dx$

t	$-\frac{\pi}{2}$	$\frac{\pi}{2}$
x	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$

$$= \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} (1-x^2)\, dx = \left(x - \frac{x^3}{3}\right)\Big|_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - \frac{8\sqrt{2}}{24} - \frac{\sqrt{2}}{12}\right)$$

$$= \left(\sqrt{2} - \frac{\sqrt{2}}{6}\right) = \frac{5\sqrt{2}}{6}$$

$$A = \frac{2a^2}{3}\left(2\sqrt{2}\cdot\frac{5\sqrt{2}}{6} - \frac{\pi}{2}\right) = \frac{a^2}{3}\cdot\frac{2(20-3\pi)}{6} = \frac{a^2}{9}(20-3\pi)$$

20. Să se calculeze aria decupată de $(x^2+y^2)^2 = a^2(x^2-y^2)$ pe sfera $x^2+y^2+z^2=a^2$.

$$\begin{cases} x = x \\ y = y \\ z = \sqrt{a^2-x^2-y^2} \end{cases}$$

$$\overrightarrow{Tx} = \vec{i} + \vec{k}\cdot\frac{-2x}{2\sqrt{a^2-x^2-y^2}}$$

$$\overrightarrow{Ty} = \vec{j} + \vec{k}\cdot\frac{-2y}{2\sqrt{a^2-x^2-y^2}}$$

$$\Rightarrow \overrightarrow{Tx}\times\overrightarrow{Ty} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{-x}{\sqrt{a^2-x^2-y^2}} \\ 0 & 1 & \frac{-y}{\sqrt{a^2-x^2-y^2}} \end{vmatrix}$$

$$= \vec{k} + \vec{i}\frac{x}{\sqrt{a^2-x^2-y^2}} + \vec{j}\frac{y}{\sqrt{a^2-x^2-y^2}}$$

$$\Rightarrow \|\overrightarrow{Tu}\times\overrightarrow{Tv}\| = \sqrt{1^2 + \frac{x^2}{a^2-x^2-y^2} + \frac{y^2}{a^2-x^2-y^2}} = \sqrt{\frac{a^2-x^2-y^2+x^2+y^2}{a^2-x^2-y^2}} =$$

$$= \frac{a}{\sqrt{a^2-x^2-y^2}} \Rightarrow A = \iint_D \frac{a}{\sqrt{a^2-x^2-y^2}}\, dx\, dy$$

$$\begin{cases} (x^2+y^2)^2 \le a^2(x^2-y^2) \\ x, y \ge 0 \end{cases} \xrightarrow{A(S)=4A}$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ J = r \end{cases}$$

$$\Rightarrow r^4 \le a^2 r^2 \cos 2t \Rightarrow r \le a\sqrt{\cos 2t}$$

$$\Rightarrow \cos 2t \ge 0 \Rightarrow t \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$$

$$\Rightarrow A = 4\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\int_{0}^{a\sqrt{\cos 2t}} \frac{a}{\sqrt{a^2-r^2}}\cdot r\,dr\,dt$$

Not $a^2-r^2 = x \Leftrightarrow -2r\,dr = dx \Leftrightarrow$

r	0	$a\sqrt{\cos 2t}$
x	a^2	$a^2-a^2\cos 2t$

$$= 2a\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\int_{a^2-a^2\cos 2t}^{a^2} \frac{1}{\sqrt{x}}\,dx\,dt = 2a\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x^{\frac{1}{2}}}{\frac{1}{2}}\Big|_{\dots}^{a^2} dt = 4a\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\sqrt{a^2} - \sqrt{a^2-a^2\cos 2t}\right)dt$$

$$= 4a\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(a - a\sqrt{1-\cos 2t}\right)dt = 4a^2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(1-\sqrt{2}\,|\sin t|\right)dt$$

$$= 4a^2\left(\frac{\pi}{2} - \sqrt{2}\int_{-\frac{\pi}{4}}^{0} -\sin t\,dt - \sqrt{2}\int_{0}^{\frac{\pi}{4}} \sin t\,dt\right)$$

$$= 4a^2\left(\frac{\pi}{2} - \sqrt{2}\cdot\cos t\Big|_{-\frac{\pi}{4}}^{0} + \sqrt{2}\cos\Big|_{0}^{\frac{\pi}{4}}\right)$$

$$= 4a^2\left(\frac{\pi}{2} - \sqrt{2}\left(1-\frac{\sqrt{2}}{2}\right) + \sqrt{2}\left(\frac{\sqrt{2}}{2}-1\right)\right) = 4a^2\left(\frac{\pi}{2} - \sqrt{2}+1+1-\sqrt{2}\right)$$

$$= 4a^2\left(\frac{\pi}{2} - 2\sqrt{2}+2\right)$$

→ din rel sunt $x^2+y^2 \le$ ceva ⇒ parametrizare

21. Să se calculeze ariile suprafeţelor:

a) $S = \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 4z, z \ge x^2+y^2\}$;

$S_1: x^2+y^2+z^2 = 4z \Leftrightarrow x^2+y^2+(z-2)^2 = 4$ sfera

S_2: deasupra unui con

$$\begin{cases} x = x \\ y = y \\ z = 2+\sqrt{4-x^2-y^2} \end{cases}$$

$$\overrightarrow{T_x} = \vec{i} + \frac{\vec{k}\cdot(-2x)}{2\sqrt{4-x^2-y^2}}; \quad \overrightarrow{T_y} = \vec{j} + \vec{k}\,\frac{-y}{\sqrt{4-x^2-y^2}}$$

$$\Rightarrow \overrightarrow{T_x} \times \overrightarrow{T_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{-x}{\sqrt{4-x^2-y^2}} \\ 0 & 1 & \end{vmatrix}$$

Flux

22. Să se calculeze fluxul câmpului vectorial $\mathbf{V}(x,y,z) = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$ prin faţa exterioară a conului $z^2 = x^2 + y^2$, cu $z \in [0,1]$.

$S : z^2 = x^2 + y^2$

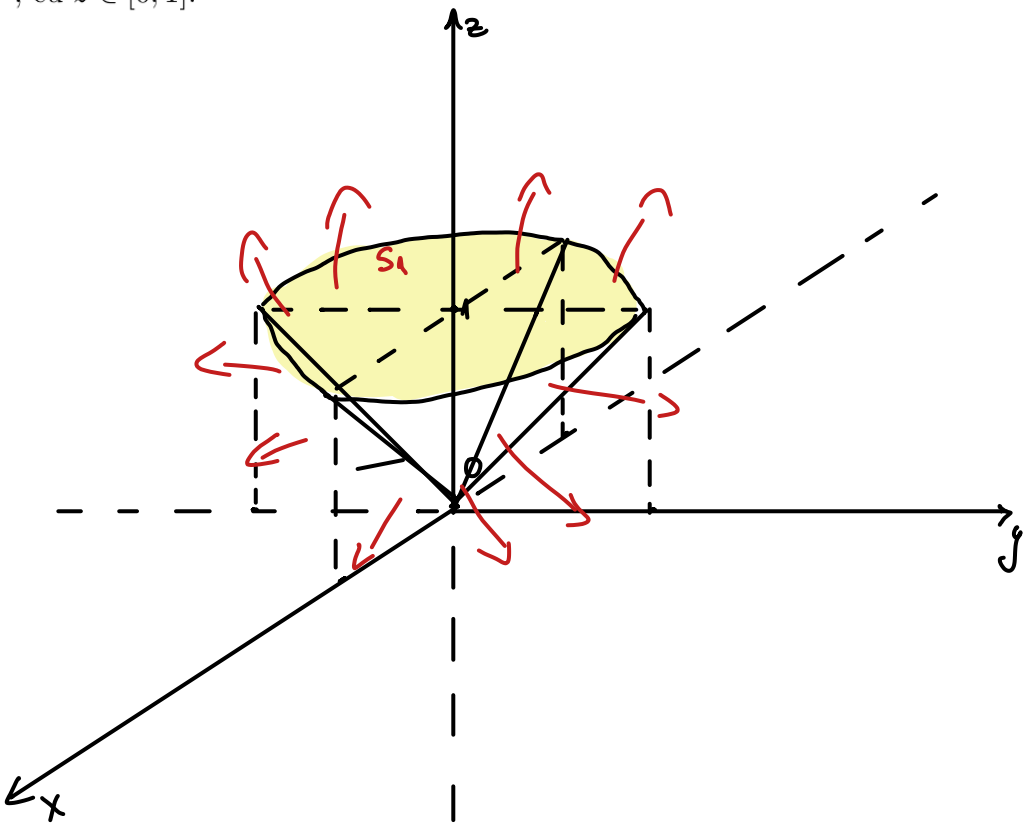
$S \cap xOy = \{O(0,0,0)\}$

$S \cap xOz \Rightarrow z = \pm x$

$S \cap yOz \Rightarrow z = \pm y$

$\Sigma = S \cup S_1$

$\mathcal{F}_\Sigma(\vec{V}) = \iint_\Sigma \vec{V} \cdot \vec{n} \cdot d\sigma$



G-O $= \iiint_K \left(\frac{dP}{dx} + \frac{dQ}{dy} + \frac{dR}{dz}\right) dx\,dy\,dz = \iiint_K 3\,dx\,dy\,dz$

$x^2 + y^2 = z^2 \\ z \in [0,1]$ $\Big| \Rightarrow x^2 + y^2 \le 1 \quad ; z = \sqrt{x^2+y^2}$

$= \iint_D \left(\int_0^{\sqrt{x^2+y^2}} 3\,dz\right) dx\,dy = 3 \iint_D \sqrt{x^2+y^2}\,dx\,dy$

$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases}$ $\Big| \Rightarrow r^2 \le 1 \Rightarrow r \in [0,1]$, $t \in [0, 2\pi]$

$= 3\int_0^{2\pi}\int_0^1 r^2\,dr\,dt = 3\int_0^{2\pi} \frac{r^3}{3}\Big|_0^1 dt = \int_0^{2\pi} dt = 2\pi$

$$F_{S_1}(\vec{V}) = \iint_{S_1} \vec{V}\vec{n}\, d\sigma = \iint_D \vec{V}(\vec{T_x} \times \vec{T_y})\, dxdy$$

$$S_1: \begin{cases} x = x \\ y = y \\ z = 1 \end{cases}$$

$$\Rightarrow \vec{T_x} = \vec{i} \quad \Big| \Rightarrow \vec{T_x} \times \vec{T_y} = \vec{k} \qquad (0,0,1)$$
$$\vec{T_y} = \vec{j}$$

$$\Rightarrow F_{S_1}(\vec{V}) = \iint_D \big(P(x,y,1), Q(x,y,1), R(x,y,1)\big)(0,0,1)\, dxdy$$

$$= \iint_D (x,y,1)(0,0,1)\, dxdy = \iint_D 1\, dxdy$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases} \Big| \Rightarrow r \in [0,1], \; t \in [0, 2\pi]$$

$$= \int_0^{2\pi}\int_0^1 r\, drdt = \int_0^{2\pi} \frac{r^2}{2}\Big|_0^1 = \frac{1}{2}\cdot 2\pi = \pi$$

$$\Rightarrow F_S(\vec{V}) = F_\Sigma(\vec{V}) - F_{S_1}(\vec{V}) =$$

$$\begin{cases} x = x \\ y = y \\ z = \sqrt{x^2+y^2} \end{cases} \Rightarrow \vec{T_x} = \vec{i} + \frac{x}{\sqrt{x^2+y^2}}\vec{k}$$

$$\vec{T_y} = \vec{j} + \frac{y}{\sqrt{x^2+y^2}}\vec{k}$$

$$\Rightarrow \vec{T_x} \times \vec{T_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{x}{\sqrt{x^2+y^2}} \\ 0 & 1 & \frac{y}{\sqrt{x^2+y^2}} \end{vmatrix} = \vec{k} - \vec{i}\frac{x}{\sqrt{x^2+y^2}} - \vec{j}\frac{y}{\sqrt{x^2+y^2}}$$

$$\Rightarrow -\vec{T_x} \times \vec{T_y} = -\vec{i}\frac{x}{\sqrt{x^2+y^2}} - \vec{j}\frac{y}{\sqrt{x^2+y^2}} + \vec{k}$$

$$\Rightarrow F(S) = \iint_D (x,y,z)(\uparrow)\, dxdy = \iint_D \left(\frac{-x^2}{\sqrt{x^2+y^2}} - \frac{y^2}{\sqrt{x^2+y^2}} + z\right) dxdy$$

$$= \iint_D \left(-\sqrt{x^2+y^2} + \sqrt{x^2+y^2}\right) dx\,dy = 0$$

23. Să se calculeze fluxul câmpului vectorial $\mathbf{V}(x, y, z) = y\,\mathbf{i} - x\,\mathbf{j} + z^2\,\mathbf{k}$ prin faţa exterioară a paraboloidului $z = x^2 + y^2$, cu $z \in [0, 1]$.

desen...

$z = x^2+y^2$, $z \in [0,1] \Rightarrow D: x^2+y^2 \le 1$

$$\begin{cases} x = x \\ y = y \\ z = x^2+y^2 \end{cases} \Rightarrow \begin{aligned} \overrightarrow{T_x} &= \vec{i} + 2x\vec{k} \\ \overrightarrow{T_y} &= \vec{j} + 2y\vec{k} \end{aligned} \Rightarrow \overrightarrow{T_x} \times \overrightarrow{T_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2x \\ 0 & 1 & 2y \end{vmatrix}$$

$$= \vec{k} - 2x\vec{i} - 2y\vec{j}$$

$$\Rightarrow F_S(\vec{V}) = \iint_D (y, -x, z^2)(-2x, -2y, 1)\,dx\,dy = \iint_D (x^2+y^2)^2\,dx\,dy$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases} \Rightarrow \begin{aligned} r &\in [0,1] \\ t &\in [0, 2\pi] \end{aligned}$$

$$= \int_0^{2\pi}\int_0^1 r^5\,dr\,dt = \int_0^{2\pi} \frac{r^6}{6}\Big|_0^1 = \frac{2\pi}{6} = \frac{\pi}{3}$$

24. Să se calculeze fluxul câmpului vectorial $\mathbf{V}(x, y, z) = \frac{1}{\sqrt{x^2+y^2}}(y\,\mathbf{i} - x\,\mathbf{j} + \mathbf{k})$ prin faţa exterioară a paraboloidului $z = 4 - x^2 - y^2$, cu $z \in [0, 1]$.

$$= \frac{y}{\sqrt{x^2+y^2}}\vec{i} - \frac{x}{\sqrt{x^2+y^2}}\vec{j} + \frac{1}{\sqrt{x^2+y^2}}\vec{k}$$

$$\begin{cases} x = x \\ y = y \\ z = 4 - x^2 - y^2 \end{cases} \Rightarrow x^2+y^2 \le 4 \quad D:$$



$$\Rightarrow \begin{aligned} \overrightarrow{T_x} &= \vec{i} - 2x\vec{k} \\ \overrightarrow{T_y} &= \vec{j} - 2y\vec{k} \end{aligned} \Rightarrow \overrightarrow{T_x} \times \overrightarrow{T_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -2x \\ 0 & 1 & -2y \end{vmatrix} = \vec{k} + 2x\vec{i} + 2y\vec{j}$$

$$\Rightarrow F_S(\vec{V}) = \iint_D \vec{V}(\overrightarrow{T_x} \times \overrightarrow{T_y})\,dx\,dy = \iint_D \left(\frac{y}{\sqrt{x^2+y^2}}, \frac{-x}{\sqrt{x^2+y^2}}, \frac{1}{\sqrt{x^2+y^2}}\right) (2x, 2y, 1)\,dx\,dy$$

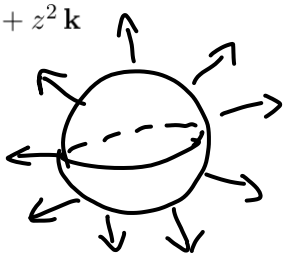
$$= \iint_D \left(\frac{2xy}{\sqrt{x^2+y^2}} - \frac{2xy}{\sqrt{x^2+y^2}} + \frac{1}{\sqrt{x^2+y^2}} \right) dx\,dy$$

$$= \iint_D \frac{1}{\sqrt{x^2+y^2}} dx\,dx$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow r^2 \le 4 \Rightarrow r \in [0,2]$$
$$t \in [0, 2\pi]$$
$$j = r$$

$$= \int_0^{2\pi}\int_0^2 \frac{1}{r} \cdot r\, dr\, dt = \int_0^{2\pi} r\Big|_0^2 dt = 4\pi$$

25. Să se calculeze fluxul câmpului vectorial $\mathbf{V}(x,y,z) = x^2\,\mathbf{i} + y^2\,\mathbf{j} + z^2\,\mathbf{k}$ prin faţa exterioară a sferei $x^2+y^2+z^2 = 1$, cu $x, y, z > 0$.



$$\begin{cases} x = x \\ y = y \\ z = \sqrt{1-x^2-y^2} \end{cases}$$

$$\vec{T_x} = \vec{i} - \frac{x}{\sqrt{1-x^2-y^2}}\vec{k}$$
$$\vec{T_y} = \vec{j} - \frac{y}{\sqrt{1-x^2-y^2}}\vec{k}$$

$$\Rightarrow \vec{T_x} \times \vec{T_y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{-x}{\sqrt{1-x^2-y^2}} \\ 0 & 1 & \frac{-y}{\sqrt{1-x^2-y^2}} \end{vmatrix} =$$

$$= \vec{k} + \frac{x}{\sqrt{1-x^2-y^2}}\vec{i} + \frac{y}{\sqrt{1-x^2-y^2}}\vec{j}$$

sfera e compact bordat

$$\Rightarrow F_S(\vec{V}) = \iint_S \vec{V}\vec{n}\,d\sigma \overset{G-O}{=} \iiint_K \left(\frac{dP}{dx} + \frac{dQ}{dy} + \frac{dR}{dz} \right) dx\,dy\,dz$$

$$= \iiint_K (2x+2y+2z)\,dx\,dy\,dz = 2\iiint_K (x+y+z)\,dx\,dy\,dz$$

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases}$$
$$j = r^2\sin\theta$$

$$x, y, z > 0 \Rightarrow \theta, \varphi \in \left[0, \frac{\pi}{2}\right]$$
$$r \in [0,1]$$

$$= 2 \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^1 (r\sin\theta\cos\varphi + r\sin\theta\sin\varphi + r\cos\theta)\, r^2 \sin\theta \; dr\, d\theta\, d\varphi$$

$$I_1 = \int_0^1 r^3 \sin^2\theta\cos\varphi \, dr = \sin^2\theta\cos\varphi \frac{r^4}{4}\Big|_0^1 = \frac{\sin^2\theta\cos\varphi}{4}$$

$$I_2 = \int_0^1 r^3 \sin^2\theta \sin\varphi \, dr = \frac{\sin^2\theta\cos\varphi}{4}$$

$$I_3 = \int_0^1 r^3 \cos\theta\sin\theta \, dr = \frac{\cos\theta\sin\theta}{4}$$

$$= 2\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \left(\cos\varphi \frac{\sin^2\theta}{2} + \frac{\cos\theta\sin\theta}{4}\right) d\theta\, d\varphi$$

$$= \int_0^{\frac{\pi}{2}} \left(\cos\varphi \int_0^{\frac{\pi}{2}} \sin^2\theta \, d\theta + \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos\theta\sin\theta \, d\theta \right) d\varphi$$

$$\int_0^{\frac{\pi}{2}} \frac{1-\cos 2\theta}{2} d\theta = \frac{1}{2}\left(\theta - \frac{\sin 2\theta}{2}\right)\Big|_0^{\frac{\pi}{2}} = \frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi}{4}$$

Not $\sin\theta = x \Leftrightarrow \cos\theta\, d\theta = dx$

θ	0	$\frac{\pi}{2}$
x	0	1

$$\frac{1}{2}\int_0^1 x\, dx = \frac{1}{2}\cdot\frac{x^2}{2}\Big|_0^1 = \frac{1}{4}$$

$$= \int_0^{\frac{\pi}{2}} \left(\cos\varphi \cdot \frac{\pi}{4} + \frac{1}{4}\right) d\varphi = \frac{\pi}{4}\int_0^{\frac{\pi}{2}} \cos\varphi\, d\varphi + \frac{\pi}{8} =$$

$$= \frac{\pi}{4}\sin\varphi\Big|_0^{\frac{\pi}{2}} + \frac{\pi}{8} = \frac{\pi}{4} + \frac{\pi}{8} = \frac{3\pi}{8}$$

26. Să se calculeze, cu ajutorul formulei Riemann–Green, valoarea integralei $\int_C (xe^{xy} + 2x\cos y)\, dx + (ye^{xy} - x^2\sin y)\, dy$, unde C este frontiera domeniului $D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le a^2\}$, cu $a > 0$.

$C = \mathrm{Fr}\, D \Rightarrow x^2 + y^2 = a^2 \qquad D: x^2 + y^2 \le a^2$

$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \iint_D (y^2 e^{xy} - 2\sin y x - x^2 e^{xy} + 2x\sin y)\, dx\,dy$$

$$= \iint_D (y^2 e^{xy} - x^2 e^{xy})\, dx\,dy = \iint_D (y^2 - x^2) e^{xy}\, dx\,dy$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases} \Rightarrow r^2 \le a^2 \Rightarrow r \in [0, a], \quad t \in [0, 2\pi]$$

$$= \int_0^{2\pi}\int_0^{a} \left(r^2\sin^2 t - r^2\cos^2 t\right) e^{r^2 \sin x \cos x} \cdot r \, dx\,dy$$

$$= \int_0^{2\pi}\int_0^{a} r^3\left(\sin^2 t - \cos^2 t\right) e^{r^2 \sin x \cos x} \, dx\,dy$$

(primitivă deci 0 (sau faci cu ω))

27. Să se calculeze, cu ajutorul formulei Riemann–Green, valoarea integralei $\int_C (x^2+y^2)\,dx + 3xy^2\,dy$, unde C este frontiera domeniului $D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le ax\}$, cu $a > 0$.

$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \iint_D (3y^2 - 2y)\,dx\,dy$$

$$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases} \Rightarrow r^2 \le ar\cos t \;|:r>0 \Leftrightarrow r \le a\cos t \Rightarrow t \in \left[-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right]$$

$$\Rightarrow r \in [0, a\cos t]$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^{a\cos t} \left(3r^3\sin^2 t - 2r^2\sin t\right) dr\,dt = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(3\sin^2 t \cdot \frac{r^4}{4}\Big|_0^{a\cos t} - 2\sin t \frac{r^3}{3}\Big|_0^{a\cos t}\right) dt$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{3}{4}\sin^2 t \cdot a^4\cos^4 t - \underbrace{\frac{2}{3}\sin t\, a^3\cos^3 t}_{=0 \text{ (f. impară)}}\right) dt$$

$$= \frac{3a^4}{4}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 t\cos^4 t\,dt = \frac{3a^4}{4}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1-\cos 2t)\cos^4 t\,dt = \frac{3a^4}{4}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 t - \cos^6 t\,dt$$

$$= \frac{3a^4}{4}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos^4 t - \cos^6 t)\,dt$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 t\,dt = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1+\cos 2t}{2}\right)^2 dt = \frac{1}{4}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + 2\cos 2t + \cos^2 2t)\,dt = \frac{1}{4}\left(t + \underbrace{\sin 2t}_{=0}\right)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \frac{1}{4}\int_0^{\frac{\pi}{2}} \frac{1+\cos 4t}{2}\,dt$$

$$= \frac{1}{4}\cdot\pi + \frac{1}{8}\cdot\left(t + \underbrace{\frac{\sin 4t}{4}}_{=0}\right)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{2\pi}{4} + \frac{\pi}{8} = \frac{3\pi}{8}$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^6 t\,dt = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1+\cos 2t}{2}\right)^3 dt = \frac{1}{8}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1+3\cos 2t+3\cos^2 2t+\cos^3 2t)\,dt$$

$$= \frac{1}{8}\left(t+3\frac{\sin 2t}{2}\right)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \frac{3}{8}\int_0^{\frac{\pi}{2}} \frac{1+\cos 4t}{2}\,dt + \frac{1}{16}\int_0^{\frac{\pi}{2}} \cos 2t\,(1-\sin^2 2t)\,dt$$

$$= \frac{\pi}{8} + \frac{3}{16}\left(t+\frac{\sin 4t}{4}\right)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{3\pi}{8} + \frac{3\pi}{16} = \frac{5\pi}{16}$$

Not $\sin 2t = x \Leftrightarrow 2\cos 2t\,dt = dx$

$\Rightarrow$

x	0	$\frac{\pi}{2}$
t	0	0

$$\Rightarrow I = \frac{3a^2}{4}\left(\frac{3\pi}{8} - \frac{5\pi}{16}\right) = \frac{3a^2\pi}{64}$$

28. Să se calculeze, cu ajutorul formulei Riemann–Green, circulaţia câmpului vectorial $\mathbf{V}(x,y) = xy\,\mathbf{i} + \frac{x^2}{2}\,\mathbf{j}$ de-a lungul curbei C, unde

$C = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 = 1, x \le 0 \le y\} \cup \{(x,y) \in \mathbb{R}^2 \mid x+y = -1, x,y \le 0\}$.

$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \iint_D (x-x)\,dx\,dy = 0$$

29. Să se calculeze, cu ajutorul formulei Riemann-Green, circulaţia câmpului vectorial $\mathbf{V}(x,y) = e^x \cos y\,\mathbf{i} - e^x \sin y\,\mathbf{j}$ de-a lungul curbei C, unde C este o curbă arbitrară conţinută în semiplanul superior, care uneşte punctele $A(1,0)$ şi $B(-1,0)$, sensul fiind de la A către B.

$$\mathcal{C}_C(\vec{V}) = \int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \iint_D (-e^x \sin y - e^x(-\sin y))\,dx\,dy = 0$$

$$\frac{\partial P}{\partial y} = -e^x \sin y$$

$$\frac{\partial Q}{\partial x} = -e^x \sin y$$

$\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \omega$ $\begin{cases} \text{închisă} \\ \mathbb{R}^2 \text{ delată} \end{cases} \Rightarrow \omega$ exactă

$$\Rightarrow \int_C \omega = f(-1,0) - f(1,0) \Rightarrow \exists f \in \mathcal{C}^1 \text{ a.î. } \begin{cases} \frac{\partial f}{\partial x} = P \\ \frac{\partial f}{\partial y} = Q \end{cases}$$

$$\begin{cases} \frac{\partial f}{\partial x} = e^x \cos y \\ \frac{\partial f}{\partial y} = -e^x \sin y \end{cases} \overset{\int dx}{\Longrightarrow} f(x,y) = \cos y\, e^x + c(y)$$

$$\Rightarrow \frac{\partial f}{\partial y} = e^x \cdot (-\sin y) + c'(y) \Rightarrow c'(y) = 0 \Rightarrow c(y) = c_1$$

$\Rightarrow f(x,y) = e^x \cos y + c_1$

$\Rightarrow \int_c \omega = f(-1,0) - f(1,0) = e^{-1} + c_1 - e - c_1 = \frac{1}{e} - e$

30. Considerăm forma diferenţială $\omega = -\frac{y}{x^2+y^2}\,dx + \frac{x}{x^2+y^2}\,dy$.

a) Să se calculeze $\int_{\mathscr{C}(O,R)} \omega$, unde $\mathscr{C}(O,R)$ este cercul de centru $O(0,0)$ şi rază $R > 0$.

b) Să se calculeze $\int_C \omega$, unde C este o curbă închisă astfel încât $O(0,0) \notin C$.

a) $\frac{\partial P}{\partial y} = -\frac{(x^2+y^2)-2y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$

$\frac{\partial Q}{\partial x} = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$

$\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow \omega$ închisă, $\mathbb{R}^2$ delată $\Rightarrow \omega$ exactă

$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases} \Rightarrow x^2+y^2 = R^2 \Rightarrow r = R \Rightarrow \begin{cases} x = R\cos t \\ y = R\sin t \end{cases}$

$t = r$

$\Rightarrow \int_c P\,dx + Q\,dy = \int_0^{2\pi} \left(\frac{-R\sin t}{R^2} \cdot R(-\sin t) + \frac{R\cos t}{R^2} \cdot R\cos t \right) dt =$

$= \int_0^{2\pi} (\sin^2 t + \cos^2 t)\,dt = \int_0^{2\pi} dt = 2\pi$

b) ω exactă $\Rightarrow \exists f \in \mathscr{C}^1$ a.î. $\begin{cases} \frac{\partial f}{\partial x} = P = -\frac{y}{x^2+y^2} \\ \frac{\partial f}{\partial y} = Q = \frac{x}{x^2+y^2} \end{cases} \Rightarrow$

$\Rightarrow f(x,y) = \int \frac{x}{y^2+x^2}\,dy = x \int \frac{1}{y^2+x^2}\,dy = x \cdot \frac{1}{x} \operatorname{arctg} \frac{y}{x} + c(x)$

$\Rightarrow \frac{\partial f}{\partial x} = \left(\operatorname{arctg} \frac{y}{x}\right)' + c'(x) = \frac{1}{\left(\frac{y}{x}\right)^2+1} \cdot \frac{-y}{x^2} + c'(x) = \frac{1}{\frac{y^2+x^2}{x^2}} \cdot \frac{-y}{x^2} = \frac{-y}{x^2+y^2} + c'(x)$

ω exactă, curbă închisă, dacă nu conţine în interior $O(0,0)$ $\Rightarrow \int_c \omega = 0$

dacă da: $\int_{\text{acea}} \omega = \int_c \omega - \int_{\mathscr{C}(0,R)} \omega$

$\Rightarrow \int_c \omega = 2\pi$

31. Considerăm forma diferenţială $\omega = \frac{x-y}{x^2+y^2}\,dx + \frac{x+y}{x^2+y^2}\,dy$. Să se calculeze integrala $\int_C \omega$, unde C este o curbă închisă astfel încât $O(0,0) \notin C$.

$$\frac{\partial P}{\partial y} = \frac{-(x^2+y^2)-(x-y)\cdot 2y}{(x^2+y^2)^2} = \frac{-x^2-y^2-2xy+2y^2}{(x^2+y^2)^2} = \frac{y^2-2xy-x^2}{(x^2+y^2)^2}$$

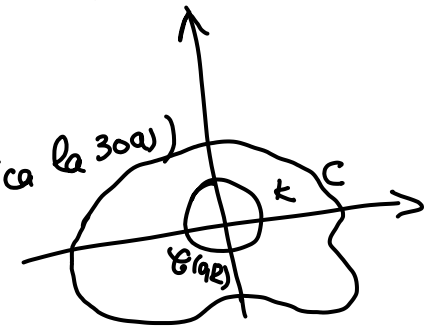
$$\frac{\partial Q}{\partial x} = \frac{(x^2+y^2)-(x+y)2x}{(x^2+y^2)^2} = \frac{x^2+y^2-2x^2-2xy}{(x^2+y^2)^2}$$

$\Rightarrow \omega$ închisă

Dacă $O(0,0) \in \text{Int } C \Rightarrow \int_C \omega = \int_\Gamma \omega + \int_{\mathcal{C}(O,R)} \omega = 2\pi$

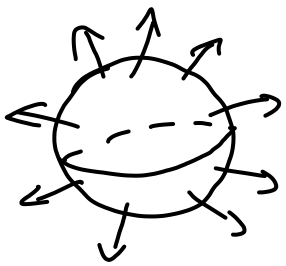
$\overset{\text{Fin K}}{=} \mathcal{C}(O,R) \simeq 2\pi$ (ca la 30a)

0 $\left(\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}\right)$



Dacă $O(0,0) \notin \text{Int } C \Rightarrow \int_C \omega = 0$

32. Să se calculeze, cu ajutorul formulei Gauss–Ostrogradski, fluxul câmpului vectorial $\mathbf{V}(x,y,z) = x^3\,\mathbf{i} + y^3\,\mathbf{j} + z^3\,\mathbf{k}$ prin faţa exterioară a sferei $x^2+y^2+z^2 = a^2$, cu $a > 0$.



$$F_S(\vec{V}) = \iint_S \vec{V}\cdot\vec{n}\,d\sigma \overset{G-O}{=}$$

$$= \iiint_K \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz$$

$$= \iiint_K (3x^2+3y^2+3z^2)\,dx\,dy\,dz =$$

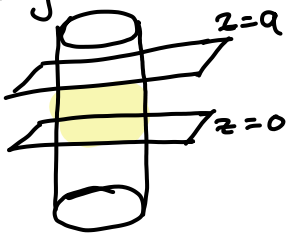
$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \\ j = r^2\sin\theta \end{cases} \Rightarrow r^2 \le a^2 \Rightarrow r \in [0,a], \quad \theta \in [0,\pi], \varphi \in [0,2\pi]$$

$$= 3\int_0^{2\pi}\int_0^{2\pi}\int_0^a r^4\sin\theta\,dr\,d\theta\,d\varphi = 3\int_0^{2\pi}\int_0^{2\pi} \sin\theta\cdot\frac{r^5}{5}\Big|_0^a d\theta\,d\varphi = \frac{3a^5}{5}\int_0^{2\pi} -\cos\theta\Big|_0^\pi d\varphi$$

$$= \frac{3a^5}{5}\int_0^{2\pi} 2\,d\varphi = \frac{12a^5\pi}{5}$$

33. Să se calculeze, cu ajutorul formulei Gauss–Ostrogradski, fluxul câmpului vectorial $\mathbf{V}(x,y,z) = x^3\,\mathbf{i} + x^2y\,\mathbf{j} + x^2z\,\mathbf{k}$ prin suprafaţa cilindrului $x^2+y^2 = r^2$, cuprinsă între planele $z=0$ şi $z=a$, cu $r, a > 0$.

$$F_S(\vec{V}) = \iint_S \vec{V}\cdot\vec{n}\,d\sigma = \iiint_K \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz$$



$$= \iiint_K (3x^2 + x^2 + x^2)\,dx\,dy\,dz = 5\iiint_K x^2\,dx\,dy\,dz$$

$D: x^2+y^2 \le r^2$

$$= 5\iint_D x^2 \left(\int_0^a dz\right) dx\,dy = 5a\iint_D x^2\,dx\,dy$$

$$\begin{cases} x = \rho\cos t \\ y = \rho\sin t \end{cases} \Rightarrow \begin{matrix} \rho \in [0, r] \\ t \in [0, 2\pi] \end{matrix}$$

$$j = \rho$$

$$= 5a\int_0^{2\pi}\int_0^r \rho^2\cos^2 t\cdot\rho\,d\rho\,dt = 5a\int_0^{2\pi} \cos^2 t\cdot\frac{\rho^4}{4}\Big|_0^r dt = \frac{5ar^4}{4}\int_0^{2\pi}\cos^2 t\,dt$$

$$= \frac{5ar^4}{4}\int_0^{2\pi}\frac{1+\cos 2t}{2}dt = \frac{5ar^4}{4}\left(\frac{1}{2}t\Big|_0^{2\pi} + \frac{1}{2}\frac{\sin 2t}{2}\Big|_0^{2\pi}\right) = \frac{5ar^4\pi}{4}$$

($\sin 2t$ evaluat între 0 și 2π este 0)

34. Să se calculeze, cu ajutorul formulei Gauss–Ostrogradski, fluxul câmpului vectorial $\mathbf{V}(x,y,z) = x^2yz\,\mathbf{i} + xy^2z\,\mathbf{j} + xyz^2\,\mathbf{k}$ prin faţa exterioară a sferei $x^2+y^2+z^2 = 1$, cu $x, y, z \ge 0$.

$\Rightarrow$ ca 32

35. Să se calculeze, cu ajutorul formulei Gauss–Ostrogradski, fluxul câmpului vectorial $\mathbf{V}(x,y,z) = xz\,\mathbf{i} + xy\,\mathbf{j} + yz\,\mathbf{k}$ prin suprafaţa S, bordul domeniului $D = \left\{(x,y,z) \in \mathbb{R}^3 \mid 2x^2+y^2 \le z \le 1-\left(x^2+2y^2\right)\right\}$, orientată după normala interioară.

$$F_S(\vec{V}) = \iint_S \vec{V}\cdot\vec{n}\,d\sigma \overset{G-O}{=} \iiint_K \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz = \iiint_K (z+x+y)\,dx\,dy\,dz$$

$$= \iint_D \left(\int_{2x^2+y^2}^{1-(x^2+2y^2)} (x+y+z)\,dz\right) dx\,dy = \iint_D \left((x+y)z\Big|_{2x^2+y^2}^{1-(x^2+2y^2)} + \frac{z^2}{2}\Big|_{2x^2+y^2}^{1-(x^2+2y^2)}\right) dx\,dy$$

$$= \iint_D \left((x+y)\left(1-x^2-2y^2-2x^2-y^2\right) + \frac{1-2x^2-2y^2-(x^2+y^2)^2-(2x^2+y^2)^2}{2}\right) dx\,dy$$

$$= \iint_D \left((x+y)\left(1-3(x^2+y^2)\right) -\right.$$

$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases} \Big| \Rightarrow 2x^2+y^2 \le 1-(x^2+2y^2) \Leftrightarrow 2x^2+y^2 \le 1-x^2-2y^2$

$\Leftrightarrow 3x^2+3y^2 \le 1 \Leftrightarrow x^2+y^2 \le \frac{1}{3} \Rightarrow r^2 \le \frac{1}{3}$

$\Rightarrow r \in \left[0, \frac{1}{\sqrt{3}}\right]$

$t \in [0, 2\pi]$

$$= \int_0^{2\pi}\int_0^{\frac{1}{\sqrt{3}}} \left((r\cos t + r\sin t)(1-3r^2) - \frac{r^2}{2}\right) r\,dr\,dt =$$

$$= \int_0^{2\pi}\int_0^{1/\sqrt{3}} \left(r^2\cos t + r^2\sin t - 3r^4\cos t - 3r^4\sin t + \frac{r^3}{2}\right) dr\,dt$$

$$= \int_0^{2\pi} \left(\cos t\,\frac{r^3}{3} + \sin t\,\frac{r^3}{3} - \cos t\cdot\frac{3r^5}{5} - \sin t\cdot\frac{3r^5}{5} + \frac{r^4}{8}\right)\Bigg|_0^{1/\sqrt{3}} dt$$

$$= \int_0^{2\pi} \left(\cos t\cdot\frac{1}{9\sqrt{3}} + \sin\cdot\frac{1}{9\sqrt{3}} - \cos t\cdot 3\cdot\frac{1}{5\cdot 9\sqrt{3}} - \sin t\cdot 3\cdot\frac{1}{5\cdot 9\sqrt{3}} + \frac{1}{8\cdot 9}\right) dt$$

$$= \int_0^{2\pi} \left((\cos t + \sin t)\left(\frac{5/1}{9\sqrt{3}} - \frac{3/1}{15\sqrt{3}}\right) + \frac{1}{72}\right) dt = \int_0^{2\pi} (\cos t + \sin t)\cdot\frac{2}{15\sqrt{3}}\,dt + \frac{\pi}{36}$$

. . . . . (hub rufaiuk calculele)

36. Să se calculeze, cu ajutorul formulei Gauss–Ostrogradski, fluxul câmpului vectorial $\mathbf{V}(x,y,z) = x(y^2-z^2)\,\mathbf{i} + y(z^2-x^2)\,\mathbf{j} + z(x^2-y^2)\,\mathbf{k}$ prin suprafaţa S, bordul domeniului $D = \{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = a^2\}$, cu $a > 0$, orientată după normala exterioară. sferă

$$F_S(\vec{V}) = \iint_S \vec{V}\vec{n}\,d\sigma = \iiint_K \left(\frac{dP}{dx} + \frac{dQ}{dy} + \frac{dR}{dz}\right) dx\,dy\,dz =$$

$$= \iiint_K (y^2 - z^2 + z^2 - x^2 + x^2 - y^2)\,dx\,dy\,dz = 0$$

37. Să se calculeze, cu ajutorul formulei Gauss–Ostrogradski, fluxul câmpului vectorial $\mathbf{V}(x,y,z) = x(z+3)\,\mathbf{i} + yz\,\mathbf{j} - (z+z^2)\,\mathbf{k}$ prin suprafaţa sferei $x^2+y^2+z^2 = 1$, cu $z \ge 0$. $\Rightarrow z = \sqrt{1-x^2-y^2}$

$$F_S(\vec{V}) = \iint_S \vec{V}\vec{n}\,d\sigma = \iiint_K \left(\frac{dP}{dx} + \frac{dQ}{dy} + \frac{dR}{dz}\right) dx\,dy\,dz = \iiint_K (z+3+z-1-2z)\,dx\,dy\,dz$$

$$= \iiint_K 2\,dx\,dy\,dz = 2\iint_D \left(\int_0^{\sqrt{1-x^2-y^2}} dz\right) dx\,dy = 2\iint_D \sqrt{1-x^2-y^2}\,dx\,dy$$

$\begin{cases} x = r\cos t \\ y = r\sin t \\ j = r \end{cases} \Big| \Rightarrow r^2 \le 1 \Rightarrow r \in [0,1]$

$t \in [0, 2\pi]$

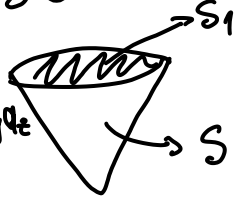
$$= \int_0^{2\pi}\int_0^1 \sqrt{1-r^2}\, 2r\, dr\, dt = \int_0^{2\pi}\int_0^1 \sqrt{x}\, dx\, dt$$

Not $1-r^2 = x \Leftrightarrow -2r\,dr = dx$ $\quad \begin{array}{c|cc} r & 0 & 1 \\ \hline x & 1 & 0 \end{array}$

$$= \int_0^{2\pi} \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\Big|_0^1 dt = \frac{2}{3}\int_0^{2\pi} dt = \frac{4\pi}{3}$$

38. Să se calculeze, cu ajutorul formulei Gauss–Ostrogradski, fluxul câmpului vectorial $\mathbf{V}(x,y,z) = x^2(y-z)\,\mathbf{i} + y^2(z-x)\,\mathbf{j} + z^2(x-y)\,\mathbf{k}$ prin suprafaţa conului $z^2 = x^2 + y^2$, cu $0 < z \leq 1$.

$\Sigma = S \cup S_1$



$$F_{\Sigma}(\vec{V}) = \iint_S \vec{V}\vec{n}\, d\sigma = \iiint_K \left(\frac{dP}{dx} + \frac{dQ}{dy} + \frac{dR}{dz}\right) dx\,dy\,dz$$

$$= \iiint_K \left(2(y-z)x + 2(z-x)y + 2(x-y)z\right) dx\,dy\,dz$$

$$= \iiint_K \left(2xy - 2xz + 2zy - 2xy + 2xz - 2yz\right) dx\,dy\,dz = 0$$

$D: x^2+y^2 \leq 1$, $z = \sqrt{x^2+y^2}$

$$F_{S_1}(\vec{F}) = \iint_{S_1} \vec{F}\cdot\vec{n}\, d\sigma = \iint_D \vec{F}\cdot(\vec{T_x} \times \vec{T_y})\, dx\,dy$$

$0 < z \leq 1 \Rightarrow \begin{cases} x = x \\ y = y \\ z = 1 \end{cases}$

$\vec{T_x} = \vec{i}$, $\vec{T_y} = \vec{j}$ $\Big| \Rightarrow \vec{T_x} \times \vec{T_y} = \vec{k}$

$$\Rightarrow F_{S_1}(\vec{F}) = \iint_D (P,Q,R)(0,0,1)\, dx\,dy = \iint_D (x-y)\, dx\,dy$$

$\begin{cases} x = r\cos t \\ y = r\sin t \end{cases}$ $\Big| \Rightarrow r^2 \leq 1 \Rightarrow r \in [0,1]$, $\Rightarrow t \in [0, 2\pi]$

$j = r$

$$= \int_0^{2\pi}\int_0^1 r^2(\cos t - \sin t)\,dr\,dt = \frac{1}{3}\int_0^{2\pi}(\cos t - \sin t)\,dt =$$

$$= \frac{1}{3}(\sin t + \cos t)\Big|_0^{2\pi} = 0$$

$$\Rightarrow F_S(\vec{v}) = F_{S_2}(\vec{v}) - F_{S_1}(\vec{v}) = 0$$

Nr 1

① min și max pt $f(x,y) = x^3 + y^2$ pe $K = \{(x,y) \in \mathbb{R} \mid x^2 + y^2 \le 1\}$

I. $\text{Int}K : x^2 + y^2 < 1$

$$\begin{cases} \frac{df}{dx} = 0 \\ \frac{df}{dy} = 0 \end{cases} \Leftrightarrow \begin{cases} 3x^2 = 0 \\ 2y = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases} \Rightarrow A \in \text{Int}K$$

$f(0,0) = 0$

II. $\text{Fr}K : x^2 + y^2 = 1$

fie $g(x,y) = x^2 + y^2 - 1 = 0$

$$\nabla f = \lambda \nabla g \Leftrightarrow (3x^2, 2y) = \lambda(2x, 2y)$$

$$\Leftrightarrow \begin{cases} 3x^2 = \lambda 2x \\ 2y = \lambda 2y \end{cases} \Leftrightarrow \begin{cases} 3x^2 - \lambda 2x = 0 \\ 2y - \lambda 2y = 0 \end{cases} \Leftrightarrow \begin{cases} x(3x - 2\lambda) = 0 \\ 2y(1 - \lambda) = 0 \end{cases}$$

I. $y = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1 \Rightarrow A(1,0), B(-1,0)$

$f(1,0) = 1$

$f(-1,0) = -1$

II. $1 - \lambda = 0 \Rightarrow \lambda = 1 \Rightarrow x(3x - 2) = 0$

$\Rightarrow$ 1. $x = 0 \Rightarrow y = \pm 1 \Rightarrow C(0,1), D(0,-1)$

$f(0,1) = 1 = f(0,-1)$

2. $x = \frac{2}{3} \Rightarrow y^2 = 1 - \frac{4}{9} \Rightarrow y = \pm\frac{\sqrt{5}}{3} \Rightarrow E\left(\frac{2}{3}, \frac{\sqrt{5}}{3}\right), F\left(\frac{2}{3}, -\frac{\sqrt{5}}{3}\right)$

$$f\left(\frac{2}{3}, \frac{\sqrt{5}}{3}\right) = f\left(\frac{2}{3}, -\frac{\sqrt{5}}{3}\right) = \frac{8}{27} + \frac{3\sqrt{5}}{9} = \frac{23}{27}$$

$\Rightarrow$ MAX = 1, MIN = -1

② $\int_0^\infty \frac{dx}{\sqrt[3]{x}\,(1+x^2)^2} = \frac{1}{2}\int_0^\infty \frac{x^{-\frac{1}{3}}}{(1+x^2)^2}\,dx = \frac{1}{2}\int_0^\infty \frac{x^{-\frac{1}{3}}}{(1+x^2)^2}\cdot 2x\,dx$

Not $x^2 = t \Leftrightarrow 2x\,dx = dt$

x	0	∞
t	0	∞

$$= \frac{1}{2}\int_0^\infty \frac{t^{-\frac{2}{3}}}{(1+t)^2}\,dt = \frac{1}{2}\,\beta\left(\frac{1}{3},\frac{5}{3}\right) = \frac{1}{2}\,\frac{\Gamma\left(\frac{1}{3}\right)\cdot\Gamma\left(\frac{5}{3}\right)}{\Gamma(2)}$$

$$\Rightarrow \begin{cases} p-1 = -\frac{2}{3} \Rightarrow p = \frac{1}{3} \\ p+q = 2 \Rightarrow q = \frac{5}{3} \end{cases}$$

$$= \frac{1}{2}\,\frac{\Gamma\left(\frac{1}{3}\right)\cdot\frac{2}{3}\cdot\Gamma\left(\frac{2}{3}\right)}{1} = \frac{1}{3}\cdot\frac{\pi}{\sin\frac{\pi}{3}} = \frac{\pi}{3}\cdot\frac{2}{\sqrt{3}} = \frac{2\pi}{3\sqrt{3}}$$

③ Vol = ? supraf mărg de $S_1: 2x^2+y^2+z^2=1$ și $S_2: 2x^2+y^2-z^2=0$

$S_1 \cap xOy \xrightarrow{z=0} \frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2} + y^2 = 1$ elipsă

$S_1 \cap xOz \Rightarrow \frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2} + z^2 = 1$ elipsă

$S_1 \cap yOz \xrightarrow{x=0} y^2+z^2=1$ cerc

$S_1 \cap Ox \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm\frac{1}{\sqrt{2}}$

$S_1 \cap Oy \Rightarrow y^2 = 1 \Rightarrow y = \pm 1$

$S_1 \cap Oz \Rightarrow z = \pm 1$

$S_2 \cap xOy \Rightarrow 2x^2+y^2=0 \Rightarrow$ pct $O(0,0,0)$

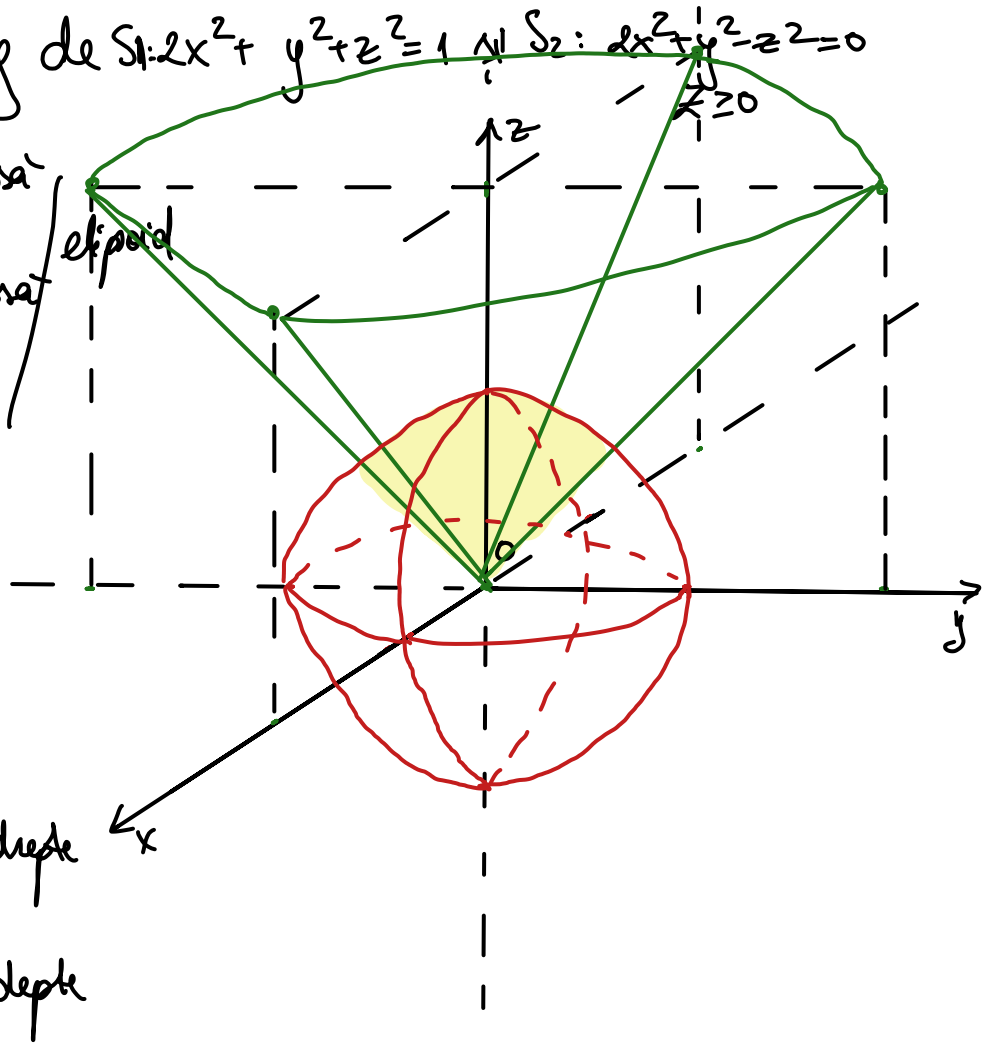
$S_2 \cap xOz \Rightarrow 2x^2 = z^2 \Big| \Rightarrow z = \sqrt{2}|x|$ ($2x^2 \geq 0$)

$z = \pm\sqrt{2}x$ drepte

ieșe pct $(\pm\sqrt{2},0,2)$

$S_2 \cap yOz \Rightarrow y^2 = z^2 \Big| \Rightarrow z = \pm y$ drepte ($z^2 \geq 0$)

ieșe pct $(0,\pm 2,2)$



$$\begin{cases} 2x^2+y^2+z^2=1 \Rightarrow z^2=1-2x^2-y^2 \\ 2x^2+y^2-z^2=0 \Rightarrow z^2=2x^2+y^2 \end{cases} \uparrow \quad 2z^2=1 \Rightarrow z^2=\frac{1}{2} \;\Big|\; z\geq 0 \Rightarrow z=\frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2}+y^2=\frac{1}{2} \quad \Rightarrow \quad D: \frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2}+y^2\leq\frac{1}{2}$$

$$\Rightarrow Vol(K)=\iiint_K dx\,dy\,dz=\iint_D\left(\int_{\sqrt{2x^2+y^2}}^{\sqrt{1-2x^2-y^2}} dz\right)dx\,dy=\iint_D\left(\sqrt{1-2x^2-y^2}-\sqrt{2x^2+y^2}\right)dx\,dy$$

$$\begin{cases} x=\frac{r}{\sqrt{2}}\cos t \\ y=r\sin t \\ j=\frac{r}{\sqrt{2}} \end{cases} \;\Big|\; \Rightarrow r^2\leq\frac{1}{2} \Rightarrow r\in\left[0,\frac{1}{\sqrt{2}}\right], \quad t\in[0,2\pi]$$

$$=\int_0^{2\pi}\int_0^{1/\sqrt{2}}\left(\sqrt{1-r^2}-r\right)\frac{r}{\sqrt{2}}\,dr\,dt=\frac{1}{\sqrt{2}}\int_0^{2\pi}\left(\frac{1}{2}\int_0^{1/\sqrt{2}}\sqrt{1-r^2}\cdot 2r\,dr-\frac{r^3}{3}\Big|_0^{\frac{1}{\sqrt{2}}}\right)$$

Not $1-r^2=x \Leftrightarrow -2r\,dr=dx \Rightarrow$

r	0	$1/\sqrt{2}$
x	1	$\frac{1}{2}$

$$=\frac{1}{\sqrt{2}}\int_0^{2\pi}\left(\frac{1}{2}\int_{\frac{1}{2}}^{1}\sqrt{x}\,dx-\frac{1}{6\sqrt{2}}\right)dt=\frac{1}{\sqrt{2}}\int_0^{2\pi}\left(\frac{1}{2}\cdot\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\Big|_{\frac{1}{2}}^{1}-\frac{1}{6\sqrt{2}}\right)dt$$

$$=\frac{1}{\sqrt{2}}\int_0^{2\pi}\left(\frac{1}{3}-\frac{1}{6\sqrt{2}}-\frac{1}{6\sqrt{2}}\right)dt=\frac{1}{\sqrt{2}}\int_0^{2\pi}\frac{\sqrt{2}-1}{3\sqrt{2}}\,dt=$$

$$=\frac{\sqrt{2}-1}{3\cdot 2}\cdot 2\pi=\frac{\pi(\sqrt{2}-1)}{3}$$

④ $A=?$ $\qquad xy=p,\ xy=q,\ y=ax,\ y=bx$

$0<p<q,\ 0<a<b,\ x>0$

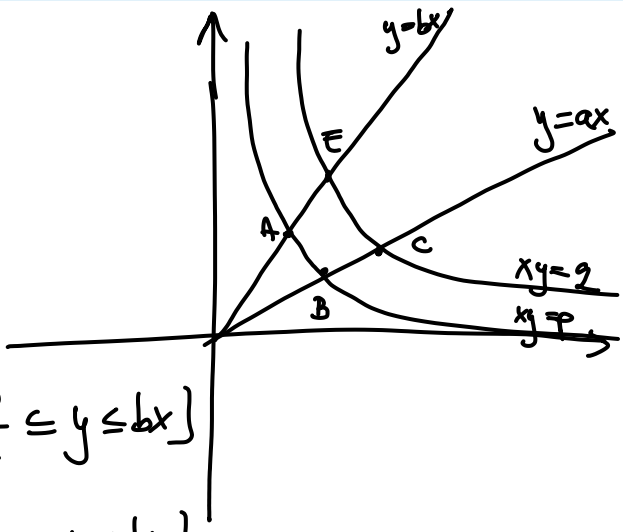
A: $\begin{cases} y=\frac{p}{x} \\ y=bx \end{cases} \Rightarrow \frac{p}{x}=bx$

$x^2=\frac{p}{b} \Rightarrow x=\sqrt{\frac{p}{b}} \Rightarrow A\left(\sqrt{\frac{p}{b}},\sqrt{pb}\right)$

B: $\begin{cases} y=\frac{p}{x} \\ y=ax \end{cases} \Rightarrow x=\sqrt{\frac{p}{a}} \Rightarrow B\left(\sqrt{\frac{p}{a}},\sqrt{pa}\right)$

$C:\ \begin{cases} y=\frac{q}{x} \\ y=ax \end{cases} \Rightarrow C\left(\sqrt{\frac{q}{a}}, \sqrt{qa}\right)$

$E:\ \begin{cases} y=\frac{q}{x} \\ y=bx \end{cases} \Rightarrow E\left(\sqrt{\frac{q}{b}}, \sqrt{qb}\right)$



$D_1 = \{(x,y)\in\mathbb{R}^2 \mid x\in\left[\sqrt{\frac{p}{b}}, \sqrt{\frac{p}{a}}\right],\ \frac{p}{x}\le y\le bx\}$

$D_2 = \{(x,y)\in\mathbb{R}^2 \mid x\in\left[\sqrt{\frac{p}{a}}, \sqrt{\frac{q}{b}}\right],\ ax\le y\le bx\}$

$D_3 = \{(x,y)\in\mathbb{R}^2 \mid x\in\left[\sqrt{\frac{q}{b}}, \sqrt{\frac{q}{a}}\right],\ ax\le y\le \frac{q}{x}\}$

$$A(D_1) = \int_{\sqrt{\frac{p}{b}}}^{\sqrt{\frac{p}{a}}} \int_{\frac{p}{x}}^{bx} 1\,dy\,dx = \dots$$

$$A(D_2) = \int_{\sqrt{\frac{p}{a}}}^{\sqrt{\frac{q}{b}}} \int_{ax}^{bx} 1\,dy\,dx = \dots$$

$$A(D_3) = \int_{\sqrt{\frac{q}{b}}}^{\sqrt{\frac{q}{a}}} \int_{ax}^{\frac{q}{x}} 1\,dy\,dx = \dots$$

$A_{tot} = A(D_1) + A(D_2) + A(D_3)$

⑤ $\vec{v} = (xy^2+x)\vec{i} + xy\vec{j}$

$C = \{(x,y)\in\mathbb{R}^2 \mid \underset{C_1}{y=x^2-4},\ \underset{C_1}{x\ge 0},\ y\le 0\} \cup \{(x,y)\in\mathbb{R}^2 \mid \underset{C_2}{\frac{x^2}{4}+y^2=1},\ x\ge 0,\ y\le 0\}$

$$\int_C \omega = \int_{C_1} \omega + \int_{C_2} \omega$$

$C_1:\ \begin{cases} x=t \\ y=t^2-4 \\ x\ge 0 \end{cases}$

$y\le 0 \Rightarrow x^2-4\le 0 \Rightarrow x^2\le 4 \mid \Rightarrow x\le 2 \quad \mid \Rightarrow t\in[0,2]$

$$\int_{C_1} \omega = \int_0^2 \left(P(t, t^2-4) \cdot 1 + Q(t, t^2-4) \cdot 2t \right) dt$$

$$= \int_0^2 \left(t(t^2-4)^2 + t + t(t^2-4) \cdot 2t \right) dt \quad \text{etc}$$

$C_2: \quad \frac{x^2}{2^2} + y^2 = 1$

$$\begin{cases} x = 2\cos t \\ y = \sin t \end{cases}$$

$x \geq 0 \mid \Rightarrow \cos t \geq 0 \quad \Rightarrow t \in \left[-\frac{\pi}{2}, 0\right]$

$y \leq 0 \quad \sin t \leq 0$

? $j = 2$

$$\int_{C_2} \omega = \int_{-\frac{\pi}{2}}^{0} \left(P(2\cos t, \sin t) \cdot (-2\sin t) + Q(2\cos t, \sin t) \cdot \cos t \right) dt$$

$$\mathcal{C}_c(\vec{v}) = \int_c \vec{V}\, d\vec{r}$$